

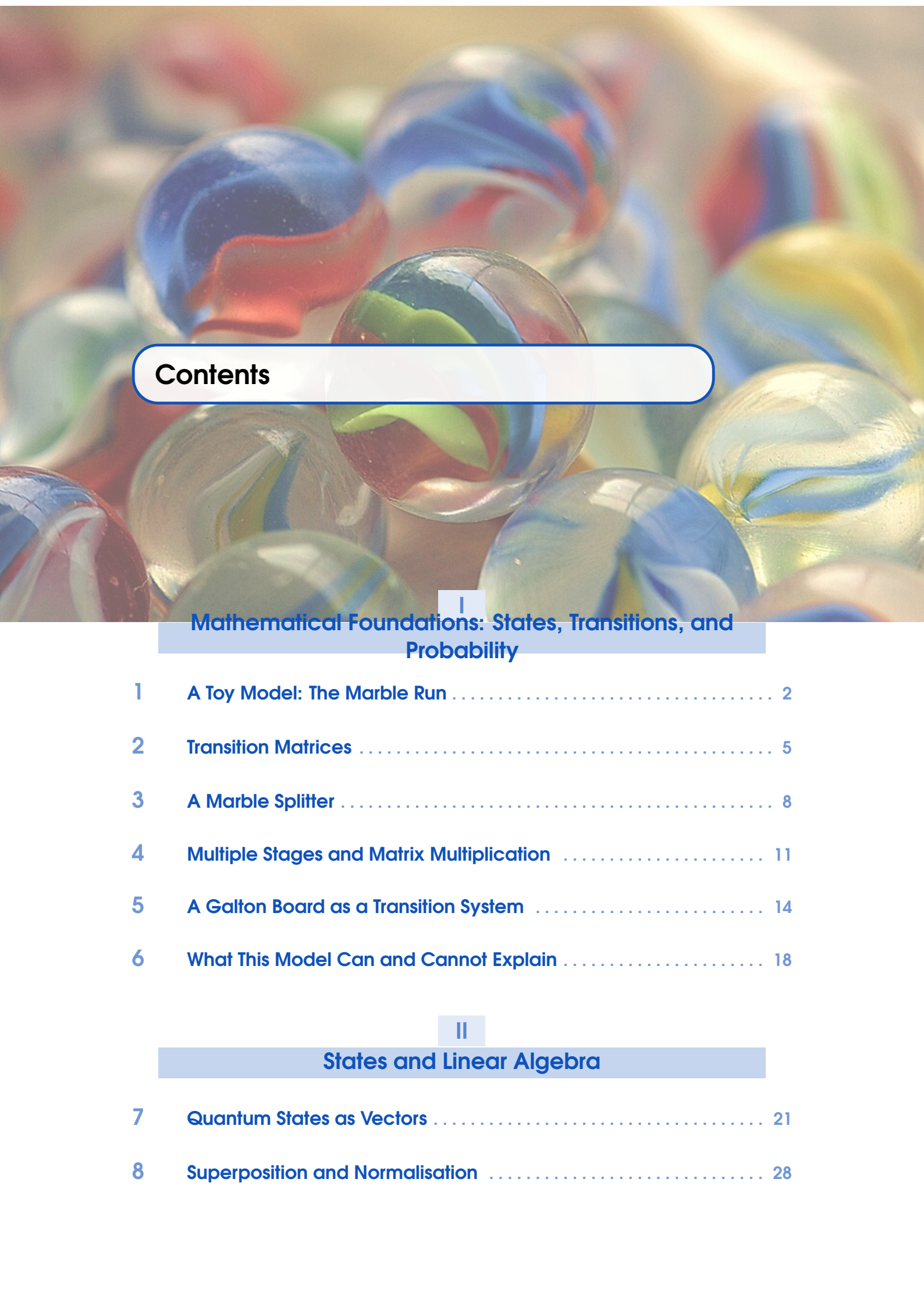
Photonic Circuits

An Introduction to Linear Optics Quantum Computing

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A close-up, slightly blurred photograph of many colorful marbles in shades of blue, red, green, and yellow. The marbles are scattered across the frame, creating a vibrant, textured background.

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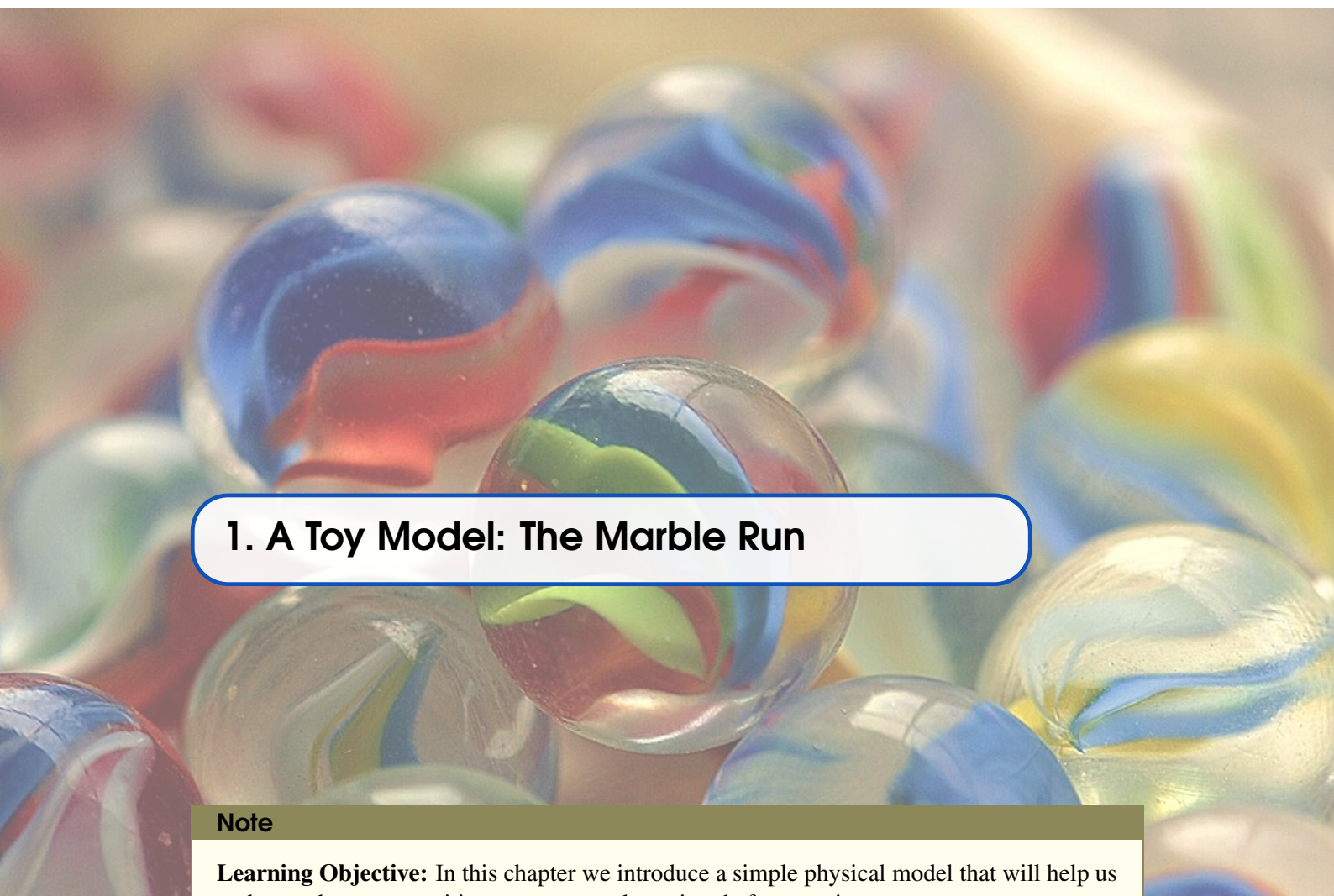
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1. A Toy Model: The Marble Run

Note

Learning Objective: In this chapter we introduce a simple physical model that will help us understand states, transitions, vectors, and matrices before moving to quantum systems.

1.1 Why Begin with a Toy Model?

Before studying photons, beam splitters, and quantum states, we begin with a simpler system: a marble run.

Imagine a marble moving through a device made from channels, splitters, and output bins. At each stage, the marble may move from one channel to another according to fixed rules.

This system is not quantum mechanical. It uses ordinary probabilities. However, it has the same basic mathematical structure that we will later use for optical circuits:

state vector $\longrightarrow$ transition matrix $\longrightarrow$ new state vector

Note

In this part, all numbers are real and non-negative. We are describing probability, not quantum amplitudes.

1.2 States as Probability Vectors

Suppose a marble can be in one of three channels:

$A, \quad B, \quad C.$

If we know the marble is definitely in channel A , we write:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

If it is definitely in channel B , we write:

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

If it is definitely in channel C , we write:

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Definition

Definition 1.1. A **probability vector** is a vector whose entries are probabilities and whose entries add to 1.

For example,

$$\begin{pmatrix} 0.2 \\ 0.5 \\ 0.3 \end{pmatrix}$$

means:

$$P(A) = 0.2, \quad P(B) = 0.5, \quad P(C) = 0.3.$$

1.3 A State Is a Snapshot

A state vector tells us what we know about the system at one moment.

It does not tell us how the system changes. It only tells us the current probabilities.

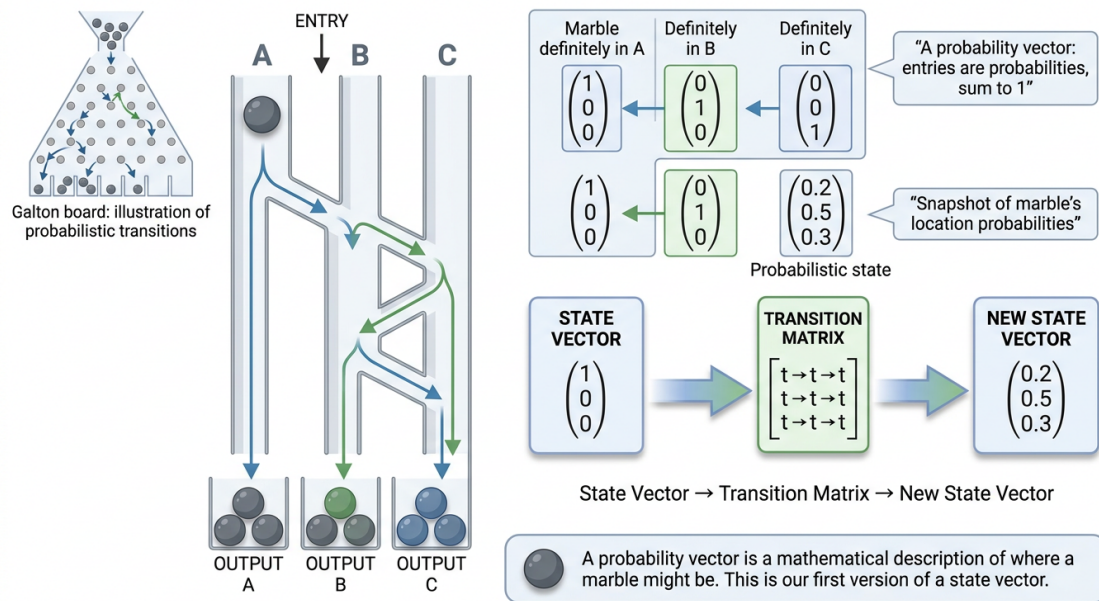
Example

Example 1.2. If

$$\mathbf{p} = \begin{pmatrix} 0.7 \\ 0.2 \\ 0.1 \end{pmatrix},$$

then the marble is most likely to be in channel A , but it is not certain.

1.4 Exercises



Exercise

Exercise 1.1. Which of the following are valid probability vectors?

$$\begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix}, \quad \begin{pmatrix} 0.2 \\ 0.2 \\ 0.7 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

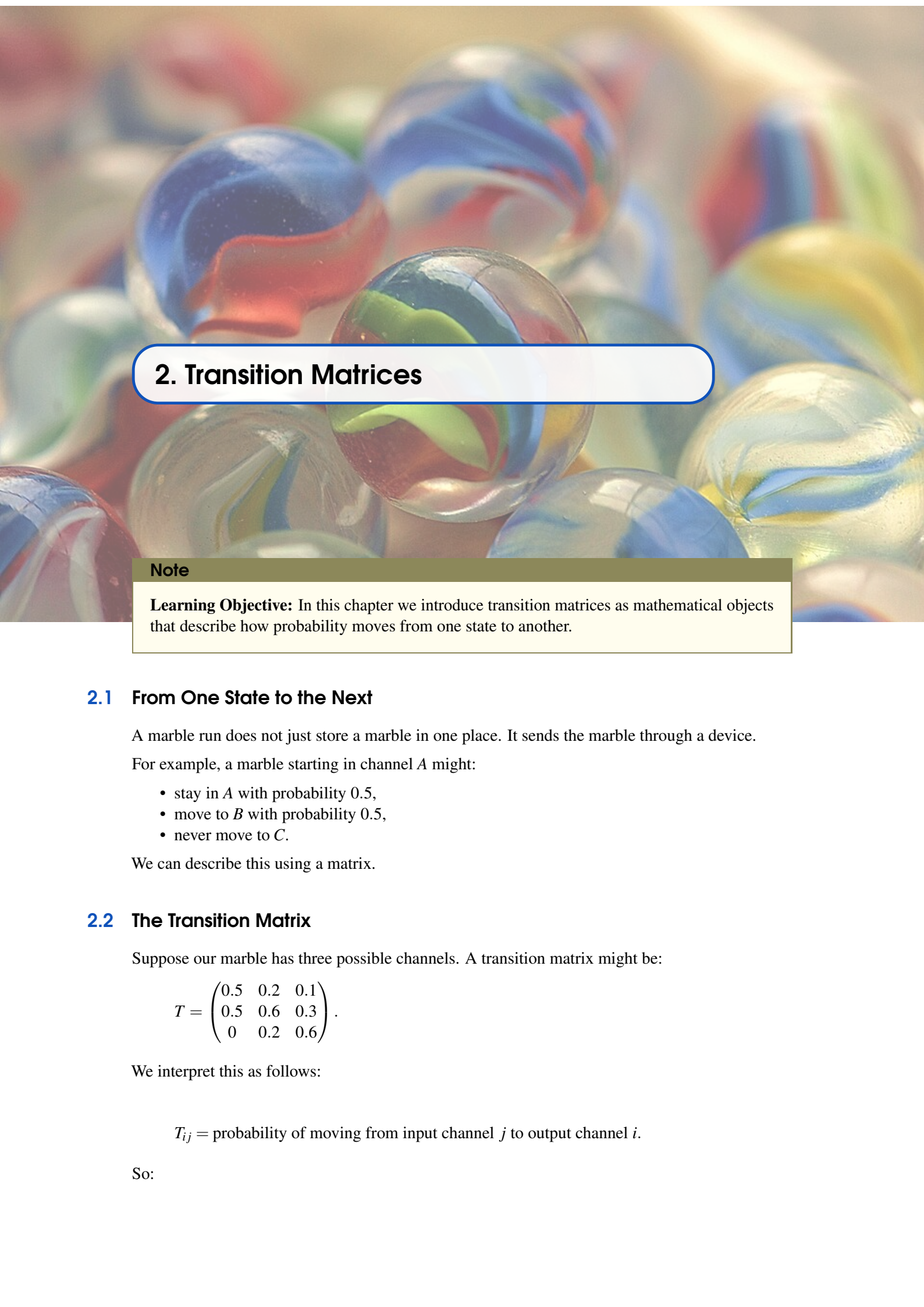
Explain your answer.

Exercise

Exercise 1.2. Write a probability vector for a marble that has probability 0.25 of being in channel A, probability 0.25 of being in channel B, and probability 0.50 of being in channel C.

Summary

A probability vector is a mathematical description of where a marble might be. This gives us our first version of a state vector.

A close-up, shallow depth-of-field photograph of numerous colorful marbles. The marbles are in various colors including blue, red, green, yellow, and white, with some showing internal patterns or swirls. They are scattered across the frame, with some in sharp focus and others blurred in the background.

2. Transition Matrices

Note

Learning Objective: In this chapter we introduce transition matrices as mathematical objects that describe how probability moves from one state to another.

2.1 From One State to the Next

A marble run does not just store a marble in one place. It sends the marble through a device.

For example, a marble starting in channel A might:

- stay in A with probability 0.5,
- move to B with probability 0.5,
- never move to C .

We can describe this using a matrix.

2.2 The Transition Matrix

Suppose our marble has three possible channels. A transition matrix might be:

$$T = \begin{pmatrix} 0.5 & 0.2 & 0.1 \\ 0.5 & 0.6 & 0.3 \\ 0 & 0.2 & 0.6 \end{pmatrix}.$$

We interpret this as follows:

T_{ij} = probability of moving from input channel j to output channel i .

So:

- columns correspond to the starting channel,
- rows correspond to the ending channel.

Note

This column convention is chosen because it matches the convention we will later use for optical circuits.

2.3 Columns Must Add to One

Each column describes all possible destinations from a particular starting channel.

For example, the first column is:

$$\begin{pmatrix} 0.5 \\ 0.5 \\ 0 \end{pmatrix}.$$

This means that a marble starting in channel A has:

$$0.5 + 0.5 + 0 = 1$$

total probability of ending somewhere.

Definition

Definition 2.1. A transition matrix is **column-stochastic** if each column adds to 1.

2.4 Applying a Transition Matrix

If the initial state is $\mathbf{p}_{\text{in}}$, then the output state is:

$$\mathbf{p}_{\text{out}} = T\mathbf{p}_{\text{in}}.$$

Worked Example

Let

$$T = \begin{pmatrix} 0.5 & 0.2 \\ 0.5 & 0.8 \end{pmatrix}, \quad \mathbf{p}_{\text{in}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then

$$\mathbf{p}_{\text{out}} = T\mathbf{p}_{\text{in}} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

So a marble starting definitely in the first channel has equal probability of ending in either channel.

2.5 Exercises

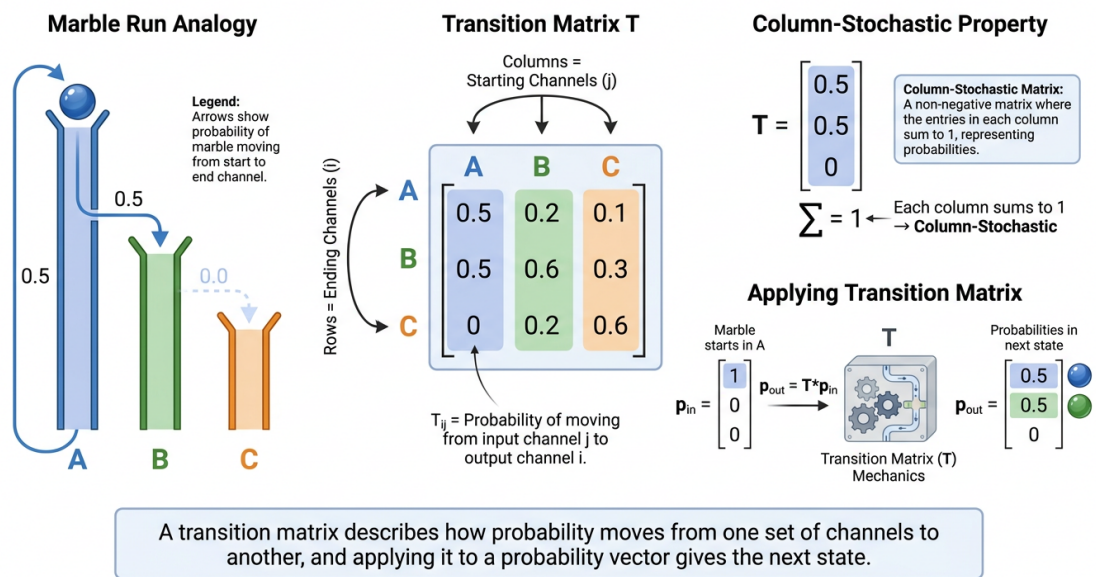


Figure 2.1: Enter Caption

Exercise**Exercise 2.1.** Check whether the matrix

$$T = \begin{pmatrix} 0.3 & 0.4 \\ 0.7 & 0.6 \end{pmatrix}$$

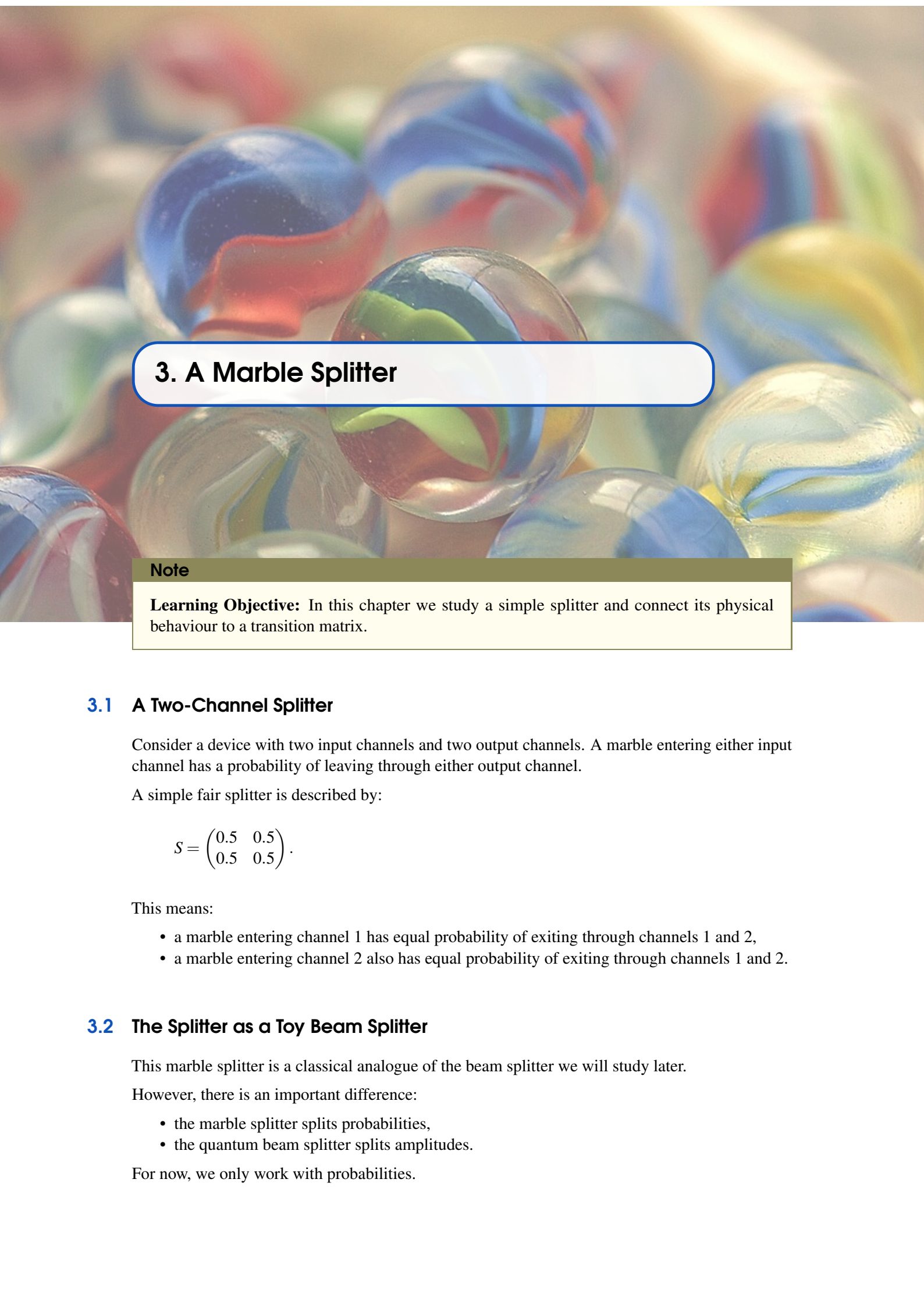
is a valid transition matrix.

Exercise**Exercise 2.2.** Calculate $T\mathbf{p}$ for

$$T = \begin{pmatrix} 0.3 & 0.4 \\ 0.7 & 0.6 \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Summary

A transition matrix describes how probability moves from one set of channels to another. Applying the matrix to a probability vector gives the next state.

A close-up, slightly blurred photograph of many colorful marbles in shades of blue, red, green, and yellow. The marbles are scattered across the frame, creating a vibrant, textured background.

3. A Marble Splitter

Note

Learning Objective: In this chapter we study a simple splitter and connect its physical behaviour to a transition matrix.

3.1 A Two-Channel Splitter

Consider a device with two input channels and two output channels. A marble entering either input channel has a probability of leaving through either output channel.

A simple fair splitter is described by:

$$S = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}.$$

This means:

- a marble entering channel 1 has equal probability of exiting through channels 1 and 2,
- a marble entering channel 2 also has equal probability of exiting through channels 1 and 2.

3.2 The Splitter as a Toy Beam Splitter

This marble splitter is a classical analogue of the beam splitter we will study later.

However, there is an important difference:

- the marble splitter splits probabilities,
- the quantum beam splitter splits amplitudes.

For now, we only work with probabilities.

3.3 Example: Definite Input

Let

$$\mathbf{p}_{\text{in}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then

$$S\mathbf{p}_{\text{in}} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

So a marble entering channel 1 has equal probability of leaving through either output channel.

3.4 Example: Uncertain Input

Let

$$\mathbf{p}_{\text{in}} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}.$$

Then

$$S\mathbf{p}_{\text{in}} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

The fair splitter erases the original imbalance.

3.5 Exercises

Exercise

Exercise 3.1. Apply

$$S = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$$

to the state

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Exercise

Exercise 3.2. Explain why this splitter is not reversible.

Exercise

Exercise 3.3. Compare this marble splitter with a quantum beam splitter. What idea is similar, and what idea is different?

Classical Marble Splitter: Probability Flow and Transition Matrix

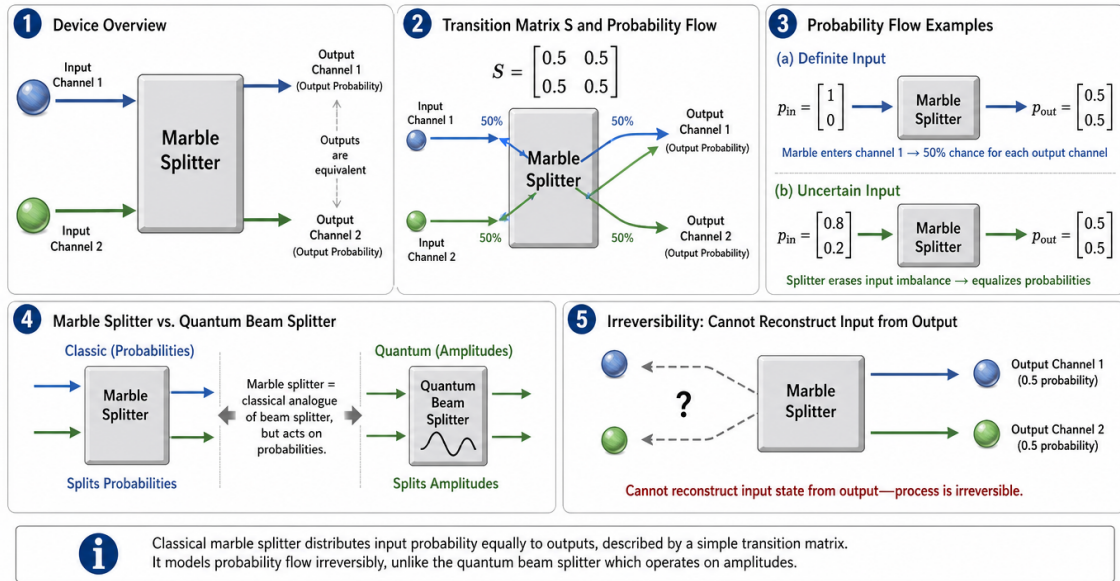
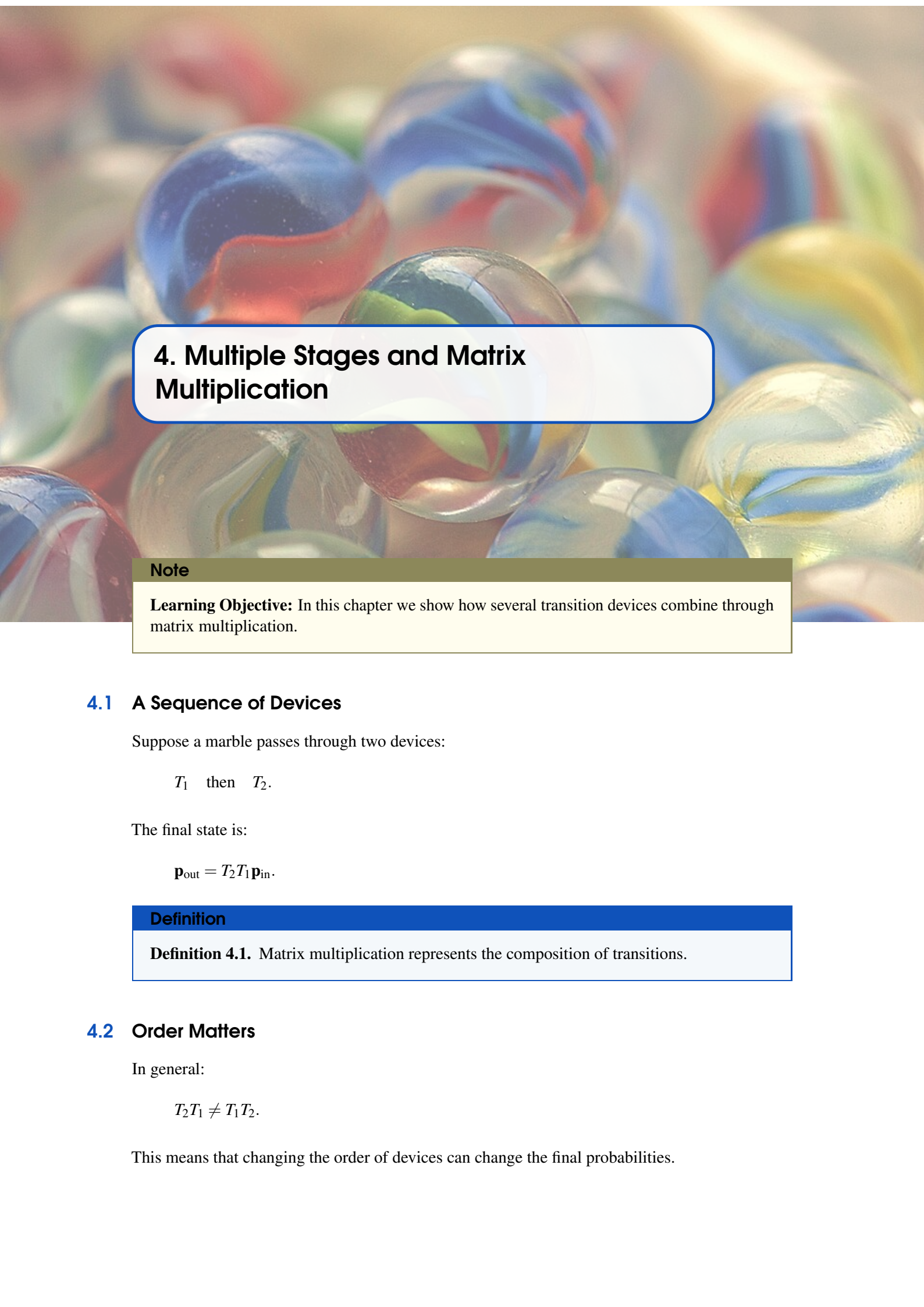


Figure 3.1: Enter Caption

Summary

The marble splitter gives us a simple transition matrix model for probability flow. It resembles a beam splitter, but it acts on probabilities rather than amplitudes.



4. Multiple Stages and Matrix Multiplication

Note

Learning Objective: In this chapter we show how several transition devices combine through matrix multiplication.

4.1 A Sequence of Devices

Suppose a marble passes through two devices:

$$T_1 \text{ then } T_2.$$

The final state is:

$$\mathbf{p}_{\text{out}} = T_2 T_1 \mathbf{p}_{\text{in}}.$$

Definition

Definition 4.1. Matrix multiplication represents the composition of transitions.

4.2 Order Matters

In general:

$$T_2 T_1 \neq T_1 T_2.$$

This means that changing the order of devices can change the final probabilities.

Worked Example

Let

$$T_1 = \begin{pmatrix} 0.5 & 0 \\ 0.5 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 0.4 \\ 0 & 0.6 \end{pmatrix}.$$

If the input is

$$\mathbf{p}_{\text{in}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

then

$$T_1 \mathbf{p}_{\text{in}} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

Then:

$$T_2 T_1 \mathbf{p}_{\text{in}} = T_2 \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}.$$

4.3 Connection to Circuits

A marble run with several stages is mathematically similar to an optical circuit with several optical components.

In both cases:

$$\text{final state} = \text{last device} \times \cdots \times \text{first device} \times \text{initial state}.$$

4.4 Exercises**Exercise**

Exercise 4.1. Calculate $T_2 T_1$ using the matrices in the worked example.

Exercise

Exercise 4.2. Calculate $T_1 T_2$. Is it the same?

Exercise

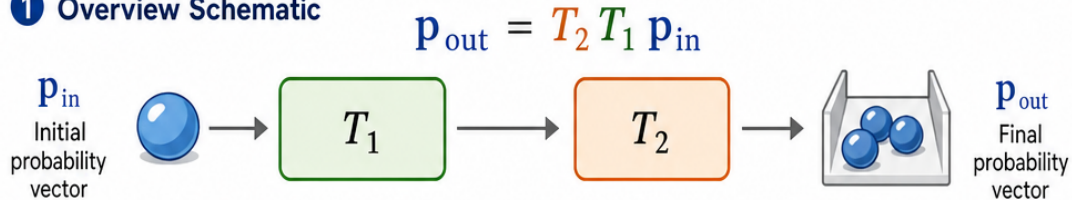
Exercise 4.3. Explain why the matrix closest to the vector acts first.

Summary

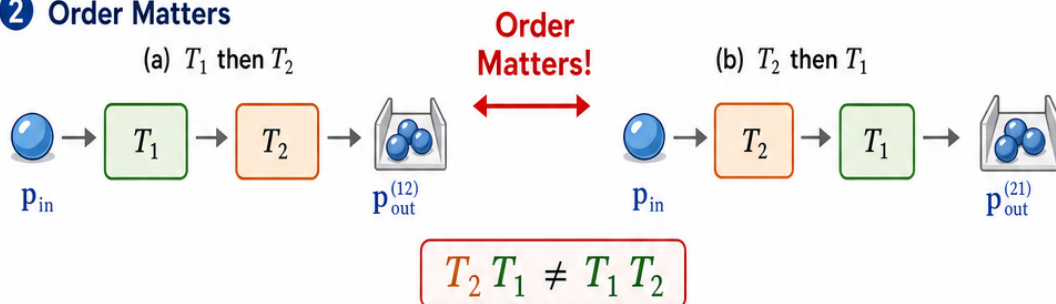
Several transition devices combine through matrix multiplication. This prepares us for optical circuits, where each optical component is represented by a matrix.

Composing Sequential Transitions via Matrix Multiplication

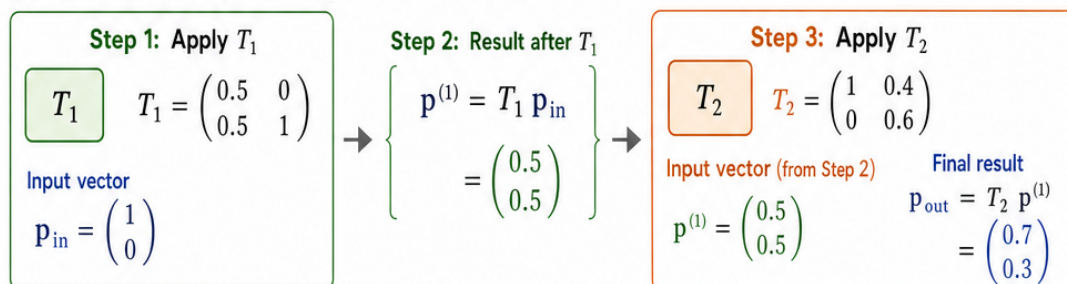
1 Overview Schematic



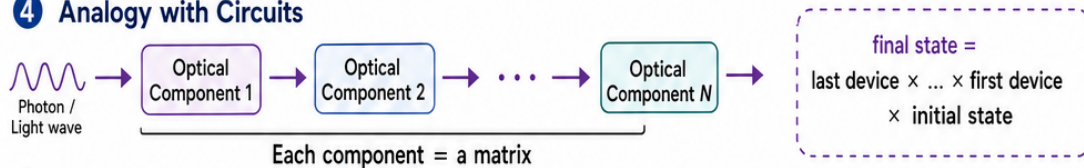
2 Order Matters



3 Worked Example—Stepwise Calculation



4 Analogy with Circuits

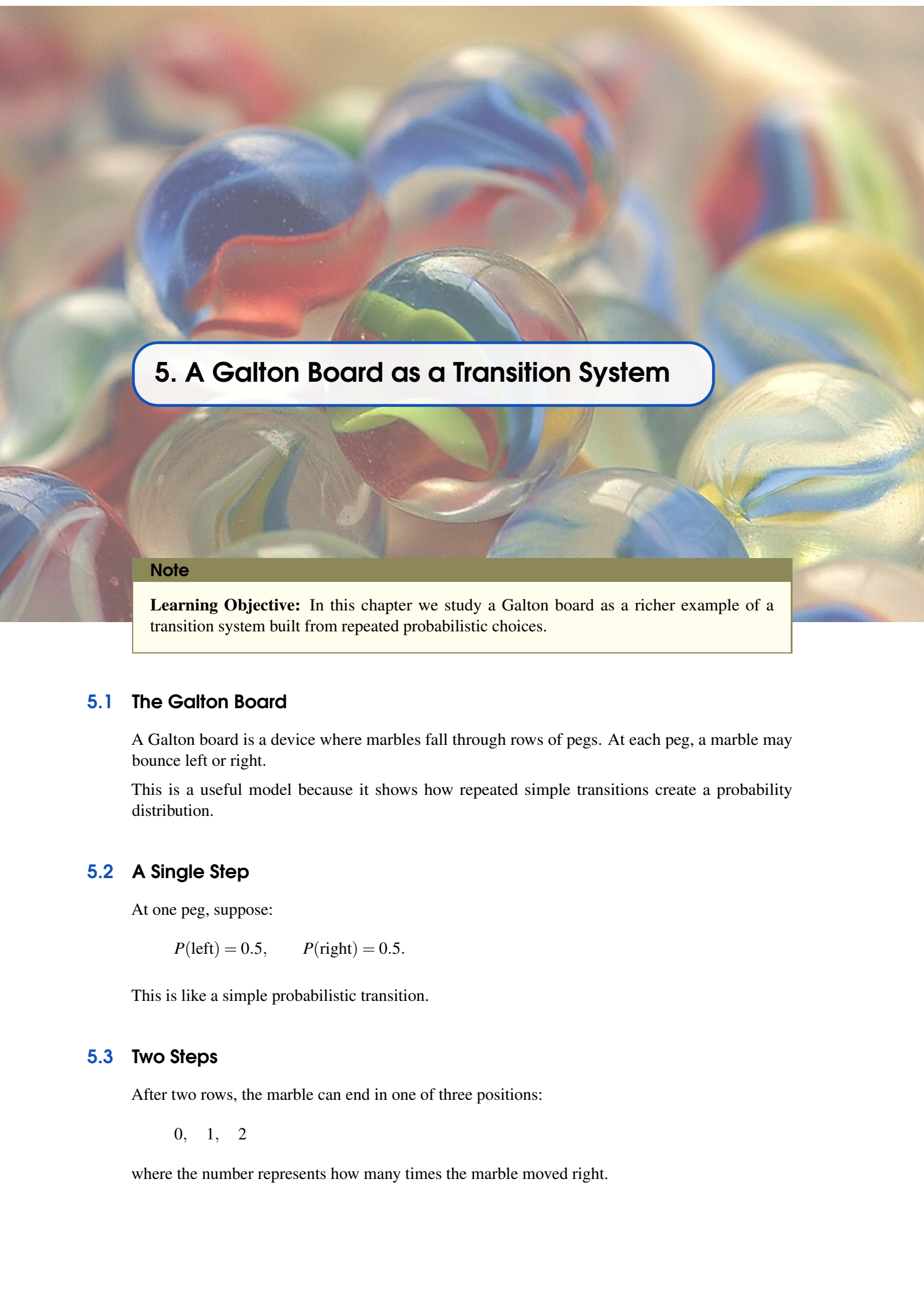


5 Summary



Several transition devices combine through matrix multiplication, just like components in an optical circuit.
The right-most matrix acts first.

Figure 4.1: Enter Caption



5. A Galton Board as a Transition System

Note

Learning Objective: In this chapter we study a Galton board as a richer example of a transition system built from repeated probabilistic choices.

5.1 The Galton Board

A Galton board is a device where marbles fall through rows of pegs. At each peg, a marble may bounce left or right.

This is a useful model because it shows how repeated simple transitions create a probability distribution.

5.2 A Single Step

At one peg, suppose:

$$P(\text{left}) = 0.5, \quad P(\text{right}) = 0.5.$$

This is like a simple probabilistic transition.

5.3 Two Steps

After two rows, the marble can end in one of three positions:

$$0, \quad 1, \quad 2$$

where the number represents how many times the marble moved right.

The probabilities are:

$$\frac{1}{4}, \quad \frac{1}{2}, \quad \frac{1}{4}.$$

This is the familiar binomial pattern.

5.4 Transition Matrix Form

The transition from one row to the next can be represented by a matrix.

For example, from two positions to three positions:

$$G = \begin{pmatrix} 0.5 & 0 \\ 0.5 & 0.5 \\ 0 & 0.5 \end{pmatrix}.$$

If

$$\mathbf{p} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix},$$

then

$$G\mathbf{p} = \begin{pmatrix} 0.25 \\ 0.50 \\ 0.25 \end{pmatrix}.$$

5.5 Why This Matters

The Galton board shows us that:

- simple local rules can generate complex distributions,
- repeated transitions are naturally described by matrices,
- the final state depends on the entire sequence of transitions.

This prepares us for optical networks, where repeated optical components create complex output distributions.

5.6 Exercises

Exercise

Exercise 5.1. Write the probability vector after one row of a fair Galton board.

Exercise

Exercise 5.2. Use the matrix

$$G = \begin{pmatrix} 0.5 & 0 \\ 0.5 & 0.5 \\ 0 & 0.5 \end{pmatrix}$$

to calculate the probability vector after two rows.

Exercise

Exercise 5.3. Explain why the centre output is more likely than the edge outputs after two rows.

Summary

A Galton board is a visual example of repeated probabilistic transitions. It gives a strong classical foundation for understanding later optical circuits.

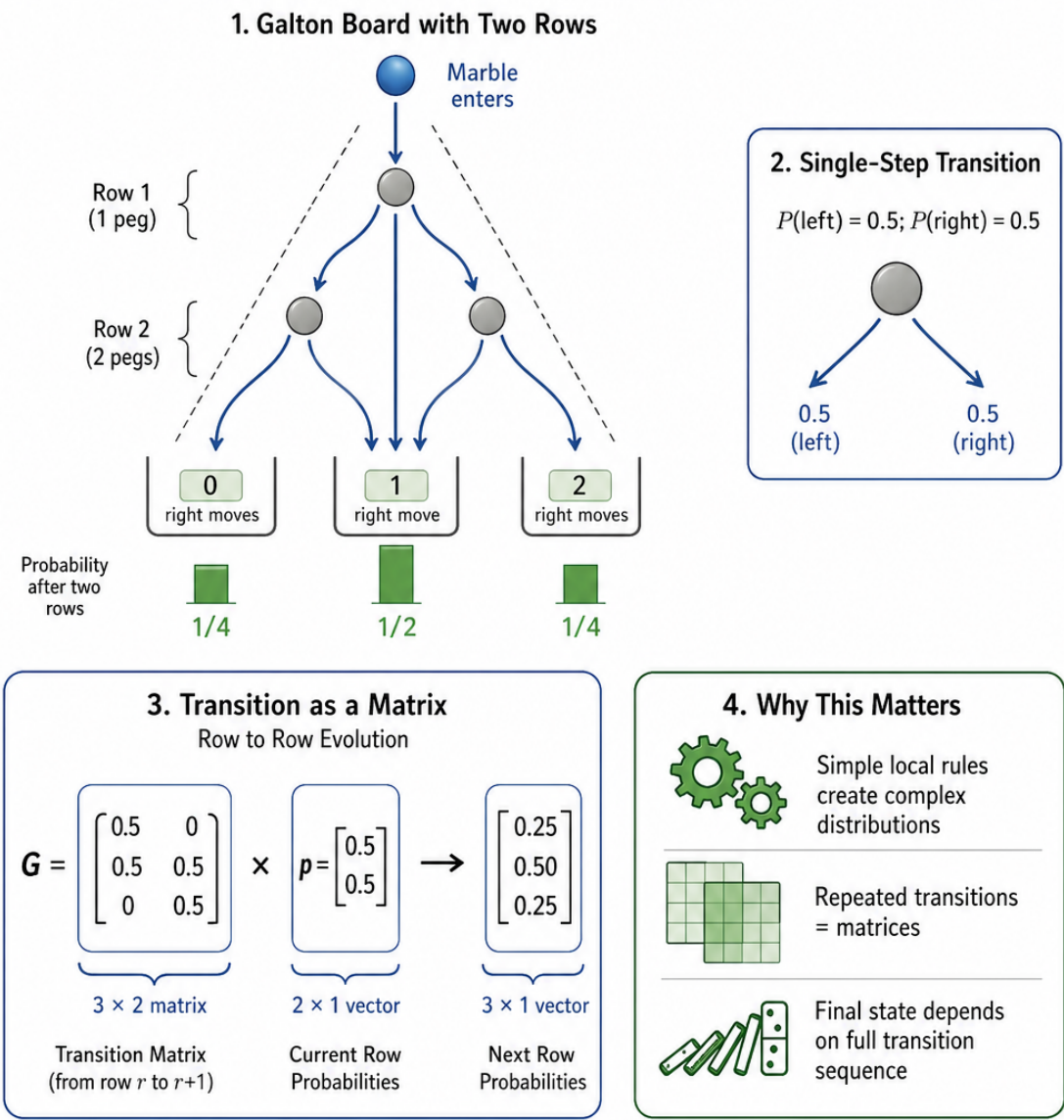
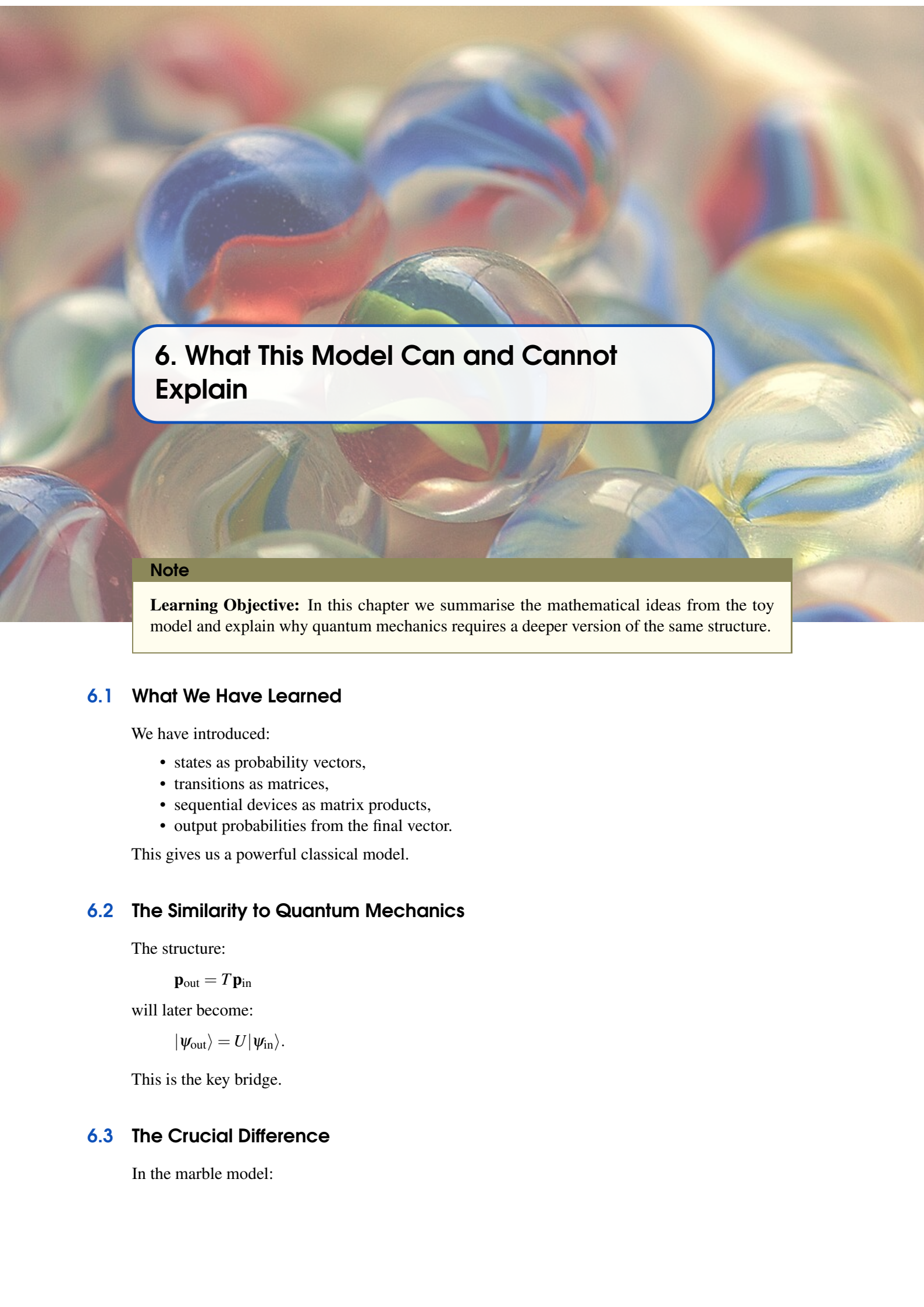


Figure X. A Galton Board as a Transition System. (Top) Marbles fall through two rows of pegs, each bounce representing a probabilistic choice (left or right, $P = 0.5$ each). The resulting probability distribution after two rows matches the binomial pattern (1/4, 1/2, 1/4). (Left) The process is formalized using a *transition matrix* acting on probability vectors. (Right) Key takeaways: simple repeated rules create complex outcomes, can be modeled as matrices, and depend on the full sequence of transitions—a foundation for understanding more complex stochastic systems.

Figure 5.1: Enter Caption

A background image showing a close-up of several colorful marbles in shades of blue, red, green, and yellow, some with swirl patterns. The marbles are slightly out of focus, creating a bokeh effect.

6. What This Model Can and Cannot Explain

Note

Learning Objective: In this chapter we summarise the mathematical ideas from the toy model and explain why quantum mechanics requires a deeper version of the same structure.

6.1 What We Have Learned

We have introduced:

- states as probability vectors,
- transitions as matrices,
- sequential devices as matrix products,
- output probabilities from the final vector.

This gives us a powerful classical model.

6.2 The Similarity to Quantum Mechanics

The structure:

$$\mathbf{p}_{\text{out}} = T \mathbf{p}_{\text{in}}$$

will later become:

$$|\psi_{\text{out}}\rangle = U |\psi_{\text{in}}\rangle.$$

This is the key bridge.

6.3 The Crucial Difference

In the marble model:

- vectors contain probabilities,
- matrices contain transition probabilities.

In quantum mechanics:

- vectors contain amplitudes,
- matrices contain transition amplitudes,
- probabilities are obtained only at measurement.

Note

The mathematics looks similar, but the meaning of the entries changes.

6.4 Why the Toy Model Is Still Useful

The toy model gives us intuition for:

- basis states,
- transitions,
- matrix multiplication,
- multi-stage circuits.

These ideas carry directly into the photonic quantum model.

6.5 Final Exercises

Exercise

Exercise 6.1. Explain the difference between a probability vector and the quantum state vectors used later in the text.

Exercise

Exercise 6.2. Why is a marble splitter only a classical analogy for a beam splitter?

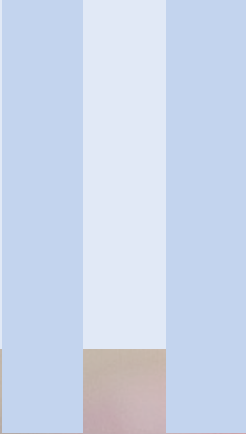
Exercise

Exercise 6.3. Describe the meaning of each part of the equation

$$\mathbf{p}_{\text{out}} = T\mathbf{p}_{\text{in}}.$$

Summary

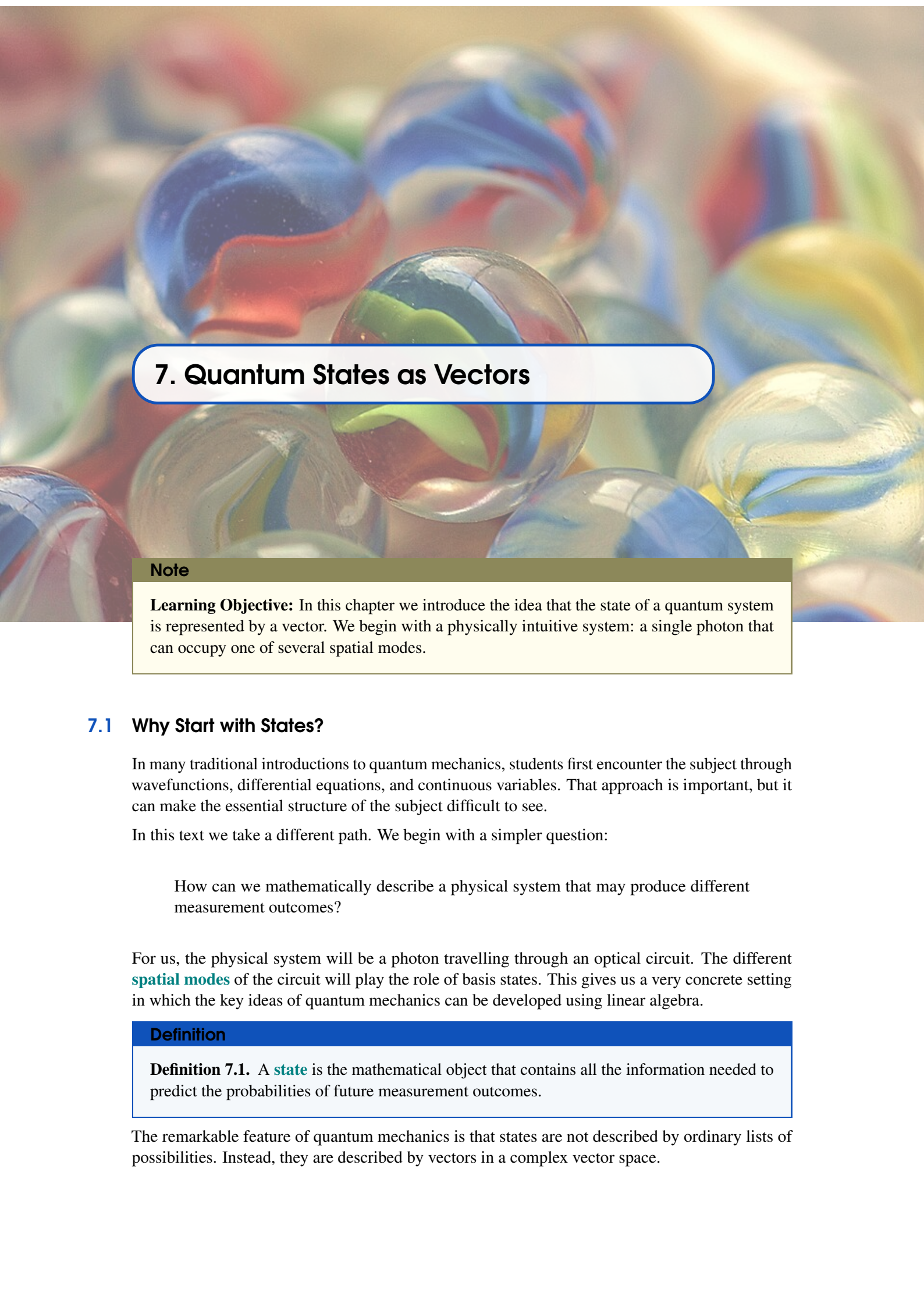
The marble-run model introduces the mathematical structure of states and transitions. Quantum mechanics will keep this structure but replace probabilities with amplitudes, leading to interference and genuinely non-classical behaviour.



States and Linear Algebra



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A background image showing a close-up of several colorful marbles in shades of blue, red, green, and yellow, some with swirl patterns. The marbles are slightly out of focus, creating a bokeh effect.

7. Quantum States as Vectors

Note

Learning Objective: In this chapter we introduce the idea that the state of a quantum system is represented by a vector. We begin with a physically intuitive system: a single photon that can occupy one of several spatial modes.

7.1 Why Start with States?

In many traditional introductions to quantum mechanics, students first encounter the subject through wavefunctions, differential equations, and continuous variables. That approach is important, but it can make the essential structure of the subject difficult to see.

In this text we take a different path. We begin with a simpler question:

How can we mathematically describe a physical system that may produce different measurement outcomes?

For us, the physical system will be a photon travelling through an optical circuit. The different **spatial modes** of the circuit will play the role of basis states. This gives us a very concrete setting in which the key ideas of quantum mechanics can be developed using linear algebra.

Definition

Definition 7.1. A **state** is the mathematical object that contains all the information needed to predict the probabilities of future measurement outcomes.

The remarkable feature of quantum mechanics is that states are not described by ordinary lists of possibilities. Instead, they are described by vectors in a complex vector space.

Note

In this first part, we will work only with finite-dimensional systems. This means that every state can be written as a finite column vector.

7.2 The Physical System Behind the Mathematics

Before developing the mathematics further, it is important to describe the physical system that this text is built around.

The basic picture is simple. We imagine a source that emits individual photons, one at a time. These photons are directed into an optical network made from devices such as optical fibres, beam splitters, and phase shifters. At the end of the network, photon detectors record where the photon is found.

In this text, the most important physical idea is that a photon may have several possible **spatial modes** available to it. A spatial mode is a physically distinct path that the photon could occupy. In a laboratory, these paths might be realised by separate optical fibres, separate waveguides on a photonic chip, or separate free-space paths defined by mirrors and beam splitters.

To make the system easy to visualise, we will usually draw these spatial modes as **rails**. Each rail represents one possible path that the photon may travel along.

For example, in a two-rail system:

- rail 1 might correspond to the upper optical fibre,
- rail 2 might correspond to the lower optical fibre.

If a photon is definitely in the upper fibre, we represent that by one basis state. If it is definitely in the lower fibre, we represent that by another basis state.

Note

The rails in our diagrams are not merely abstract lines. They represent real physical channels through which light can travel.

The role of the state vector is to describe the photon relative to these possible paths. In the simplest two-rail case, the basis states

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

mean:

- the photon is definitely in rail 1,
- the photon is definitely in rail 2.

A more general state such as

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

means that the photon is in a superposition of these two possible paths.

So when we write a state vector, we are not inventing an abstract mathematical game disconnected from reality. We are describing a real optical experiment in which a single photon can be prepared, guided through different possible paths, and eventually detected.

This is one of the great advantages of linear optics as a teaching model for quantum mechanics. The physical system is concrete and intuitive, while the mathematics is exactly the linear algebra that quantum theory requires.

7.3 A Physical System: One Photon in Two Rails

Consider an optical device with two possible paths, or **rails**, that a single photon may occupy. We label the two basis states by

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These are the simplest possible states: the photon is definitely in rail 1, or definitely in rail 2.

Definition

Definition 7.2. A set of states is called a **basis** if every allowed state can be written as a linear combination of those basis states.

In this case, the basis is

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$$

Any state of a single photon in this two-rail system can therefore be written as

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

where a and b are, in general, complex numbers.

R At this stage, it is tempting to think that a is the probability of finding the photon in rail 1 and b is the probability of finding it in rail 2. That is not correct. The entries a and b are called **amplitudes**. Probabilities will be obtained from these amplitudes later.

7.4 Basis States and Definite Outcomes

If the state is

$$|\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

then a measurement of the rail occupation will definitely find the photon in rail 1.

Similarly, if

$$|\psi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

then the photon will definitely be found in rail 2.

These basis states correspond to measurement outcomes that are perfectly definite. Quantum mechanics becomes interesting when the state is not one of these basis vectors.

Worked Example

Consider the state

$$|\psi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

What does this state mean physically?

Solution. This is the basis state corresponding to a photon definitely in rail 2. If we measure which rail contains the photon, the outcome is certain: rail 2.

7.5 Bra-Ket Notation and Inner Products

Throughout this text we will use both ordinary column-vector notation and the standard notation of quantum mechanics.

If

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

then we may also write

$$|\psi\rangle = a|1\rangle + b|2\rangle.$$

This symbol $|\psi\rangle$ is called a **ket**. A ket is simply a state vector written in the standard notation of quantum mechanics.

Definition

Definition 7.3. A **ket** is a column vector representing the state of a quantum system.

Associated with every ket is a corresponding **bra**, written $\langle\psi|$. The bra is obtained by taking the transpose of the ket and also taking the complex conjugate of each entry.

So if

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

then the corresponding bra is

$$\langle\psi| = (a^* \quad b^*),$$

where the star $*$ denotes complex conjugation.

Definition**Definition 7.4.** If

$$|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix},$$

then the corresponding bra is

$$\langle\psi| = (a_1^* \ a_2^* \ \cdots \ a_n^*).$$

When a bra multiplies a ket, the result is a single number:

$$\langle\phi|\psi\rangle.$$

This is called the **inner product**.

If

$$|\phi\rangle = \begin{pmatrix} c \\ d \end{pmatrix}, \quad |\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

then

$$\langle\phi|\psi\rangle = (c^* \ d^*) \begin{pmatrix} a \\ b \end{pmatrix} = c^*a + d^*b.$$

This is very similar to the ordinary dot product, except that in quantum mechanics we must take the complex conjugate of the first vector.

Note

For real vectors, the inner product is just the ordinary dot product. For complex vectors, the conjugate is essential.

A particularly important case is when a bra multiplies its own ket:

$$\langle\psi|\psi\rangle.$$

For the state

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

we get

$$\langle\psi|\psi\rangle = a^*a + b^*b = |a|^2 + |b|^2.$$

This quantity gives the squared length of the vector, and the normalisation condition for a valid quantum state is

$$\langle\psi|\psi\rangle = 1.$$

Worked Example

Let

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Find the corresponding bra and calculate $\langle\psi|\psi\rangle$.

Solution. The bra is

$$\langle\psi| = \frac{1}{\sqrt{2}} (1 \quad -i).$$

So

$$\langle\psi|\psi\rangle = \frac{1}{2} (1 \quad -i) \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{2} (1 + (-i)(i)).$$

Since $(-i)(i) = 1$, this becomes

$$\langle\psi|\psi\rangle = \frac{1}{2} (1 + 1) = 1.$$

So the state is normalised.

R The notation $\langle\phi|\psi\rangle$ may be read as “the bra phi acting on the ket psi”. In this text, you can safely think of it as the complex version of the dot product.

7.6 Exercises**Exercise**

Exercise 7.1. Write each of the following states as a column vector:

1. $|1\rangle$
2. $|2\rangle$
3. $3|1\rangle - 2|2\rangle$
4. $\frac{1}{2}|1\rangle + \frac{1}{2}|2\rangle$

Exercise

Exercise 7.2. A photon in a two-rail system is in the state

$$|\psi\rangle = \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$$

State whether this means the photon is definitely in one rail, or not.

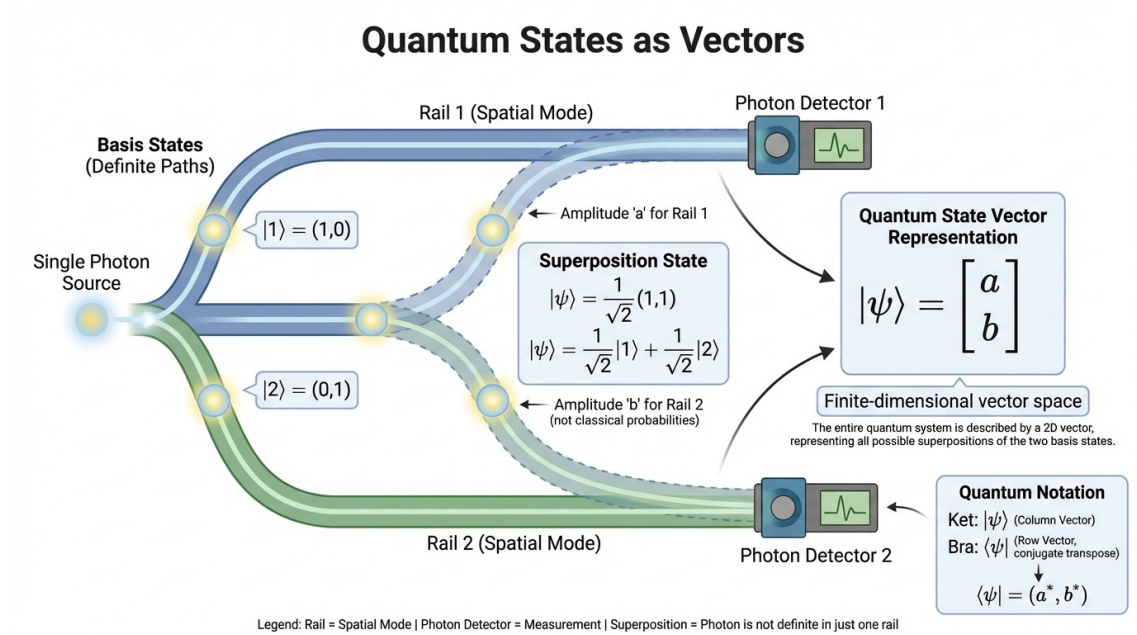


Figure 7.1: Enter Caption

Exercise**Exercise 7.3.** Express the vector

$$\begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

in ket notation using the basis states $|1\rangle$ and $|2\rangle$.**Summary**

In this chapter we introduced the central idea that a quantum state is represented by a vector. For a single photon in two rails, the basis states correspond to definite rail occupation, and more general states are linear combinations of those basis states.

8. Superposition and Normalisation

Note

Learning Objective: In this chapter we introduce superposition, explain why amplitudes must be normalised, and connect amplitudes to probabilities.

8.1 Superposition

Suppose a photon is prepared in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

This is not a basis state. It is a **superposition** of the two basis states:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle.$$

Definition

Definition 8.1. A **superposition** is a state formed by adding basis states together with complex coefficients.

This does *not* mean the photon is partly in one rail and partly in the other in an ordinary classical sense. Instead, quantum mechanics assigns amplitudes to the different possibilities, and measurement outcomes are obtained from those amplitudes.

R Superposition is one of the defining features of quantum mechanics. The state vector contains more structure than a simple statement of ignorance about which outcome will occur.

8.2 From Amplitudes to Probabilities

If the state is

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

then the probability of finding the photon in rail 1 is

$$P(1) = |a|^2,$$

and the probability of finding it in rail 2 is

$$P(2) = |b|^2.$$

This is the simplest version of the **Born rule**.

Definition

Definition 8.2. The **Born rule** states that the probability of a measurement outcome is the modulus squared of the corresponding amplitude.

So amplitudes are not probabilities themselves. They are more fundamental quantities from which probabilities are calculated.

8.3 Why States Must Be Normalised

Probabilities must add to 1. Therefore, for a valid state

$$|a|^2 + |b|^2 = 1.$$

Definition

Definition 8.3. A state is called **normalised** if the sum of the squared moduli of its amplitudes is 1.

For a two-component state

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

normalisation means

$$\langle\psi|\psi\rangle = |a|^2 + |b|^2 = 1.$$

At this level, you may simply interpret this as the mathematical condition required so that the total probability is 1.

Worked Example

Determine whether the state

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

is normalised.

Solution.

$$\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

So the state is normalised.

Worked Example

Consider the state

$$|\phi\rangle = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Is this a valid quantum state as written?

Solution. We compute

$$|2|^2 + |1|^2 = 4 + 1 = 5.$$

So this vector is not normalised. It can be turned into a valid state by dividing by its length:

$$|\phi_{\text{norm}}\rangle = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

8.4 Complex Amplitudes

Quantum amplitudes may be complex. For example,

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

is a perfectly valid state, because

$$\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

The probabilities of the two rail outcomes are still

$$P(1) = \frac{1}{2}, \quad P(2) = \frac{1}{2}.$$

So the phase information carried by complex amplitudes is not always visible from a single measurement of the basis states. It becomes important when states evolve and interfere.

Note

Two states can give the same measurement probabilities in one basis, yet behave differently when passed through an optical circuit. This is one reason amplitudes are more informative than probabilities alone.

8.5 A First Look at Measurement

If a photon is prepared in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

then measuring which rail contains the photon gives:

$$P(1) = \frac{1}{2}, \quad P(2) = \frac{1}{2}.$$

A single measurement produces only one result. But if the same state is prepared many times and measured repeatedly, we expect about half the outcomes in rail 1 and half in rail 2.

This is one of the first important distinctions between classical and quantum descriptions:

- the *state* evolves deterministically,
- but a *measurement outcome* is probabilistic.

8.6 Exercises**Exercise**

Exercise 8.1. Determine whether each state is normalised:

1. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
2. $\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$
3. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$
4. $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$

Exercise

Exercise 8.2. For each of the following normalised states, find the probability of detecting the photon in each rail:

1. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
2. $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

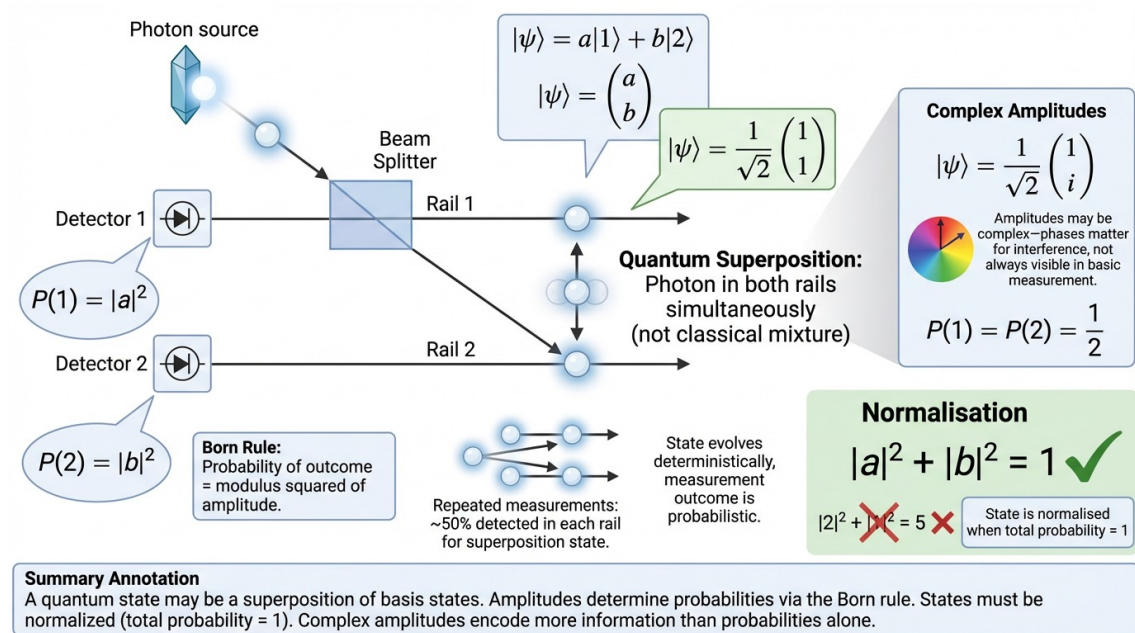


Figure 8.1: Enter Caption

$$3. \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Exercise

Exercise 8.3. Normalise the state

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Then determine the probability of finding the photon in rail 1.

Exercise

Exercise 8.4. Show that the states

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

have the same total probability but different amplitudes.

Summary

A quantum state may be a superposition of basis states. The coefficients in the state vector are amplitudes, and probabilities are obtained from their squared moduli. A valid state must be normalised so that the total probability is 1.

9. States of Photons in Spatial Modes

Note

Learning Objective: In this chapter we connect the vector formalism directly to optical systems with more than two rails, and prepare the language needed for operators and optical components in the next part of the text.

9.1 From Two Rails to Many Rails

There is no reason to stop at two rails. If a single photon can occupy one of N spatial modes, then the state space has dimension N .

For example, for three rails we use the basis states

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

A general state is

$$|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1|1\rangle + a_2|2\rangle + a_3|3\rangle,$$

with

$$|a_1|^2 + |a_2|^2 + |a_3|^2 = 1.$$

Definition

Definition 9.1. A **mode** is one of the distinguishable paths, channels, or field configurations available to the photon.

In this text we mainly use spatial modes, which can be pictured as different rails in an optical circuit diagram.

9.2 A Concrete Three-Mode Example

Suppose a photon is prepared in the state

$$|\psi\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Then the probabilities of detecting the photon in the three rails are

$$P(1) = \frac{1}{6}, \quad P(2) = \frac{1}{6}, \quad P(3) = \frac{4}{6} = \frac{2}{3}.$$

So the photon is most likely to be detected in rail 3, but not with certainty.

Worked Example

Verify that the state

$$|\psi\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

is normalised.

Solution.

$$\left| \frac{1}{\sqrt{6}} \right|^2 + \left| \frac{1}{\sqrt{6}} \right|^2 + \left| \frac{2}{\sqrt{6}} \right|^2 = \frac{1}{6} + \frac{1}{6} + \frac{4}{6} = 1.$$

9.3 Dual-Rail Encoding as a First Quantum Bit

A particularly important case is the two-rail system. In quantum information, a two-dimensional quantum system is called a **qubit**. One physical way to encode a qubit is to use two rails and one photon:

$$|0\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

This is called **dual-rail encoding**.

Note

Different texts choose different conventions for which rail corresponds to $|0\rangle$ and which corresponds to $|1\rangle$. What matters is that the choice is made clearly and used consistently.

A general qubit state is then

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

This is one of the main reasons linear optics is such a good teaching platform: the mathematics of qubits appears naturally from a physically intuitive system.

9.4 What the State Does and Does Not Tell Us

The state vector tells us the probabilities for measurement outcomes in a given basis, but it is more than a probability list. It also contains phase information, and that phase information affects how the state changes when it passes through an optical device.

So the state serves two purposes:

1. it predicts measurement probabilities;
2. it determines how the system responds to transformations.

The second point is what makes the formalism powerful. Once we know how an optical element acts on basis states, linear algebra tells us how it acts on every possible state.

9.5 Preparing for Operators

In the next part of the text we will study **operators**: matrices that act on states. In the optical setting these operators correspond to devices such as

- beam splitters,
- phase shifters,
- swaps of spatial modes.

The central rule will be very simple:

$$\text{new state} = \text{operator} \times \text{old state}.$$

This means that once we represent states as vectors and optical components as matrices, the whole system becomes a problem in linear algebra.

Investigation

Open the [Linear Optics Lab Bench](#) and set up a system with two rails and a single photon in one input rail. Before running any circuit, write down the corresponding basis state vector. Then change the input rail and write down the new basis state.

9.6 Exercises

Exercise

Exercise 9.1. Write the basis states for a four-rail single-photon system as column vectors.

Exercise

Exercise 9.2. A single photon in three rails is in the state

$$|\psi\rangle = \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Show that this state is normalised, and determine the probability of detection in each rail.

Exercise

Exercise 9.3. Express the state

$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

in ket notation for a four-mode system.

Exercise

Exercise 9.4. In a dual-rail encoding, explain why the two basis states form a two-dimensional vector space.

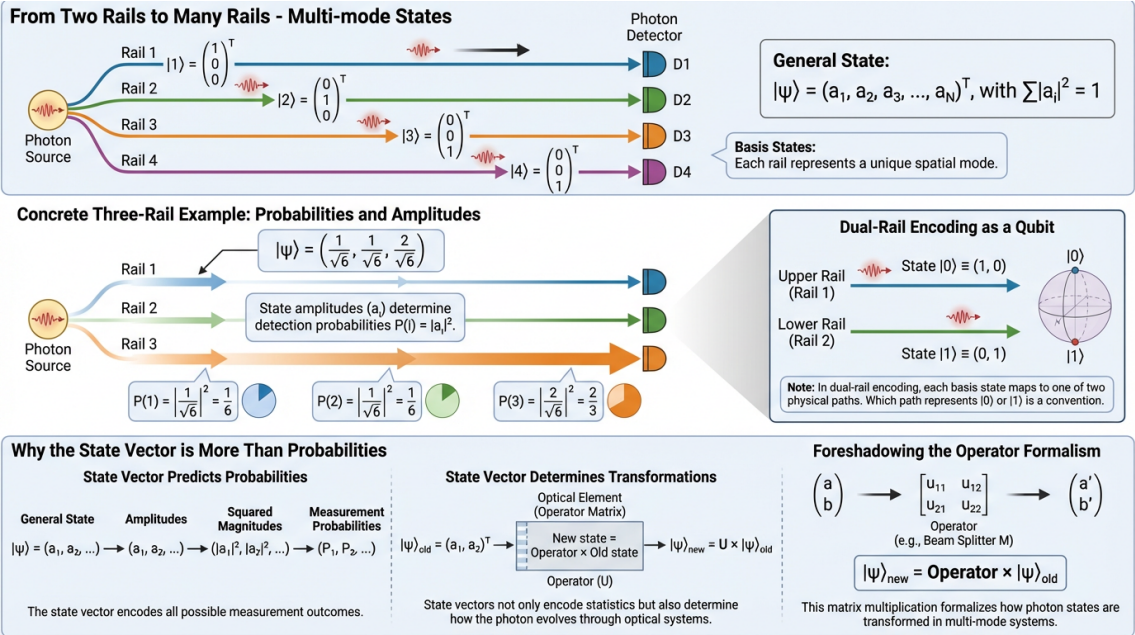
Exercise

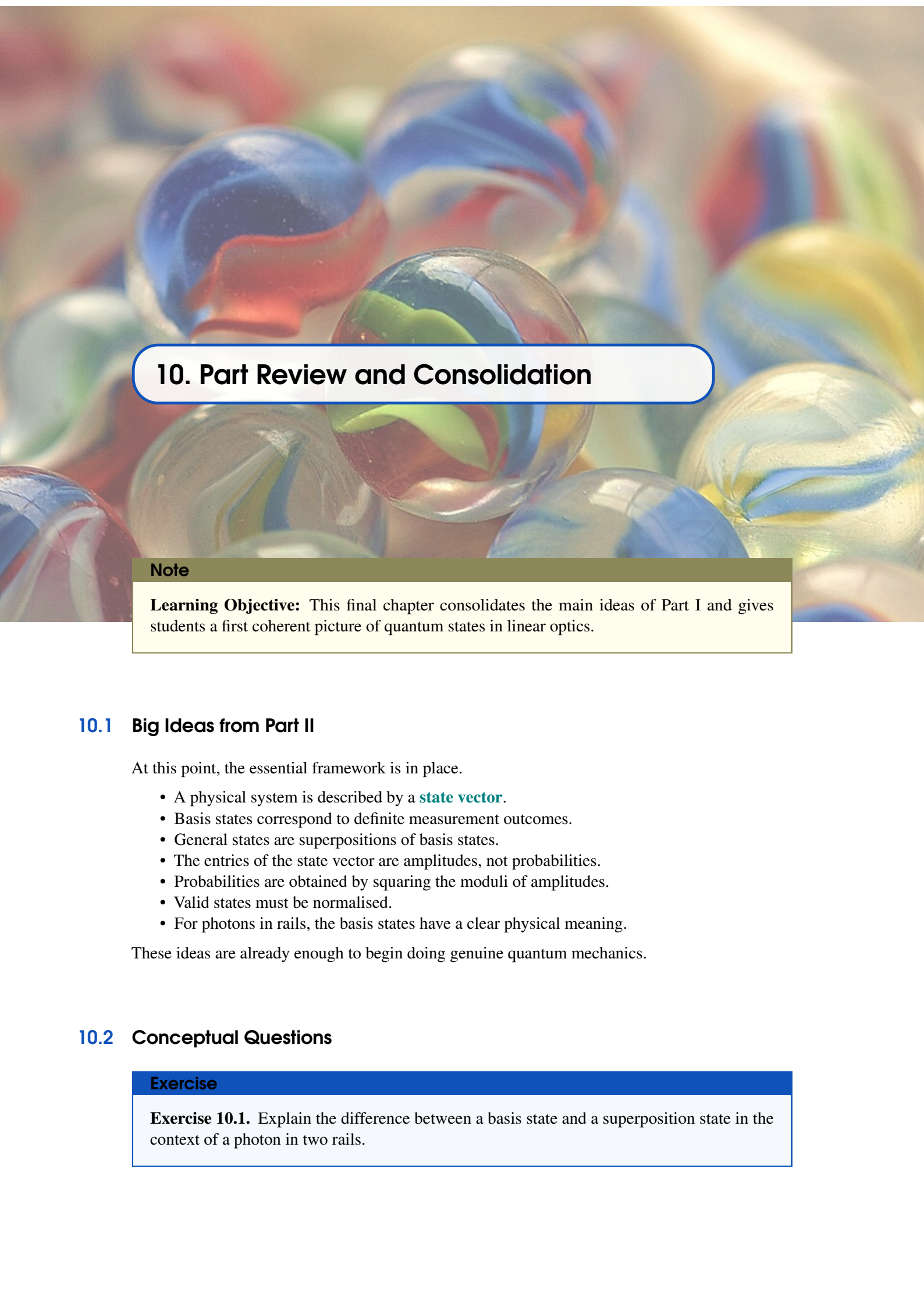
Exercise 9.5. A student claims that the state vector is just a shorthand way of listing probabilities. Give one reason why this statement is incomplete.

9.7 Chapter Review

Summary

In this chapter we extended the state-vector picture from two rails to many spatial modes. We introduced dual-rail encoding as a natural physical realisation of a qubit, and saw that the state vector contains both probability information and the structure needed for later transformations by optical devices.



A close-up, slightly blurred image of many colorful marbles in shades of blue, red, green, and yellow, creating a vibrant, textured background.

10. Part Review and Consolidation

Note

Learning Objective: This final chapter consolidates the main ideas of Part I and gives students a first coherent picture of quantum states in linear optics.

10.1 Big Ideas from Part II

At this point, the essential framework is in place.

- A physical system is described by a **state vector**.
- Basis states correspond to definite measurement outcomes.
- General states are superpositions of basis states.
- The entries of the state vector are amplitudes, not probabilities.
- Probabilities are obtained by squaring the moduli of amplitudes.
- Valid states must be normalised.
- For photons in rails, the basis states have a clear physical meaning.

These ideas are already enough to begin doing genuine quantum mechanics.

10.2 Conceptual Questions

Exercise

Exercise 10.1. Explain the difference between a basis state and a superposition state in the context of a photon in two rails.

Exercise

Exercise 10.2. Why must the amplitudes in a quantum state satisfy a normalisation condition?

Exercise

Exercise 10.3. A photon is prepared many times in the state

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

What would you expect to observe after many measurements of rail occupation?

Exercise

Exercise 10.4. Why are complex numbers useful in quantum mechanics, even when a single measurement only produces real probabilities?

10.3 Mixed Practice**Exercise**

Problem 10.1. A photon in a two-rail system is prepared in the state

$$|\psi\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

1. Show that the state is normalised.
2. Find the probability of detecting the photon in each rail.
3. State whether the photon is definitely in one rail before measurement.

Exercise

Problem 10.2. A photon in a three-rail system is prepared in the state

$$|\phi\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}.$$

1. Show that the state is normalised.
2. Find the probability of detection in each rail.
3. Explain why the presence of i does not change the total probability.

Exercise

Problem 10.3. A student writes the state

$$\begin{pmatrix} 0.6 \\ 0.6 \end{pmatrix}$$

for a two-rail system and claims it is valid because the numbers add to 1.2, which is close to 1. Explain the error and correct the state by normalising it.

10.4 Looking Ahead

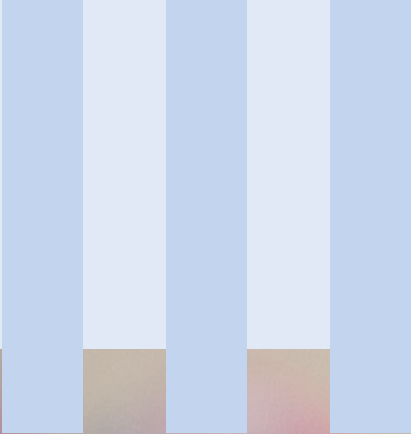
In Part II we turn from states to **operators**. Once optical elements are represented as matrices, we will be able to calculate how a state changes as it passes through a circuit.

That is the moment where linear optics becomes a true quantum laboratory on paper:

state $\longrightarrow$ matrix action $\longrightarrow$ new state $\longrightarrow$ measurement probabilities.

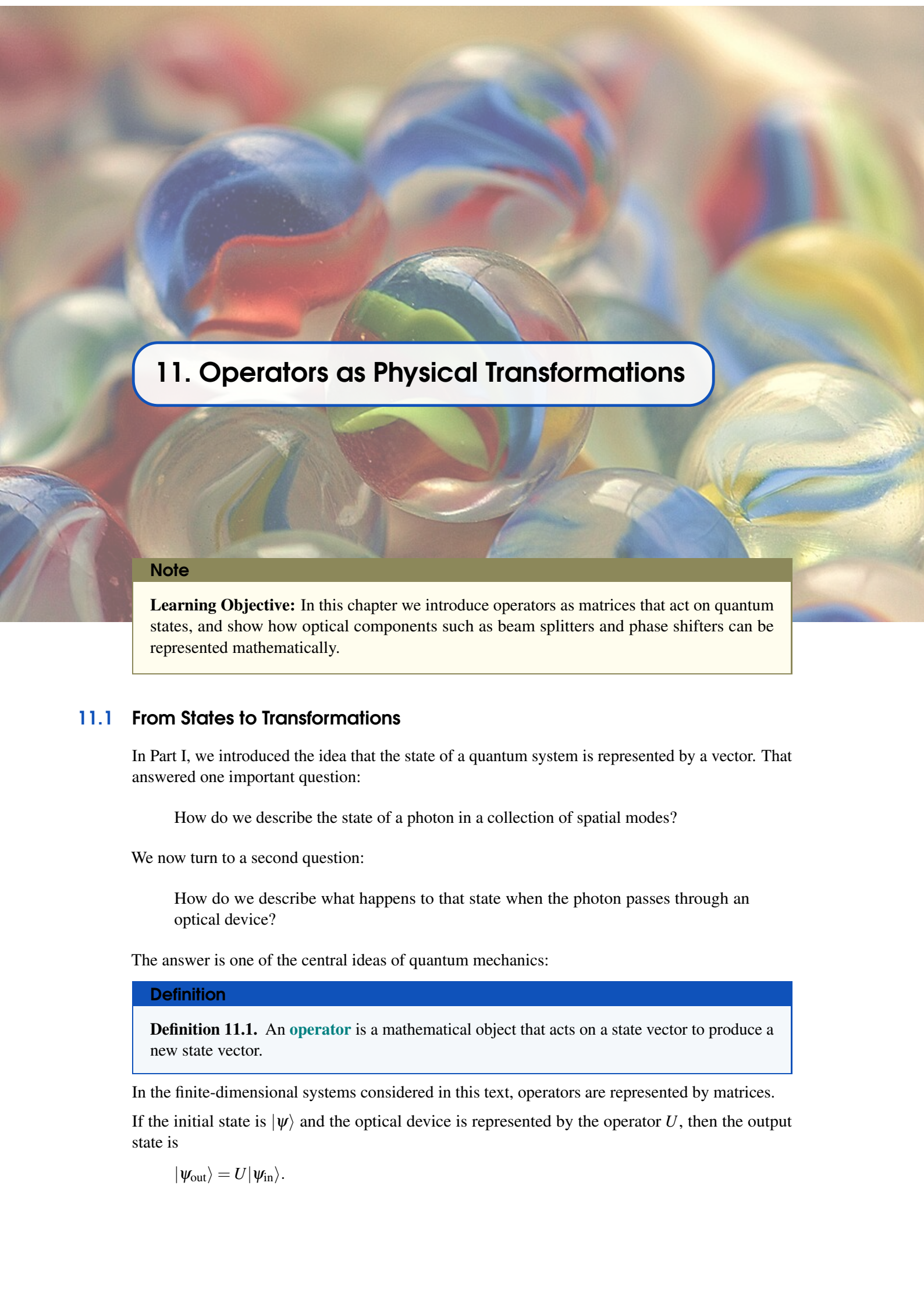
Summary

Part I established the language of the course. We now know what a state is, how to represent it mathematically, and how to interpret it physically for a photon in spatial modes. The next step is to understand how optical devices act on those states.



Operators and Optical Components

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11. Operators as Physical Transformations

Note

Learning Objective: In this chapter we introduce operators as matrices that act on quantum states, and show how optical components such as beam splitters and phase shifters can be represented mathematically.

11.1 From States to Transformations

In Part I, we introduced the idea that the state of a quantum system is represented by a vector. That answered one important question:

How do we describe the state of a photon in a collection of spatial modes?

We now turn to a second question:

How do we describe what happens to that state when the photon passes through an optical device?

The answer is one of the central ideas of quantum mechanics:

Definition

Definition 11.1. An **operator** is a mathematical object that acts on a state vector to produce a new state vector.

In the finite-dimensional systems considered in this text, operators are represented by matrices.

If the initial state is $|\psi\rangle$ and the optical device is represented by the operator U , then the output state is

$$|\psi_{\text{out}}\rangle = U|\psi_{\text{in}}\rangle.$$

This rule is simple, but powerful. Once we know how an operator acts on the basis states, linear algebra tells us how it acts on every superposition.

11.2 Operators in a Two-Rail System

Consider a photon in a two-rail system. A general state is

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}.$$

An operator on this system must take a two-component vector to another two-component vector, so it is represented by a 2×2 matrix:

$$U = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}.$$

Applying the operator gives

$$U|\psi\rangle = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} u_{11}a + u_{12}b \\ u_{21}a + u_{22}b \end{pmatrix}.$$

So the operator mixes the amplitudes of the basis states.

R This is exactly why matrix multiplication is so important in quantum mechanics. A matrix tells us how amplitudes are redistributed between the different basis states.

11.3 Linearity

The key property of quantum evolution in this setting is **linearity**.

Suppose

$$|\psi\rangle = a|1\rangle + b|2\rangle.$$

Then

$$U|\psi\rangle = U(a|1\rangle + b|2\rangle).$$

Because matrix multiplication is linear,

$$U|\psi\rangle = aU|1\rangle + bU|2\rangle.$$

Definition

Definition 11.2. An operator U is **linear** if

$$U(\alpha|\phi\rangle + \beta|\psi\rangle) = \alpha U|\phi\rangle + \beta U|\psi\rangle$$

for all states $|\phi\rangle, |\psi\rangle$ and all complex numbers α, β .

This means that if we know how an optical component acts on the basis states, we automatically know how it acts on any superposition.

Worked Example

Suppose an operator U acts on the basis states as

$$U|1\rangle = |1\rangle, \quad U|2\rangle = -|2\rangle.$$

Find $U|\psi\rangle$ for

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle).$$

Solution. By linearity,

$$U|\psi\rangle = \frac{1}{\sqrt{2}}(U|1\rangle + U|2\rangle) = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle).$$

11.4 The Identity Operator

The simplest operator is the one that does nothing:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition

Definition 11.3. The **identity operator** I leaves every state unchanged:

$$I|\psi\rangle = |\psi\rangle.$$

Physically, this represents no change to the system.

11.5 Exercises**Exercise**

Exercise 11.1. Let

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Find $U|1\rangle$ and $U|2\rangle$.

Exercise

Exercise 11.2. If

$$|\psi\rangle = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix},$$

calculate $U|\psi\rangle$.

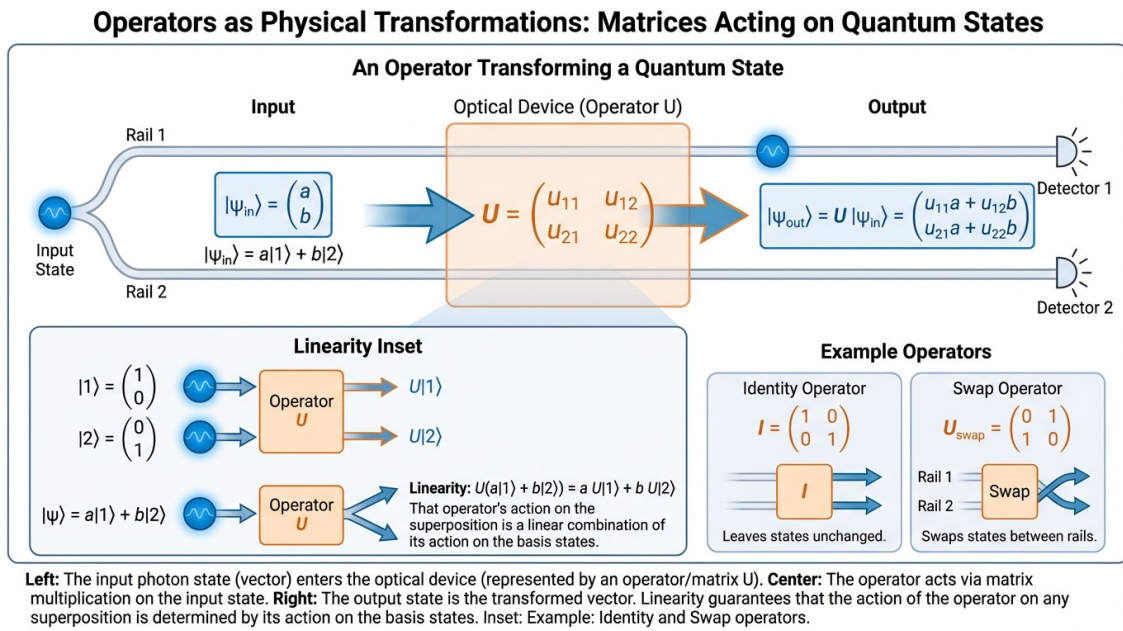


Figure 11.1: Enter Caption

Exercise

Exercise 11.3. Explain in words what linearity means for the action of an optical component on a superposition state.

Exercise

Exercise 11.4. Show that the identity operator leaves the state

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

unchanged.

Summary

Operators describe how quantum states change. In a finite-dimensional system, operators are matrices, and the evolution of the state is found by matrix multiplication. Because the evolution is linear, knowing the action of an operator on basis states is enough to determine its action on every state.

12. Unitary Operators and Probability Conservation

Note

Learning Objective: In this chapter we explain why physically allowed optical transformations must be unitary, and how unitarity ensures that total probability is conserved.

12.1 Why Not Every Matrix Is Allowed

A matrix can act on a vector, but not every matrix represents a physically possible optical transformation.

Suppose a photon begins in a normalised state:

$$\langle \psi | \psi \rangle = 1.$$

After passing through a physical optical device, the output state must also be a valid quantum state, so it must still satisfy

$$\langle \psi_{\text{out}} | \psi_{\text{out}} \rangle = 1.$$

This means the transformation must preserve the inner product structure of the space.

Definition

Definition 12.1. An operator U is **unitary** if

$$U^\dagger U = I,$$

where $U^\dagger$ is the conjugate transpose of U .

For the systems in this text, the physically allowed lossless optical transformations are unitary.

12.2 The Adjoint of an Operator

If

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then the adjoint is

$$U^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}.$$

So to form $U^\dagger$, we:

1. transpose the matrix;
2. take the complex conjugate of every entry.

Note

For matrices with only real entries, the adjoint is just the transpose.

12.3 Why Unitarity Preserves Normalisation

Suppose

$$|\psi_{\text{out}}\rangle = U|\psi_{\text{in}}\rangle.$$

Then

$$\langle\psi_{\text{out}}|\psi_{\text{out}}\rangle = \langle\psi_{\text{in}}|U^\dagger U|\psi_{\text{in}}\rangle.$$

If U is unitary, then $U^\dagger U = I$, so

$$\langle\psi_{\text{out}}|\psi_{\text{out}}\rangle = \langle\psi_{\text{in}}|I|\psi_{\text{in}}\rangle = \langle\psi_{\text{in}}|\psi_{\text{in}}\rangle.$$

So the length of the state vector is unchanged, and total probability is conserved.

 This is the mathematical reason unitary matrices are so important in quantum mechanics. They preserve the total probability of all possible measurement outcomes.

12.4 A Simple Example of a Unitary Operator

Consider

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This operator swaps the two rails.

Since the entries are real,

$$U^\dagger = U^T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Now compute

$$U^\dagger U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

So U is unitary.

12.5 Columns of a Unitary Matrix

A useful fact is that the columns of a unitary matrix form an orthonormal set.

For a 2×2 matrix

$$U = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix},$$

this means:

- each column has length 1;
- the two columns are orthogonal under the complex inner product.

Physically, this means the transformed basis states remain properly normalised and distinct.

12.6 Exercises

Exercise

Exercise 12.1. Find the adjoint of

$$U = \begin{pmatrix} 1 & i \\ 2 & 0 \end{pmatrix}.$$

Exercise

Exercise 12.2. Show that

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is unitary.

Exercise

Exercise 12.3. Determine whether

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

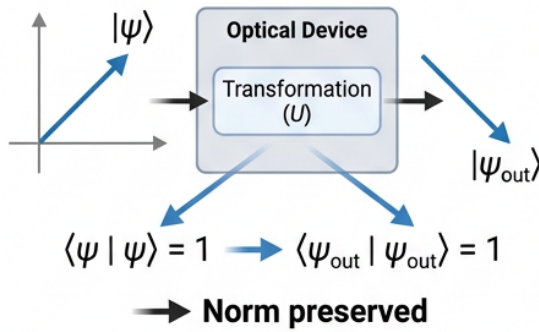
is unitary.

Exercise

Exercise 12.4. Explain why a physical optical transformation must preserve the normalisation of a state.

Summary

Not every matrix represents a possible quantum evolution. Physically allowed lossless optical transformations are unitary, meaning that they preserve inner products and therefore preserve total probability.

A What Makes a Physical Transformation?

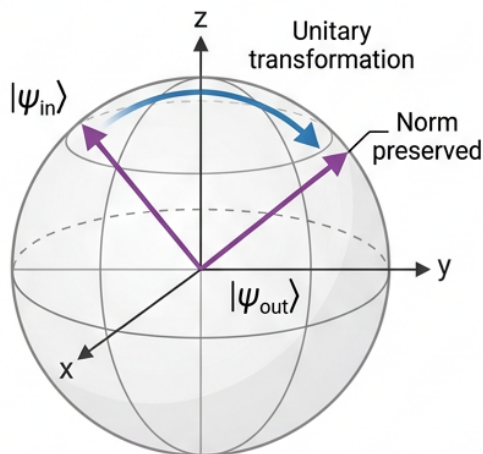
Physical optical transformations must preserve the norm (total probability).

B Definition of a Unitary Operator

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad U^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

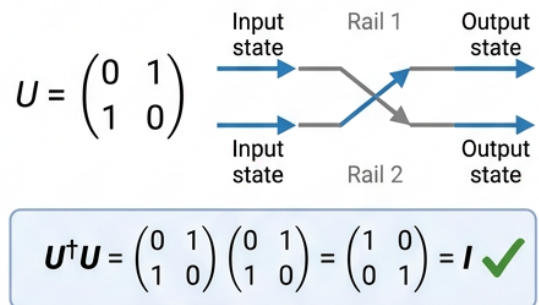
$$U = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \quad \text{Unitary: } U^\dagger U = I$$

* For real matrices, $U^\dagger = U^T$

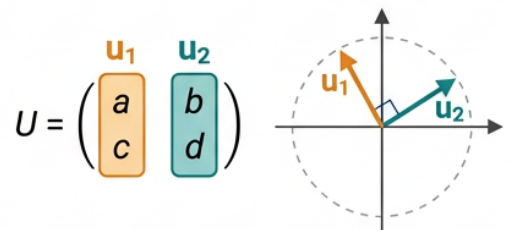
C Unitary Transformation Preserves Probability

$$\begin{aligned} \langle\psi_{\text{out}}|\psi_{\text{out}}\rangle &= \langle\psi_{\text{in}}|U^\dagger U|\psi_{\text{in}}\rangle \\ &= \langle\psi_{\text{in}}|I|\psi_{\text{in}}\rangle = \langle\psi_{\text{in}}|\psi_{\text{in}}\rangle \end{aligned}$$

Unitarity ensures total probability is unchanged during quantum evolution.

D Example – Swap Operator

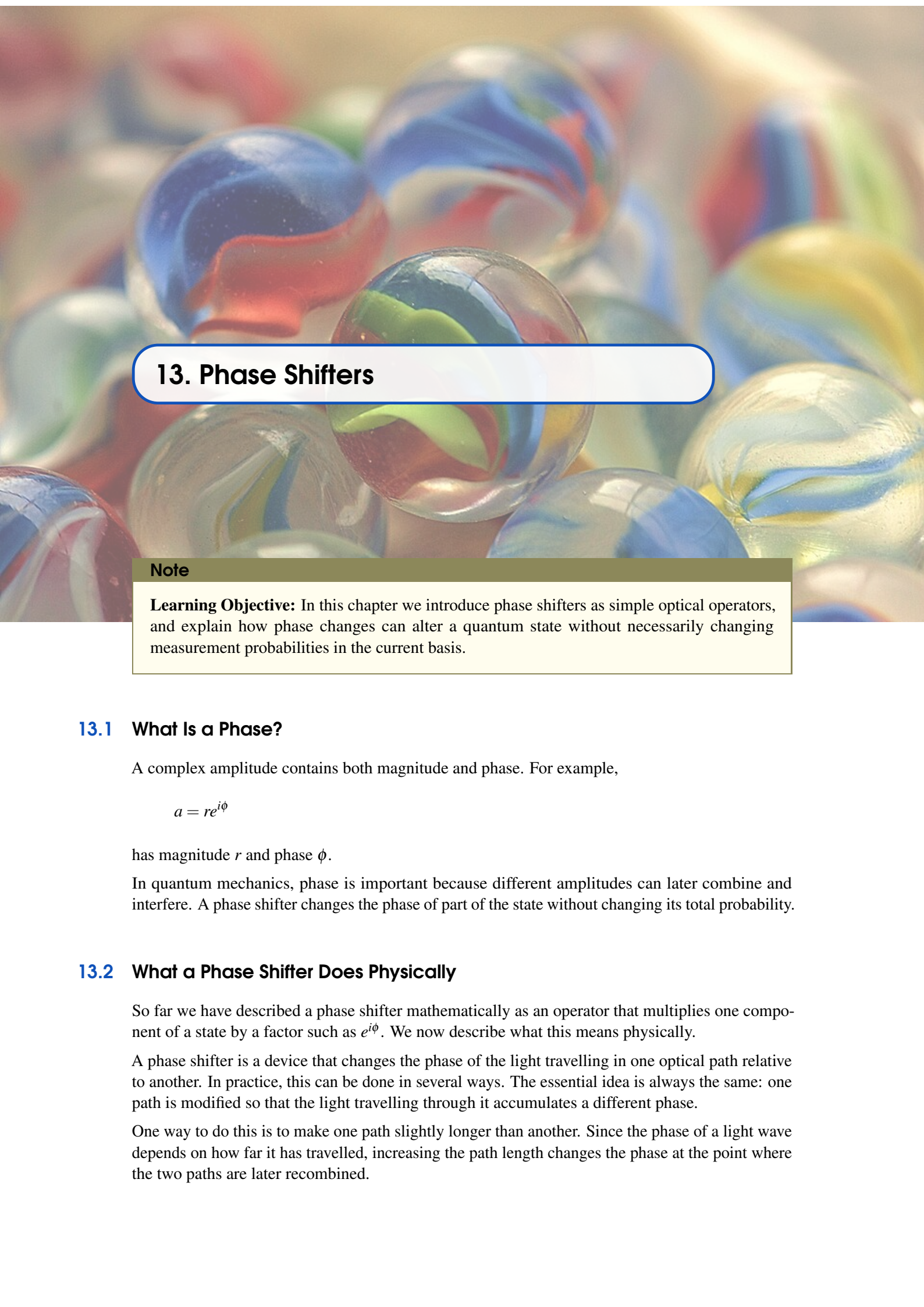
This swap operation is unitary—it preserves probability.

E Orthonormal Columns of a Unitary

Column vectors: Orthonormal set ($\langle\mathbf{u}_i, \mathbf{u}_j\rangle = \delta_{ij}$)

Unitary = Probability Conservation

Figure 12.1: Enter Caption

A background image of many colorful marbles in shades of blue, red, green, and yellow, some with swirl patterns. A white rounded rectangle with a blue border is overlaid on the image, containing the chapter title.

13. Phase Shifters

Note

Learning Objective: In this chapter we introduce phase shifters as simple optical operators, and explain how phase changes can alter a quantum state without necessarily changing measurement probabilities in the current basis.

13.1 What Is a Phase?

A complex amplitude contains both magnitude and phase. For example,

$$a = re^{i\phi}$$

has magnitude r and phase ϕ .

In quantum mechanics, phase is important because different amplitudes can later combine and interfere. A phase shifter changes the phase of part of the state without changing its total probability.

13.2 What a Phase Shifter Does Physically

So far we have described a phase shifter mathematically as an operator that multiplies one component of a state by a factor such as $e^{i\phi}$. We now describe what this means physically.

A phase shifter is a device that changes the phase of the light travelling in one optical path relative to another. In practice, this can be done in several ways. The essential idea is always the same: one path is modified so that the light travelling through it accumulates a different phase.

One way to do this is to make one path slightly longer than another. Since the phase of a light wave depends on how far it has travelled, increasing the path length changes the phase at the point where the two paths are later recombined.

Another method is to pass the light through a material with a different refractive index. Light travels more slowly in a medium than in vacuum, so changing the material, or changing its effective optical thickness, changes the phase accumulated by the wave.

On integrated photonic chips, phase shifts are often produced by small electrical or thermal elements that slightly change the refractive index of a waveguide. This changes the optical path length without needing to move any mechanical parts.

R The important quantity is not usually the absolute phase of one path by itself, but the *relative phase* between two possible paths.

This is why phase shifters are so important in interferometry. If two amplitudes later meet at a beam splitter, their relative phase determines whether they interfere constructively or destructively.

At the single-photon level, the same idea applies. Even though a photon is not described as a tiny classical particle moving along one definite path, the amplitudes associated with different possible paths still accumulate phases. When those amplitudes are recombined, the relative phase affects the final measurement probabilities.

So physically, a phase shifter does not usually change where the photon is immediately detected if we measure in the same basis straight away. Instead, it changes the relationship between the amplitudes of different paths, and this becomes observable when those paths are later mixed together.

Note

A phase shifter is therefore best understood as a device that changes the *timing relationship* or *optical path length relationship* between different components of a quantum state.

13.3 A One-Mode Phase Shift

If a single basis state picks up a phase $e^{i\phi}$, then

$$|1\rangle \mapsto e^{i\phi}|1\rangle.$$

Since

$$|e^{i\phi}|^2 = 1,$$

this does not change the probability associated with that basis state by itself.

13.4 A Two-Rail Phase Shifter

In a two-rail system, a phase shifter on the second rail can be represented by

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}.$$

Applying this to the state

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$$

Phase, Phase Shifts, and Physical Phase Shifters in Quantum Optics

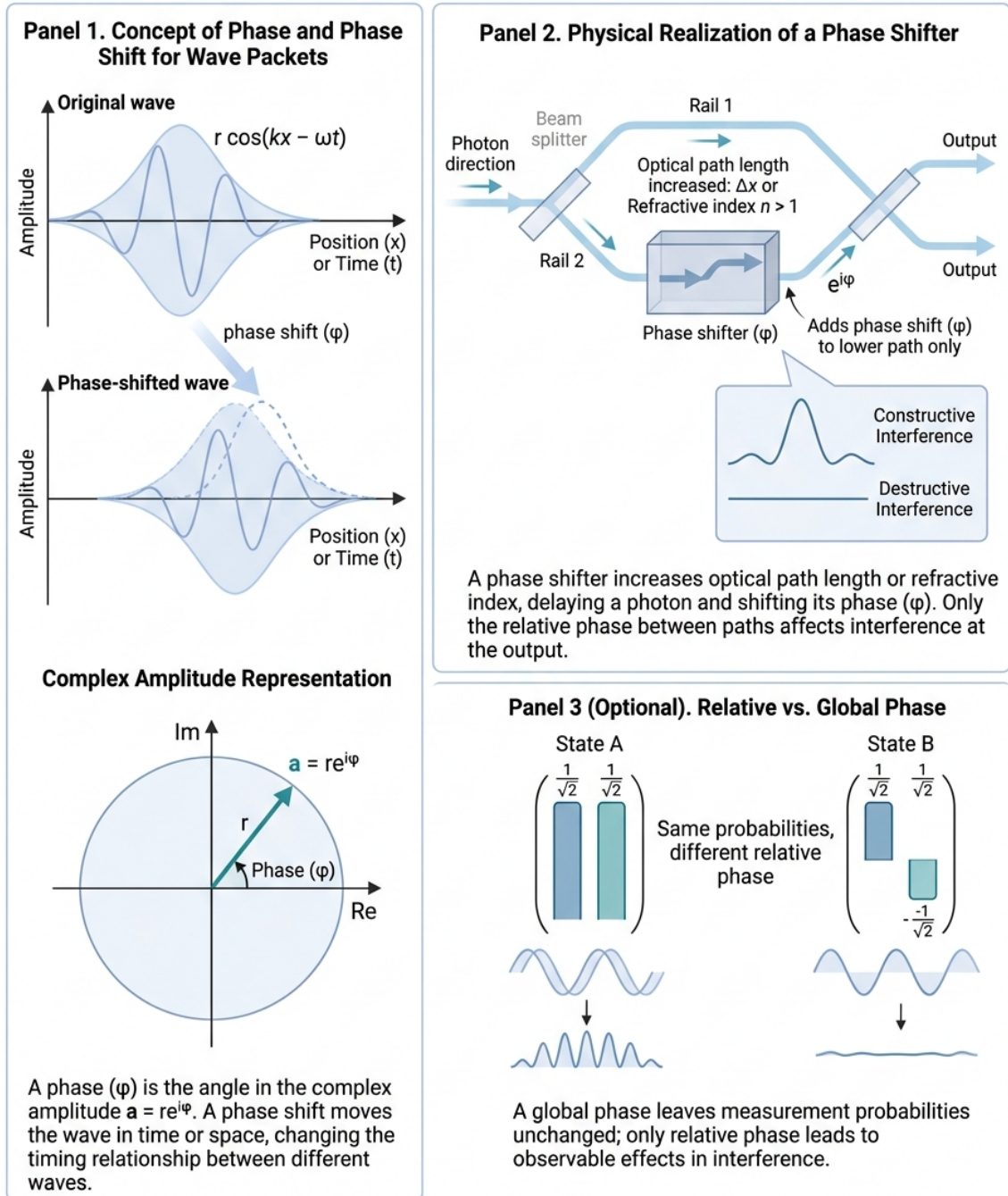


Figure 13.1: Enter Caption

gives

$$P(\phi)|\psi\rangle = \begin{pmatrix} a \\ e^{i\phi}b \end{pmatrix}.$$

So the amplitude in rail 1 is unchanged, while the amplitude in rail 2 acquires a phase.

Worked Example

Apply the phase shifter

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

to the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Solution.

$$P(\phi)|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\phi} \end{pmatrix}.$$

13.5 Global Phase and Relative Phase

Not all phases are physically important.

If the whole state is multiplied by the same phase factor,

$$|\psi\rangle \mapsto e^{i\gamma}|\psi\rangle,$$

then all measurement probabilities remain unchanged.

Definition

Definition 13.1. A **global phase** is an overall phase factor multiplying the entire state. Global phase does not affect measurement probabilities.

By contrast, a phase difference between components of a superposition is physically meaningful.

Definition

Definition 13.2. A **relative phase** is a phase difference between different components of a quantum state.

For example,

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

have the same basis-state probabilities, but a different relative phase.

Note

A phase shifter may appear to do nothing if we immediately measure in the same basis. Its effect becomes visible when the state later passes through another optical element and interference occurs.

13.6 Phase Shifters Are Unitary

To check that

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

is unitary, we compute

$$P(\phi)^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\phi} \end{pmatrix}.$$

Then

$$P(\phi)^\dagger P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\phi} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

So the phase shifter is unitary, as expected for a lossless optical element.

13.7 Exercises**Exercise**

Exercise 13.1. Apply

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

to the state

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Exercise

Exercise 13.2. Apply $P(\phi)$ to

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Exercise

Exercise 13.3. Explain the difference between a global phase and a relative phase.

Exercise

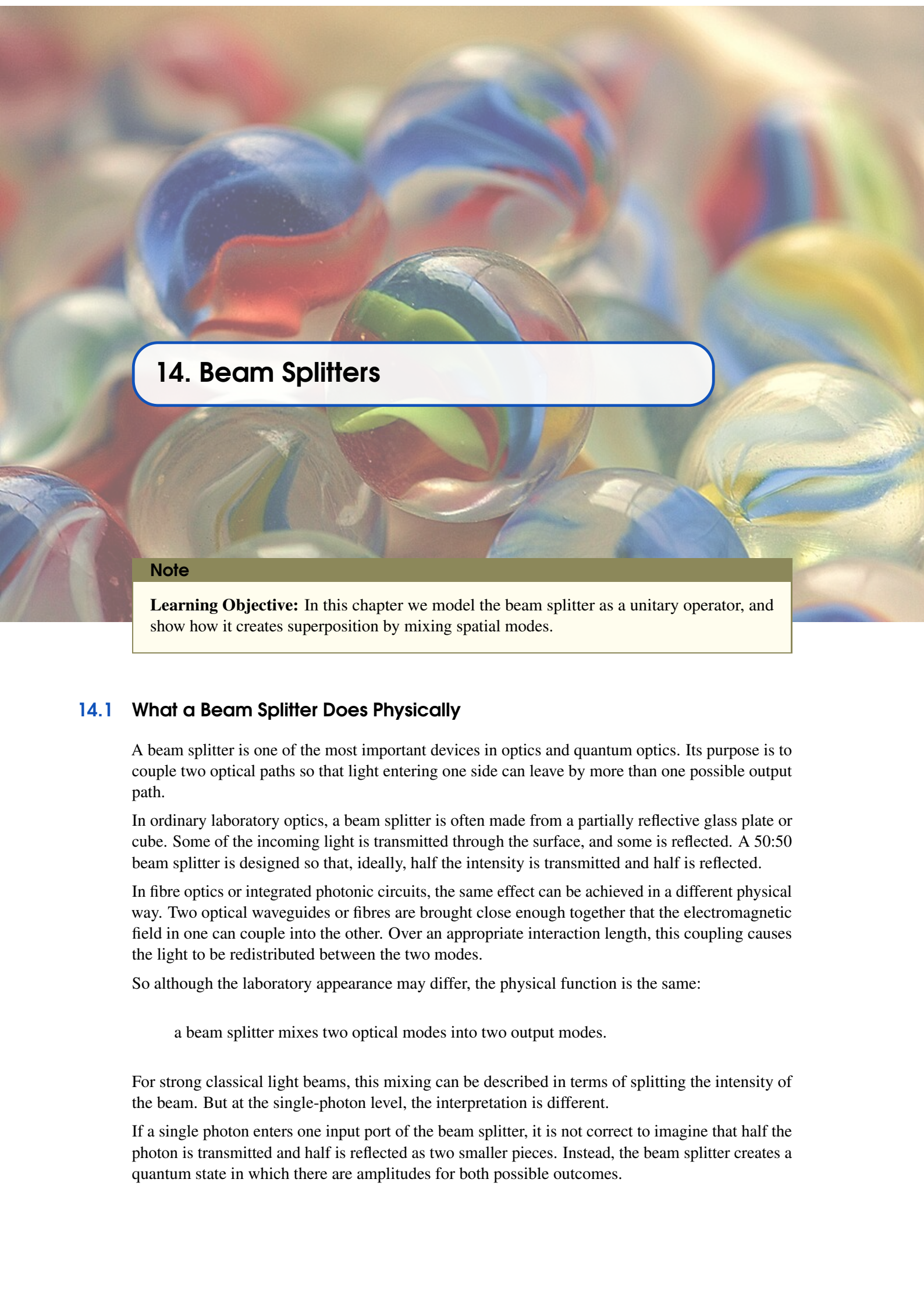
Exercise 13.4. Show that multiplying a normalised state by an overall phase $e^{i\gamma}$ does not change its normalisation.

Investigation

Use the [Linear Optics Lab Bench](#) to place a phase shifter on one rail of a two-rail system. Start with a single input photon. Compare the displayed state before and after the phase shifter. Does the output probability change immediately, or only after further mixing by another component?

Summary

A phase shifter changes the phase of part of a quantum state. Although this may leave immediate measurement probabilities unchanged in the current basis, the relative phase becomes important when amplitudes later interfere.

A background image of many colorful marbles in shades of blue, red, green, and yellow, some with swirl patterns. A white rounded rectangle with a blue border is overlaid on the image, containing the chapter title.

14. Beam Splitters

Note

Learning Objective: In this chapter we model the beam splitter as a unitary operator, and show how it creates superposition by mixing spatial modes.

14.1 What a Beam Splitter Does Physically

A beam splitter is one of the most important devices in optics and quantum optics. Its purpose is to couple two optical paths so that light entering one side can leave by more than one possible output path.

In ordinary laboratory optics, a beam splitter is often made from a partially reflective glass plate or cube. Some of the incoming light is transmitted through the surface, and some is reflected. A 50:50 beam splitter is designed so that, ideally, half the intensity is transmitted and half is reflected.

In fibre optics or integrated photonic circuits, the same effect can be achieved in a different physical way. Two optical waveguides or fibres are brought close enough together that the electromagnetic field in one can couple into the other. Over an appropriate interaction length, this coupling causes the light to be redistributed between the two modes.

So although the laboratory appearance may differ, the physical function is the same:

a beam splitter mixes two optical modes into two output modes.

For strong classical light beams, this mixing can be described in terms of splitting the intensity of the beam. But at the single-photon level, the interpretation is different.

If a single photon enters one input port of the beam splitter, it is not correct to imagine that half the photon is transmitted and half is reflected as two smaller pieces. Instead, the beam splitter creates a quantum state in which there are amplitudes for both possible outcomes.

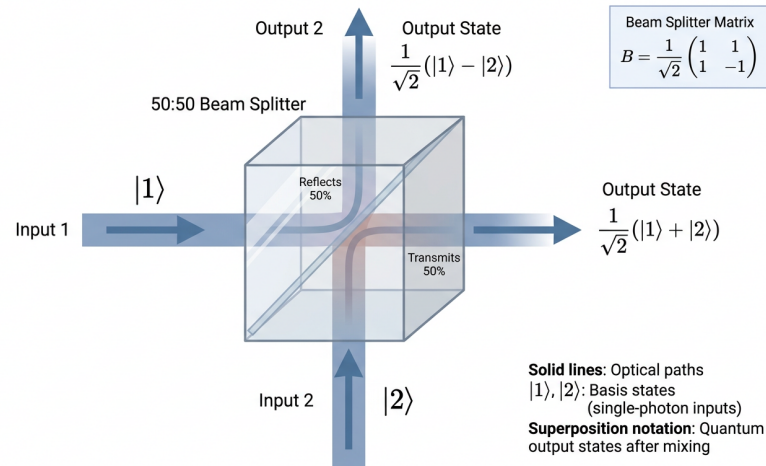


Figure 14.1: Enter Caption

This is why the output is described by a superposition. The photon has two possible paths available after the beam splitter, and the state vector assigns amplitudes to those possibilities.

R The beam splitter does not force the photon to “choose” a path immediately. Rather, it creates a state in which both output paths are part of the quantum description until a measurement is made.

This makes the beam splitter the natural optical device for introducing superposition. A basis state corresponding to a definite input path is turned into a state with amplitudes in more than one output path.

Later, when different paths are recombined, those amplitudes can interfere. This is why beam splitters are central to experiments such as the Mach–Zehnder interferometer and the Hong–Ou–Mandel effect.

Note

A beam splitter is therefore not just a passive optical junction. In quantum mechanics, it is a device that creates and mixes probability amplitudes.

14.2 A Simple 50:50 Beam Splitter Matrix

A convenient idealised model for a 50:50 beam splitter is

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

This matrix acts on the basis states as

$$B|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad B|2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

So a photon entering either input rail leaves in a superposition of the two output rails.

R Different phase conventions are used in the literature for beam splitters. The exact signs and factors of i may vary. What matters is that the transformation is unitary and captures the mixing of the two modes.

14.3 Beam Splitter Action on a General State

Let

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Then

$$B|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ a-b \end{pmatrix}.$$

So the beam splitter mixes the amplitudes from the two rails.

Worked Example

A photon enters rail 1, so the input state is

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Find the output state after a 50:50 beam splitter.

Solution.

$$B|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

So the output state is an equal superposition of the two output rails.

14.4 Checking Unitarity

We verify that B is unitary. Since the entries are real,

$$B^\dagger = B^T.$$

Also,

$$B^T B = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

So the beam splitter preserves total probability.

14.5 Beam Splitters Create Superposition

One of the most important roles of the beam splitter is that it takes basis states to superposition states.

This means that the beam splitter is not simply redirecting a photon along a definite classical path. Instead, it creates a state in which both output possibilities carry amplitudes.

This is one of the first places where the linear optics system clearly displays quantum behaviour in mathematical form.

14.6 Exercises

Exercise

Exercise 14.1. Calculate $B|2\rangle$ for

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Exercise

Exercise 14.2. Apply the beam splitter to the state

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Exercise

Exercise 14.3. Show directly that the columns of B are orthonormal.

Exercise

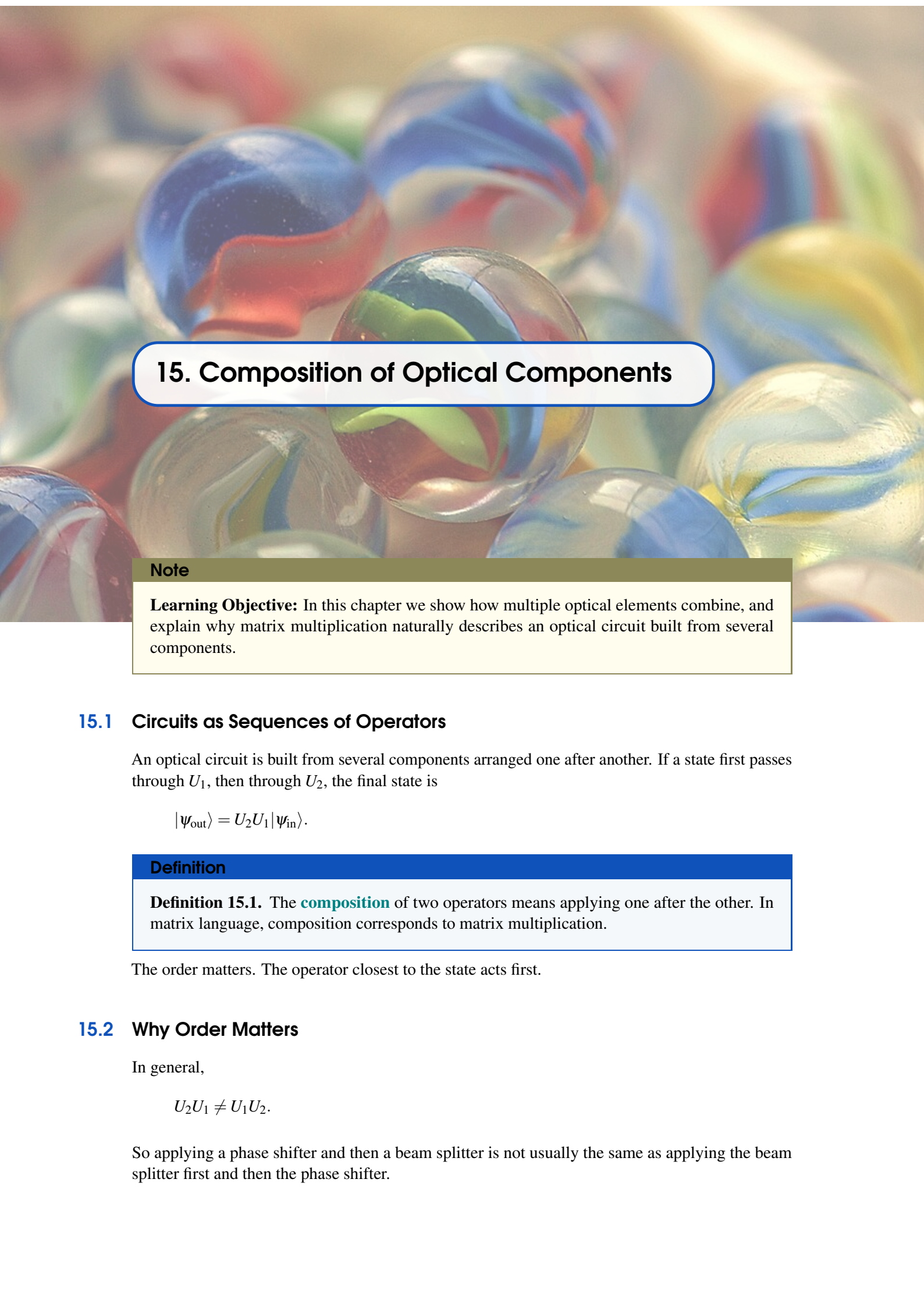
Exercise 14.4. Explain why a beam splitter can create superposition even when the input is a basis state.

Investigation

In the [Linear Optics Lab Bench](#), prepare a single photon in one rail and place a 50:50 beam splitter across two rails. Write down the predicted output state and compare it with the displayed amplitudes or probabilities.

Summary

A beam splitter mixes spatial modes and can turn a definite input path into a superposition of output paths. Mathematically, it is represented by a unitary matrix acting on the state vector.



15. Composition of Optical Components

Note

Learning Objective: In this chapter we show how multiple optical elements combine, and explain why matrix multiplication naturally describes an optical circuit built from several components.

15.1 Circuits as Sequences of Operators

An optical circuit is built from several components arranged one after another. If a state first passes through U_1 , then through U_2 , the final state is

$$|\psi_{\text{out}}\rangle = U_2 U_1 |\psi_{\text{in}}\rangle.$$

Definition

Definition 15.1. The **composition** of two operators means applying one after the other. In matrix language, composition corresponds to matrix multiplication.

The order matters. The operator closest to the state acts first.

15.2 Why Order Matters

In general,

$$U_2 U_1 \neq U_1 U_2.$$

So applying a phase shifter and then a beam splitter is not usually the same as applying the beam splitter first and then the phase shifter.

This is one reason quantum circuits are interesting: the physical order of the devices matters mathematically.

Worked Example

Let

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}, \quad B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$BP(\phi) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{i\phi} \\ 1 & -e^{i\phi} \end{pmatrix},$$

while

$$P(\phi)B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ e^{i\phi} & -e^{i\phi} \end{pmatrix}.$$

These are not equal in general, so the order matters.

15.3 Successive Beam Splitters

Consider a photon entering rail 1 and passing through two identical 50:50 beam splitters in succession. Using

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

we compute

$$B^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = I.$$

So two such beam splitters in a row give the identity transformation in this convention.

This is a simple but important reminder that interference between amplitudes can reconstruct the original state.

15.4 A First Circuit Calculation

Let the input state be

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Now apply:

1. a beam splitter B ,
2. then a phase shifter $P(\phi)$,
3. then another beam splitter B .

The output state is

$$|\psi_{\text{out}}\rangle = BP(\phi)B|1\rangle.$$

First,

$$B|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$P(\phi)B|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\phi} \end{pmatrix}.$$

Finally,

$$BP(\phi)B|1\rangle = \frac{1}{2} \begin{pmatrix} 1 + e^{i\phi} \\ 1 - e^{i\phi} \end{pmatrix}.$$

This is one of the most important expressions in elementary linear optics. It shows how a relative phase introduced between two paths changes the final output amplitudes after recombination.

15.5 Toward Interference

If we now calculate output probabilities, we obtain

$$P(1) = \left| \frac{1 + e^{i\phi}}{2} \right|^2, \quad P(2) = \left| \frac{1 - e^{i\phi}}{2} \right|^2.$$

These depend on ϕ , which means the phase shifter has changed the observable outcomes after the second beam splitter.

This is interference: amplitudes combine, and their relative phase affects the probabilities.

Note

A phase shift may appear invisible when looking only at one stage of a circuit, but become observable after amplitudes are mixed again by a beam splitter.

15.6 Exercises

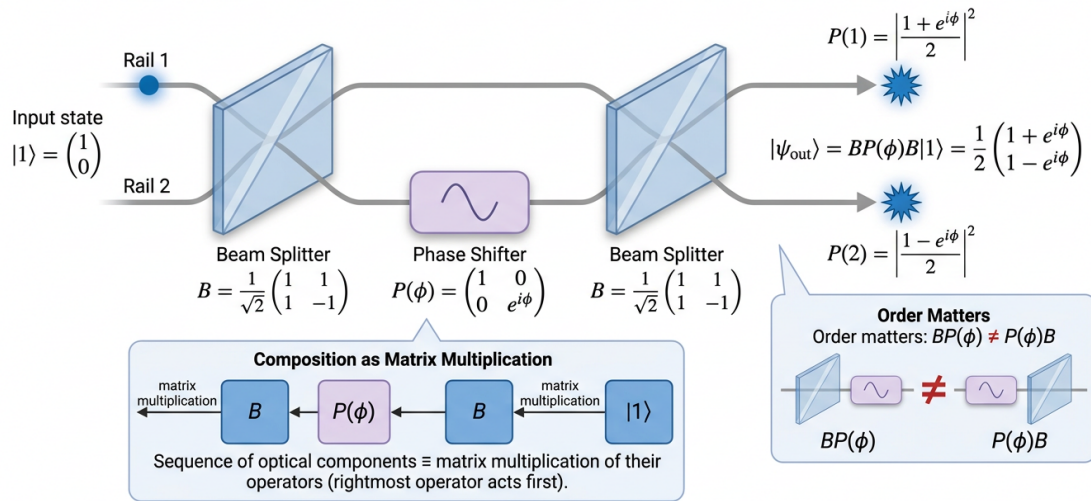
Exercise

Exercise 15.1. Let

$$U = BP(\phi).$$

Calculate $U|1\rangle$.

Composition of Optical Components as Matrix Multiplication



"Optical circuits consist of a sequence of components (beam splitters, phase shifters, etc.), each represented by a matrix. The transformation of an input state is described by the product of these matrices in the order the components are encountered—demonstrating why physical composition of devices is modeled by matrix multiplication, and why the order is crucial."

Figure 15.1: Enter Caption

Exercise

Exercise 15.2. Show directly that $BP(\phi) \neq P(\phi)B$ in general.

Exercise

Exercise 15.3. Verify that $B^2 = I$ for the beam splitter convention used in this text.

Exercise

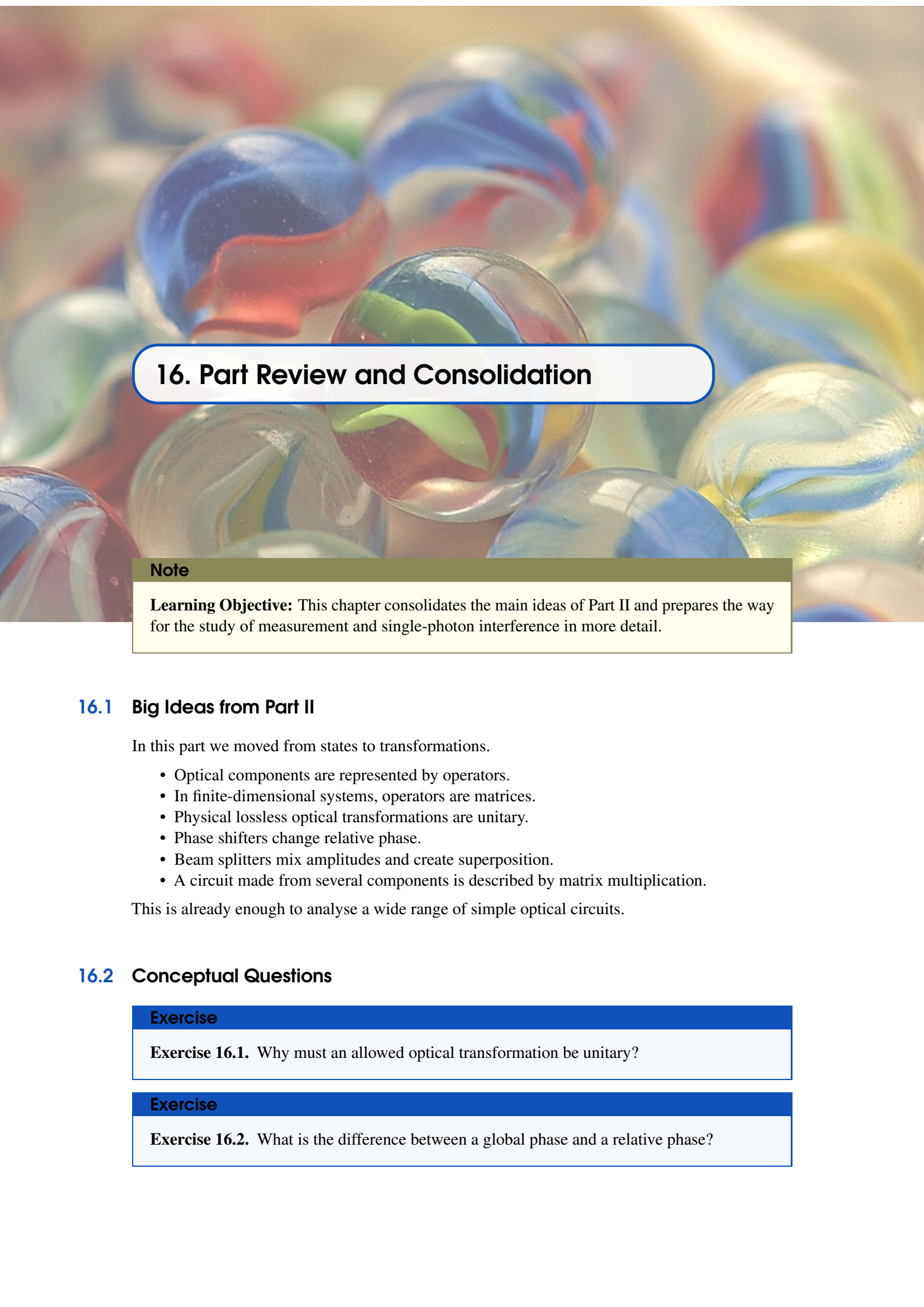
Exercise 15.4. For the circuit $BP(\phi)B$, calculate the output state when the input is $|2\rangle$.

Investigation

Use the [Linear Optics Lab Bench](#) to build a two-rail circuit with beam splitter, phase shifter, then beam splitter. Vary the phase and compare the predicted output amplitudes with the observed output probabilities.

Summary

Optical circuits are described by composing operators, which corresponds to matrix multiplication. The order of components matters, and combining beam splitters with phase shifters leads naturally to interference effects.

A close-up, slightly blurred photograph of numerous colorful marbles in shades of blue, red, green, and yellow, creating a vibrant, textured background.

16. Part Review and Consolidation

Note

Learning Objective: This chapter consolidates the main ideas of Part II and prepares the way for the study of measurement and single-photon interference in more detail.

16.1 Big Ideas from Part II

In this part we moved from states to transformations.

- Optical components are represented by operators.
- In finite-dimensional systems, operators are matrices.
- Physical lossless optical transformations are unitary.
- Phase shifters change relative phase.
- Beam splitters mix amplitudes and create superposition.
- A circuit made from several components is described by matrix multiplication.

This is already enough to analyse a wide range of simple optical circuits.

16.2 Conceptual Questions

Exercise

Exercise 16.1. Why must an allowed optical transformation be unitary?

Exercise

Exercise 16.2. What is the difference between a global phase and a relative phase?

Exercise

Exercise 16.3. Why does the order of optical components matter in general?

Exercise

Exercise 16.4. Explain why a beam splitter is able to create superposition.

16.3 Mixed Practice**Exercise**

Problem 16.1. A photon in a two-rail system is prepared in the state

$$|\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

It then passes through the beam splitter

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

1. Find the output state.
2. Show that the output state is normalised.
3. Determine the probability of detecting the photon in each output rail.

Exercise

Problem 16.2. A photon in the state

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

passes through the phase shifter

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}.$$

1. Find the output state.
2. Show that the probabilities in the rail basis are unchanged immediately after this phase shifter.
3. Explain why the phase is still physically important.

Exercise

Problem 16.3. Consider the circuit $BP(\phi)B$ acting on the input state $|1\rangle$.

1. Show that the output state is

$$\frac{1}{2} \begin{pmatrix} 1 + e^{i\phi} \\ 1 - e^{i\phi} \end{pmatrix}.$$

2. Find the output probabilities when $\phi = 0$.
3. Find the output probabilities when $\phi = \pi$.
4. Explain the physical significance of these two cases.

16.4 Looking Ahead

In Part III we will focus more carefully on measurement and single-photon interference. We will see that the Born rule, combined with operator evolution, gives a complete predictive framework for many optical experiments.

We will also begin to understand why quantum amplitudes, rather than classical probabilities, are the natural language of these systems.

Summary

Part II introduced operators as matrices representing optical components. We saw how beam splitters and phase shifters act on states, why unitarity matters, and how circuits are described through the composition of operators. This gives us a complete linear-algebraic framework for simple quantum optical systems.

IV Measurement and Single-Photon Interference

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- 17.3 Measurement as Projection
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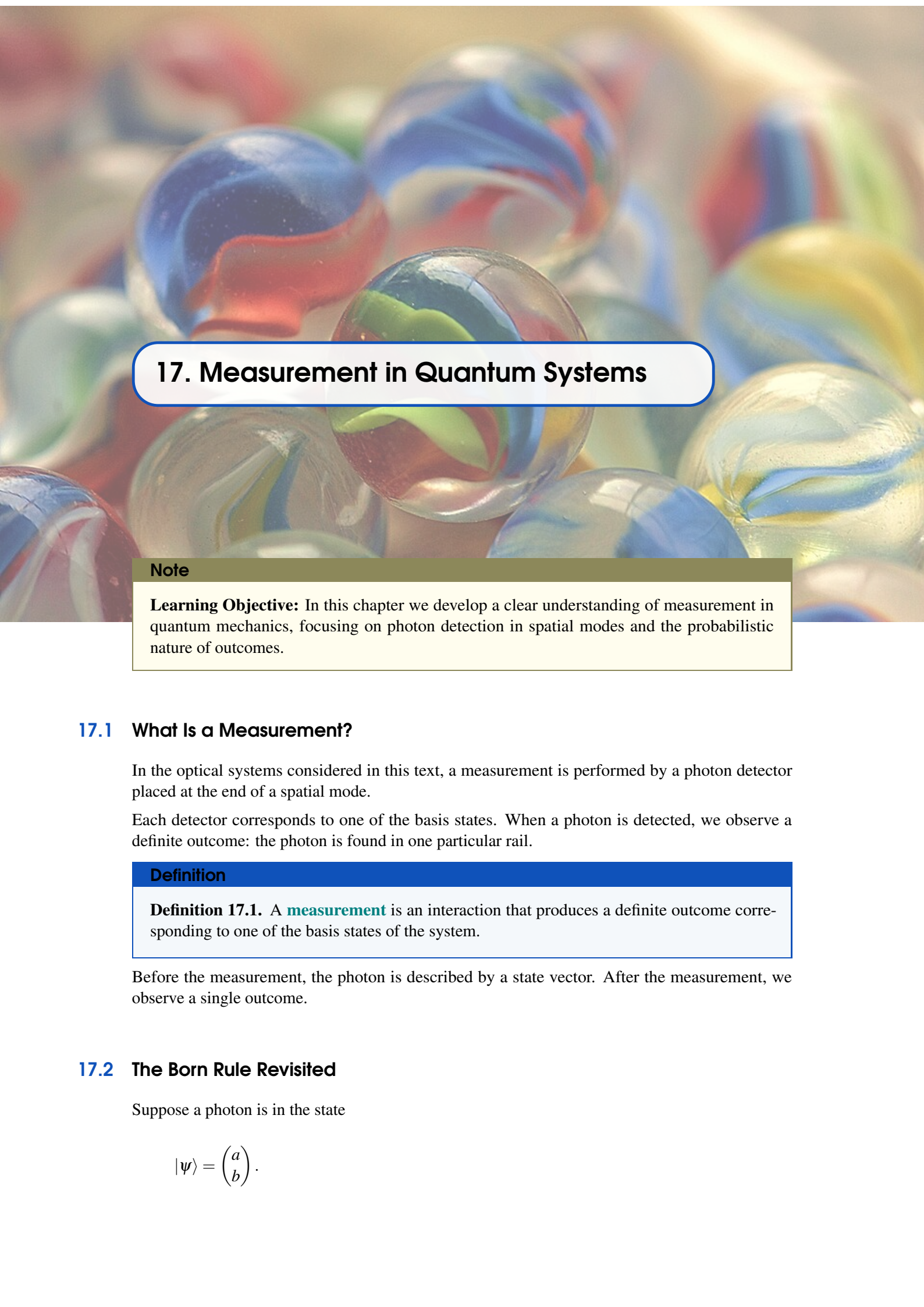
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A background image showing a close-up of several colorful marbles in shades of blue, red, green, and yellow, with some light reflecting off their glossy surfaces.

17. Measurement in Quantum Systems

Note

Learning Objective: In this chapter we develop a clear understanding of measurement in quantum mechanics, focusing on photon detection in spatial modes and the probabilistic nature of outcomes.

17.1 What Is a Measurement?

In the optical systems considered in this text, a measurement is performed by a photon detector placed at the end of a spatial mode.

Each detector corresponds to one of the basis states. When a photon is detected, we observe a definite outcome: the photon is found in one particular rail.

Definition

Definition 17.1. A **measurement** is an interaction that produces a definite outcome corresponding to one of the basis states of the system.

Before the measurement, the photon is described by a state vector. After the measurement, we observe a single outcome.

17.2 The Born Rule Revisited

Suppose a photon is in the state

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Then:

$$P(1) = |a|^2, \quad P(2) = |b|^2.$$

Definition

Definition 17.2. The probability of a measurement outcome is given by the squared modulus of the corresponding amplitude:

$$P(i) = |\langle i | \psi \rangle|^2.$$

This connects the vector formalism directly to experimental observations.

17.3 Measurement as Projection

When a measurement is performed and a particular outcome is obtained, the state of the system changes.

If the photon is detected in rail 1, the state becomes

$$|1\rangle.$$

If it is detected in rail 2, the state becomes

$$|2\rangle.$$

Definition

Definition 17.3. Measurement projects the state onto the basis state corresponding to the observed outcome.

This is sometimes called **collapse of the state**.

 The evolution of the state before measurement is deterministic. The outcome of a measurement is probabilistic.

17.4 Repeated Measurements

If we prepare the same state many times and measure repeatedly, the relative frequencies of outcomes approach the probabilities predicted by the Born rule.

For example, if

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

then after many trials we expect approximately:

- 50% detections in rail 1,
- 50% detections in rail 2.

17.5 Measurement Depends on the Basis

The probabilities depend on which measurement we perform.

If we measure in the rail basis, we use:

$$P(i) = |\langle i | \psi \rangle|^2.$$

But if the system is first transformed by an operator U , then the probabilities are determined from:

$$U|\psi\rangle.$$

This means that changing the measurement setup (for example by inserting beam splitters) changes the observed probabilities.

Note

In linear optics, changing the measurement basis is often achieved by placing optical components before the detectors.

17.6 Exercises

Exercise

Exercise 17.1. A photon is in the state

$$\frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Find the probability of detecting the photon in each rail.

Exercise

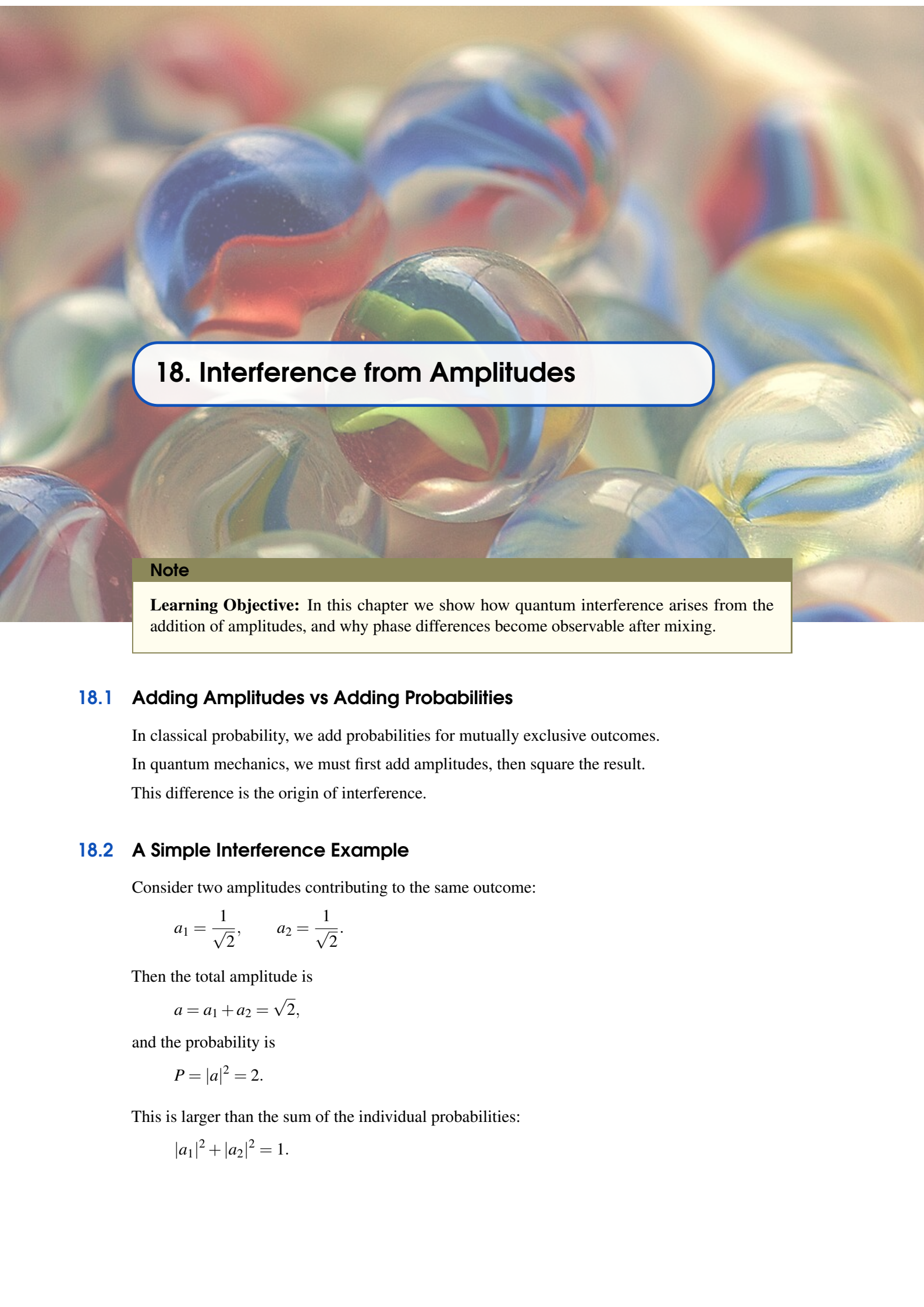
Exercise 17.2. Explain what happens to the state after a measurement outcome is observed.

Exercise

Exercise 17.3. Why must probabilities sum to 1?

Summary

Measurement produces definite outcomes, but the probabilities of those outcomes are determined by the amplitudes in the state vector. After measurement, the state is projected onto the observed basis state.

A background image showing a close-up of several colorful marbles in shades of blue, red, green, and yellow. The marbles are out of focus, creating a bokeh effect. A white rounded rectangle with a blue border is overlaid on the image, containing the chapter title.

18. Interference from Amplitudes

Note

Learning Objective: In this chapter we show how quantum interference arises from the addition of amplitudes, and why phase differences become observable after mixing.

18.1 Adding Amplitudes vs Adding Probabilities

In classical probability, we add probabilities for mutually exclusive outcomes.

In quantum mechanics, we must first add amplitudes, then square the result.

This difference is the origin of interference.

18.2 A Simple Interference Example

Consider two amplitudes contributing to the same outcome:

$$a_1 = \frac{1}{\sqrt{2}}, \quad a_2 = \frac{1}{\sqrt{2}}.$$

Then the total amplitude is

$$a = a_1 + a_2 = \sqrt{2},$$

and the probability is

$$P = |a|^2 = 2.$$

This is larger than the sum of the individual probabilities:

$$|a_1|^2 + |a_2|^2 = 1.$$

Definition

Definition 18.1. **Interference** occurs when amplitudes combine before probabilities are calculated.

18.3 Constructive and Destructive Interference

If the amplitudes have the same phase, they reinforce:

$$a_1 + a_2 \rightarrow \text{large probability.}$$

If they have opposite phase, they cancel:

$$a_1 + (-a_2) = 0.$$

So

$$P = 0.$$

 This is not possible in classical probability. It is a uniquely quantum effect.

18.4 Role of Phase

Consider

$$a_1 = \frac{1}{\sqrt{2}}, \quad a_2 = \frac{1}{\sqrt{2}}e^{i\phi}.$$

Then

$$a = a_1 + a_2 = \frac{1}{\sqrt{2}}(1 + e^{i\phi}).$$

The probability depends on ϕ , showing that phase affects observable outcomes.

18.5 Exercises**Exercise**

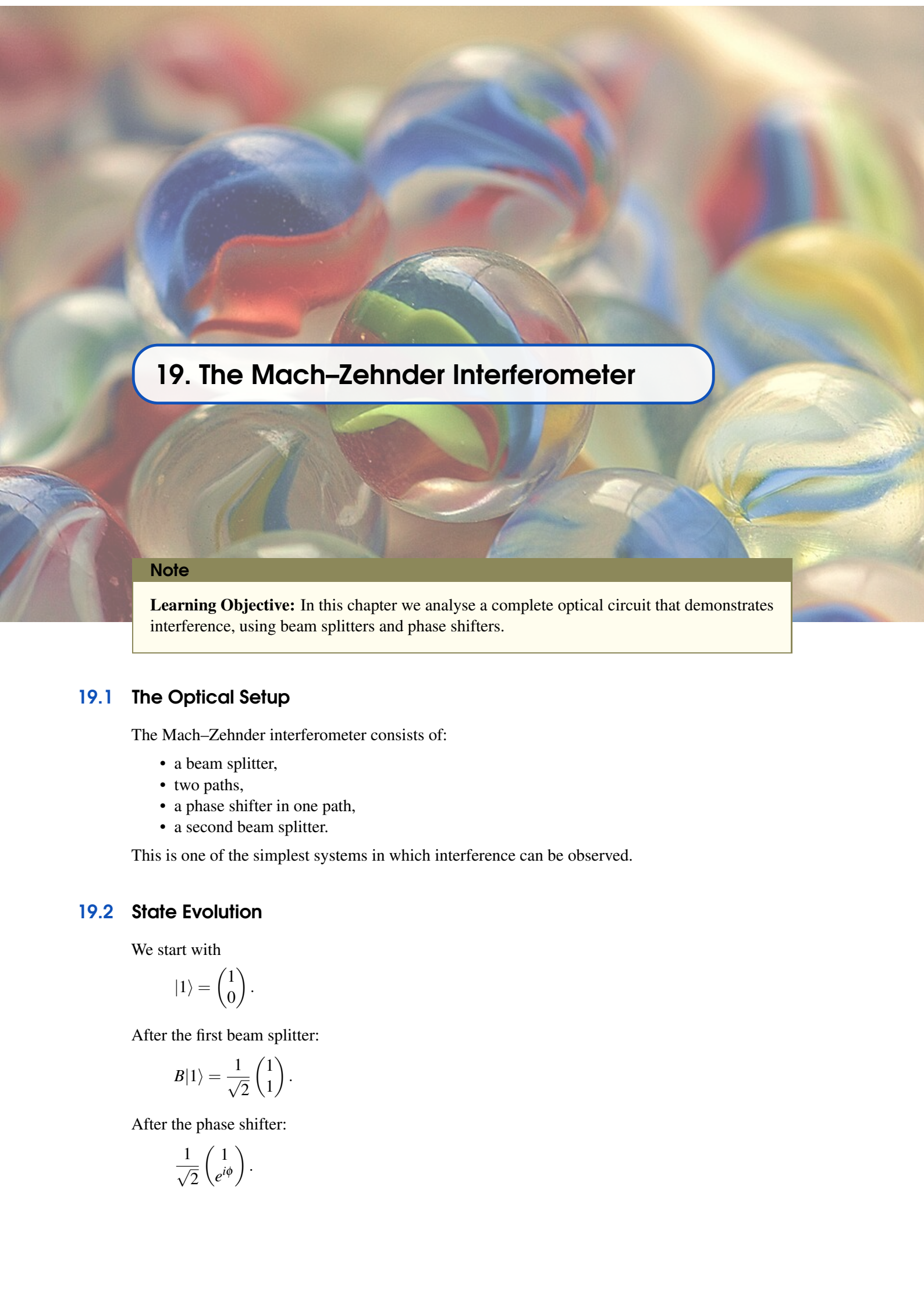
Exercise 18.1. Calculate the probability when two amplitudes $1/\sqrt{2}$ and $-1/\sqrt{2}$ combine.

Exercise

Exercise 18.2. Explain why interference cannot be explained using classical probability alone.

Summary

Interference arises because amplitudes add before probabilities are calculated. Relative phase determines whether interference is constructive or destructive.



19. The Mach-Zehnder Interferometer

Note

Learning Objective: In this chapter we analyse a complete optical circuit that demonstrates interference, using beam splitters and phase shifters.

19.1 The Optical Setup

The Mach-Zehnder interferometer consists of:

- a beam splitter,
- two paths,
- a phase shifter in one path,
- a second beam splitter.

This is one of the simplest systems in which interference can be observed.

19.2 State Evolution

We start with

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

After the first beam splitter:

$$B|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

After the phase shifter:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\phi} \end{pmatrix}.$$

After the second beam splitter:

$$\frac{1}{2} \begin{pmatrix} 1 + e^{i\phi} \\ 1 - e^{i\phi} \end{pmatrix}.$$

19.3 Output Probabilities

$$P(1) = \left| \frac{1 + e^{i\phi}}{2} \right|^2, \quad P(2) = \left| \frac{1 - e^{i\phi}}{2} \right|^2.$$

These vary with ϕ , producing an interference pattern.

19.4 Special Cases

- If $\phi = 0$, then all photons exit one port.
- If $\phi = \pi$, then all photons exit the other port.

19.5 Physical Interpretation

The photon does not travel along a single path in the classical sense. Instead, the amplitudes associated with different paths interfere when recombined.

Note

The Mach–Zehnder interferometer is a direct demonstration that phase differences affect measurable outcomes.

19.6 Exercises

Exercise

Exercise 19.1. Verify the output state for the interferometer step by step.

Exercise

Exercise 19.2. Calculate the output probabilities for $\phi = \pi/2$.

Exercise

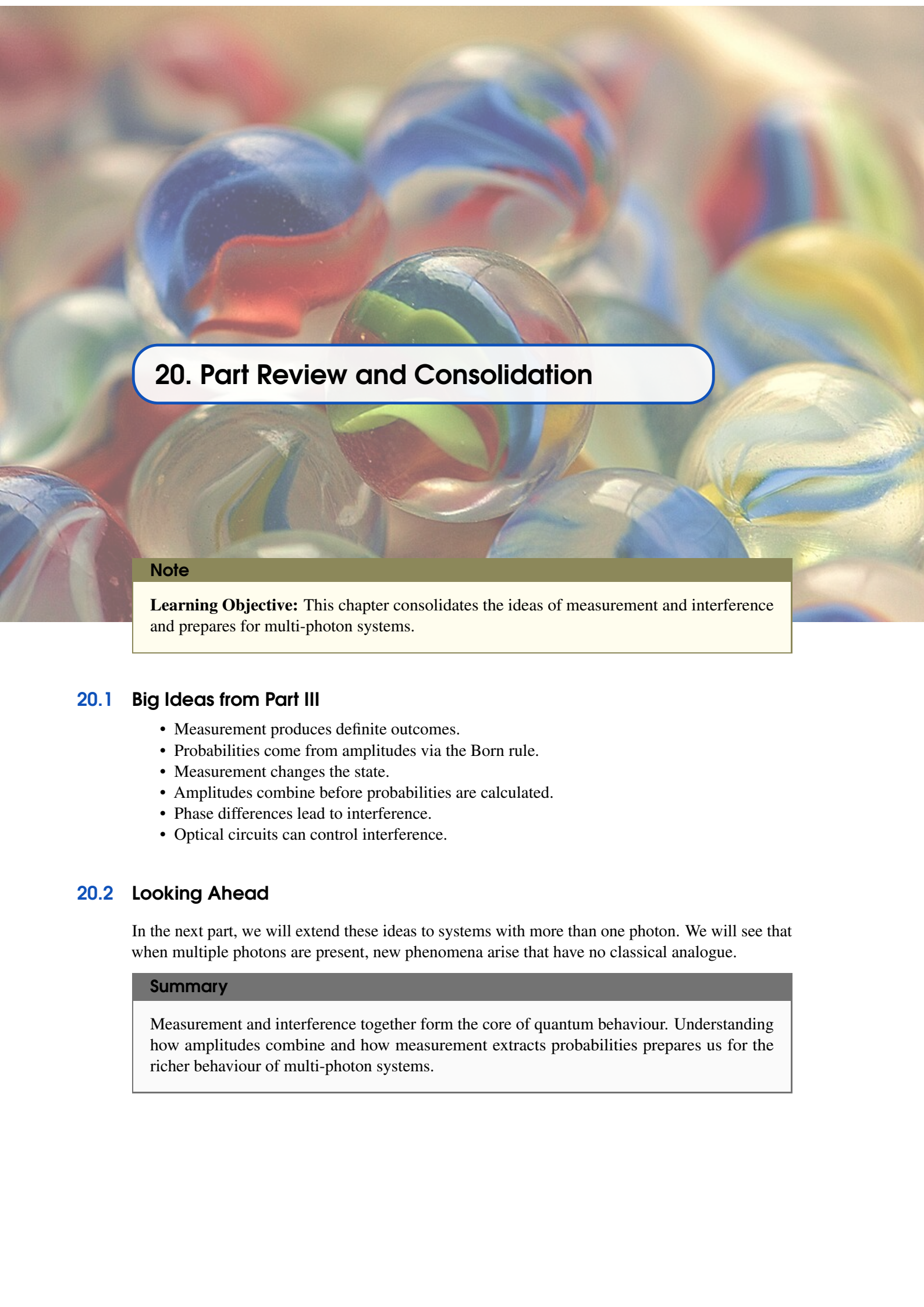
Exercise 19.3. Explain why the phase shifter has no effect if the second beam splitter is removed.

Investigation

Use the [Linear Optics Lab Bench](#) to construct a Mach–Zehnder interferometer. Vary the phase and compare the observed probabilities with the theoretical predictions.

Summary

The Mach–Zehnder interferometer shows how phase differences lead to interference. This provides a clear demonstration of how amplitudes, not probabilities, determine quantum behaviour.

A close-up, shallow depth-of-field photograph of numerous colorful marbles. The marbles are in various shades of blue, red, green, and yellow, with some showing internal patterns or swirls. They are scattered across the frame, with some in sharp focus and others blurred in the background.

20. Part Review and Consolidation

Note

Learning Objective: This chapter consolidates the ideas of measurement and interference and prepares for multi-photon systems.

20.1 Big Ideas from Part III

- Measurement produces definite outcomes.
- Probabilities come from amplitudes via the Born rule.
- Measurement changes the state.
- Amplitudes combine before probabilities are calculated.
- Phase differences lead to interference.
- Optical circuits can control interference.

20.2 Looking Ahead

In the next part, we will extend these ideas to systems with more than one photon. We will see that when multiple photons are present, new phenomena arise that have no classical analogue.

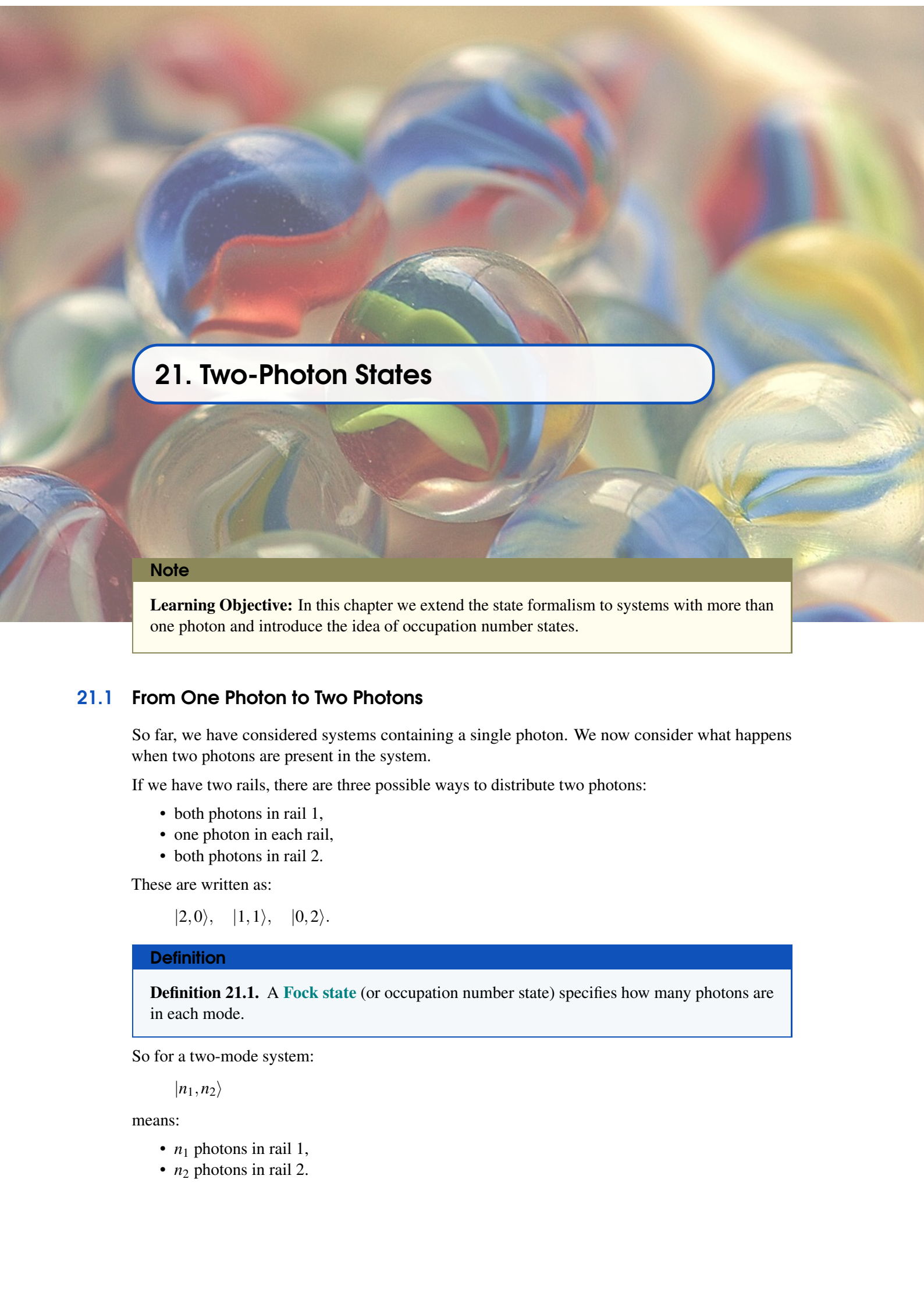
Summary

Measurement and interference together form the core of quantum behaviour. Understanding how amplitudes combine and how measurement extracts probabilities prepares us for the richer behaviour of multi-photon systems.



Two Photons and Indistinguishability

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21. Two-Photon States

Note

Learning Objective: In this chapter we extend the state formalism to systems with more than one photon and introduce the idea of occupation number states.

21.1 From One Photon to Two Photons

So far, we have considered systems containing a single photon. We now consider what happens when two photons are present in the system.

If we have two rails, there are three possible ways to distribute two photons:

- both photons in rail 1,
- one photon in each rail,
- both photons in rail 2.

These are written as:

$$|2,0\rangle, \quad |1,1\rangle, \quad |0,2\rangle.$$

Definition

Definition 21.1. A **Fock state** (or occupation number state) specifies how many photons are in each mode.

So for a two-mode system:

$$|n_1, n_2\rangle$$

means:

- n_1 photons in rail 1,
- n_2 photons in rail 2.

21.2 The State Space

For two photons in two modes, the state space is three-dimensional:

$$\text{span}\{|2,0\rangle, |1,1\rangle, |0,2\rangle\}.$$

A general state is:

$$|\psi\rangle = a|2,0\rangle + b|1,1\rangle + c|0,2\rangle,$$

with

$$|a|^2 + |b|^2 + |c|^2 = 1.$$

21.3 Physical Interpretation

Each basis state corresponds to a definite detection pattern:

- $|2,0\rangle$: both photons detected in rail 1,
- $|1,1\rangle$: one photon in each rail,
- $|0,2\rangle$: both photons in rail 2.

Note

In experiments, these outcomes are observed using photon detectors placed at each output mode.

21.4 Exercises

Exercise

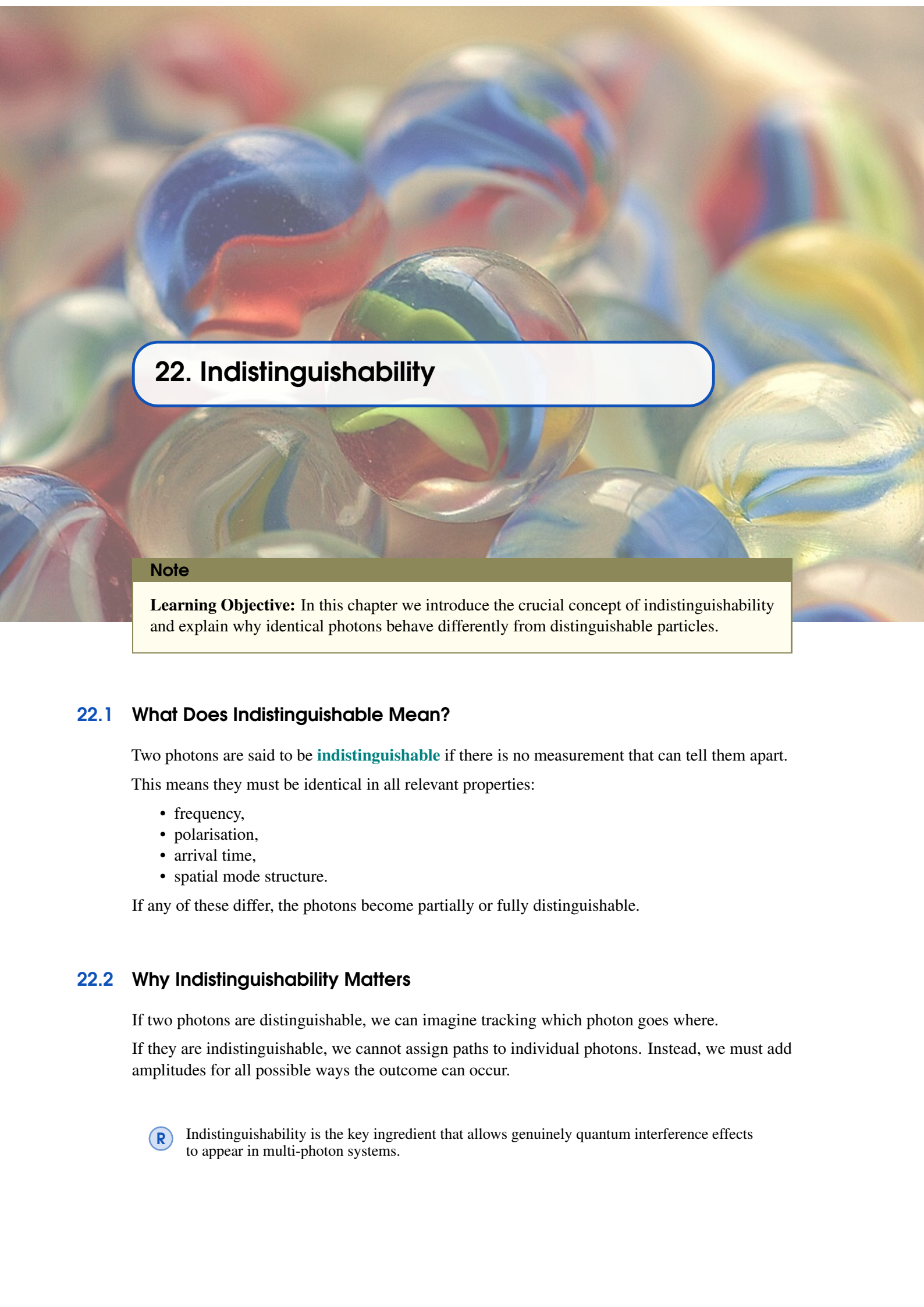
Exercise 21.1. List all possible Fock states for two photons in three modes.

Exercise

Exercise 21.2. Explain what the state $|1,1\rangle$ represents physically.

Summary

Two-photon systems are described using occupation number states. Even for two modes, the state space is already larger than in the single-photon case.

A background image showing a close-up of several colorful marbles in shades of blue, red, green, and yellow, with some light reflecting off their glossy surfaces.

22. Indistinguishability

Note

Learning Objective: In this chapter we introduce the crucial concept of indistinguishability and explain why identical photons behave differently from distinguishable particles.

22.1 What Does Indistinguishable Mean?

Two photons are said to be **indistinguishable** if there is no measurement that can tell them apart.

This means they must be identical in all relevant properties:

- frequency,
- polarisation,
- arrival time,
- spatial mode structure.

If any of these differ, the photons become partially or fully distinguishable.

22.2 Why Indistinguishability Matters

If two photons are distinguishable, we can imagine tracking which photon goes where.

If they are indistinguishable, we cannot assign paths to individual photons. Instead, we must add amplitudes for all possible ways the outcome can occur.

R Indistinguishability is the key ingredient that allows genuinely quantum interference effects to appear in multi-photon systems.

22.3 Distinguishable vs Indistinguishable Cases

Consider two photons entering a beam splitter from different input rails.

- If they are distinguishable, we treat them as independent particles.
- If they are indistinguishable, their amplitudes interfere.

This leads to dramatically different outcomes.

22.4 Exercises

Exercise

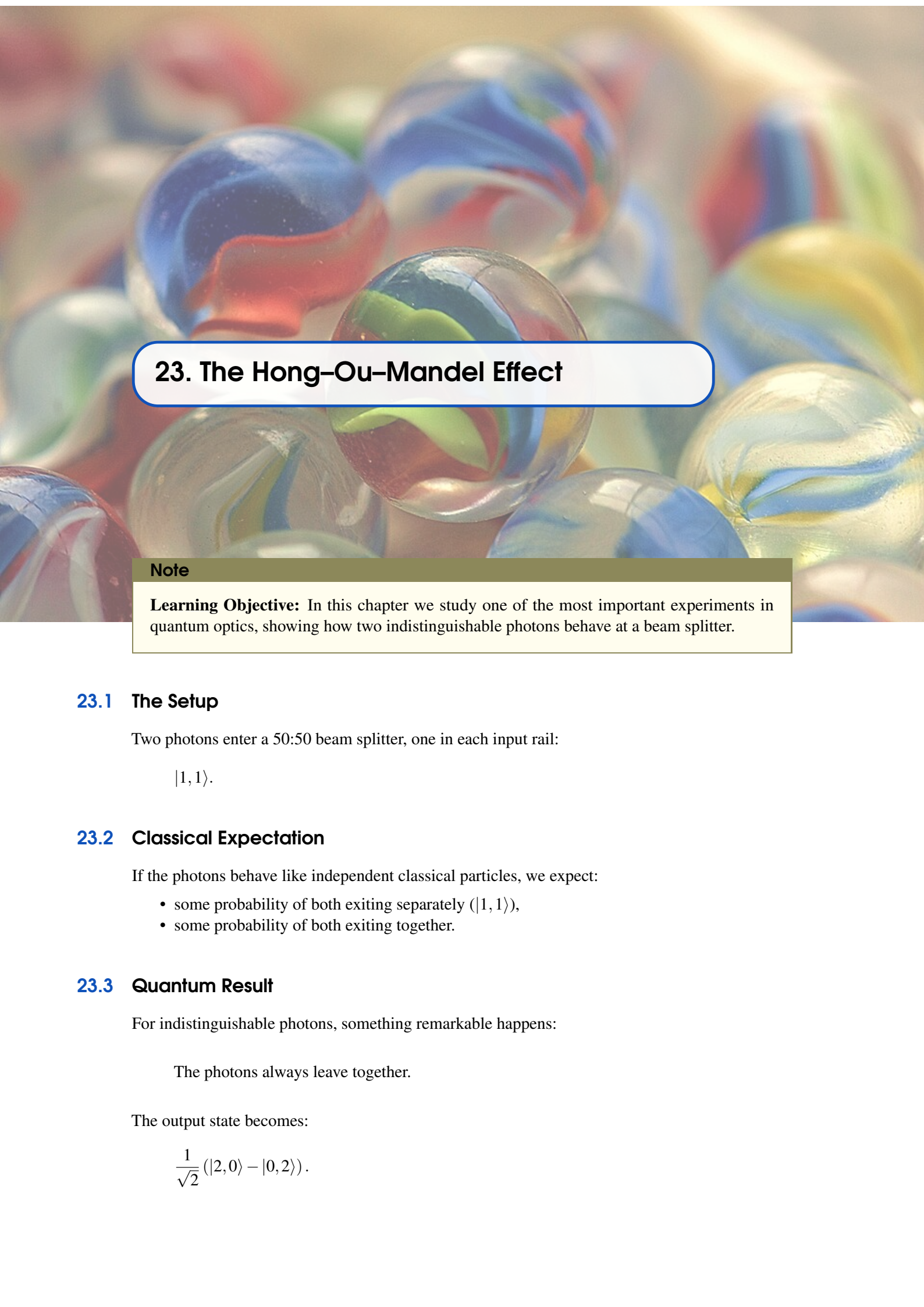
Exercise 22.1. Give two examples of physical differences that would make photons distinguishable.

Exercise

Exercise 22.2. Explain why indistinguishability prevents us from assigning definite paths to individual photons.

Summary

Indistinguishability determines whether multi-photon amplitudes interfere. This is the origin of many uniquely quantum optical effects.

A background image of many colorful marbles in shades of blue, red, green, and yellow, some with swirl patterns. A white rounded rectangle with a blue border is centered over the image, containing the chapter title.

23. The Hong–Ou–Mandel Effect

Note

Learning Objective: In this chapter we study one of the most important experiments in quantum optics, showing how two indistinguishable photons behave at a beam splitter.

23.1 The Setup

Two photons enter a 50:50 beam splitter, one in each input rail:

$$|1, 1\rangle.$$

23.2 Classical Expectation

If the photons behave like independent classical particles, we expect:

- some probability of both exiting separately ($|1, 1\rangle$),
- some probability of both exiting together.

23.3 Quantum Result

For indistinguishable photons, something remarkable happens:

The photons always leave together.

The output state becomes:

$$\frac{1}{\sqrt{2}} (|2, 0\rangle - |0, 2\rangle).$$

So:

$$P(1, 1) = 0.$$

Definition

Definition 23.1. This phenomenon is called the **Hong–Ou–Mandel effect**.

23.4 Physical Interpretation

The $|1, 1\rangle$ outcome can occur via two paths:

- both photons transmit,
- both photons reflect.

For indistinguishable photons, these amplitudes cancel exactly.

 This cancellation is an example of destructive interference between quantum amplitudes.

23.5 Exercises

Exercise

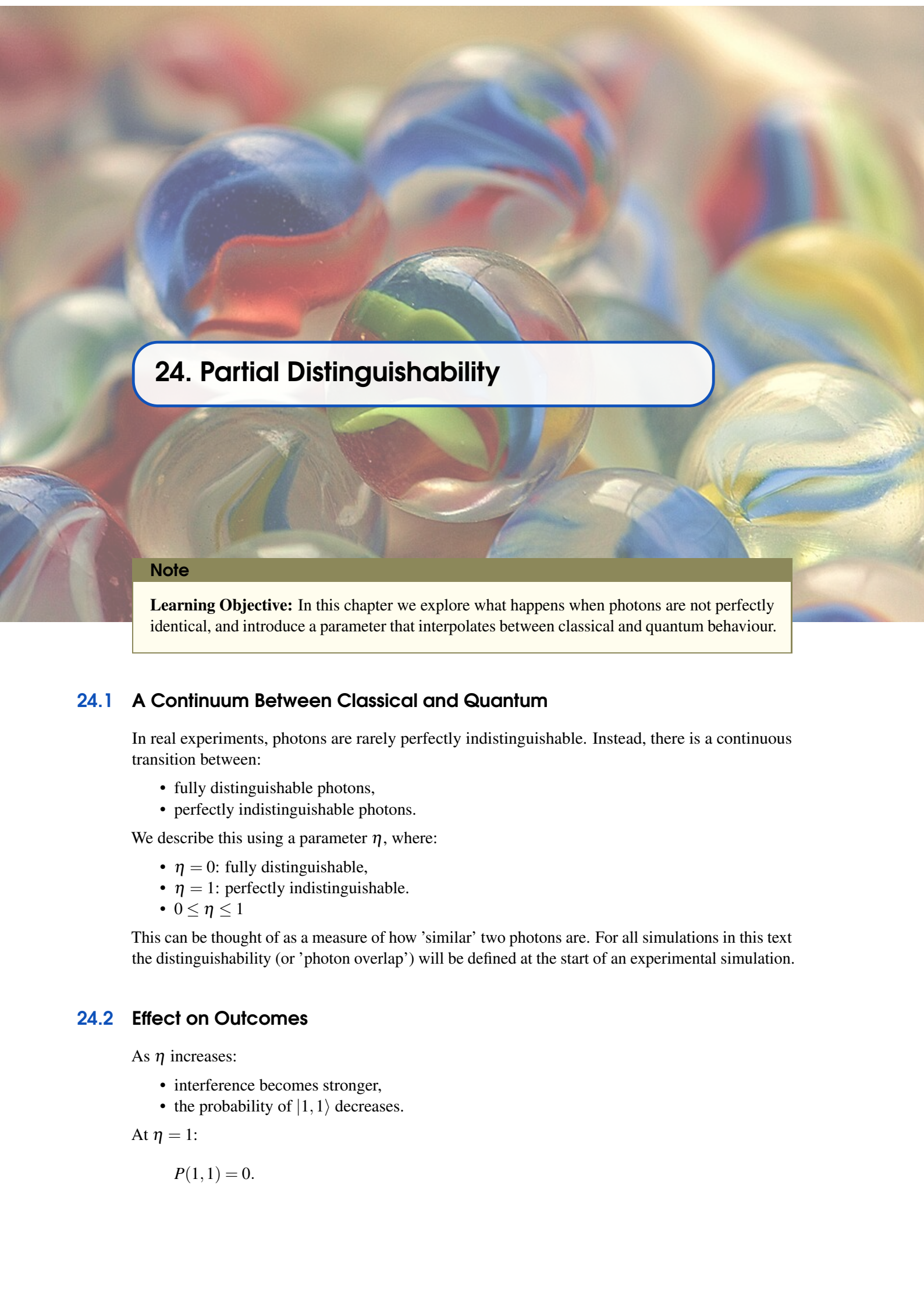
Exercise 23.1. Explain why the $|1, 1\rangle$ outcome disappears for indistinguishable photons.

Exercise

Exercise 23.2. Compare this result with what you would expect classically.

Summary

The Hong–Ou–Mandel effect shows that indistinguishable photons interfere in such a way that certain outcomes become impossible.

A close-up, slightly blurred image of several colorful marbles in shades of blue, red, green, and yellow, creating a vibrant, abstract background.

24. Partial Distinguishability

Note

Learning Objective: In this chapter we explore what happens when photons are not perfectly identical, and introduce a parameter that interpolates between classical and quantum behaviour.

24.1 A Continuum Between Classical and Quantum

In real experiments, photons are rarely perfectly indistinguishable. Instead, there is a continuous transition between:

- fully distinguishable photons,
- perfectly indistinguishable photons.

We describe this using a parameter η , where:

- $\eta = 0$: fully distinguishable,
- $\eta = 1$: perfectly indistinguishable.
- $0 \leq \eta \leq 1$

This can be thought of as a measure of how 'similar' two photons are. For all simulations in this text the distinguishability (or 'photon overlap') will be defined at the start of an experimental simulation.

24.2 Effect on Outcomes

As η increases:

- interference becomes stronger,
- the probability of $|1, 1\rangle$ decreases.

At $\eta = 1$:

$$P(1, 1) = 0.$$

At $\eta = 0$:

$$P(1,1) = \frac{1}{2}.$$

24.3 Visibility

The depth of the dip in $P(1,1)$ is called the **visibility**.

Definition

Definition 24.1. Visibility measures how strongly interference suppresses certain outcomes.

24.4 Exercises

Exercise

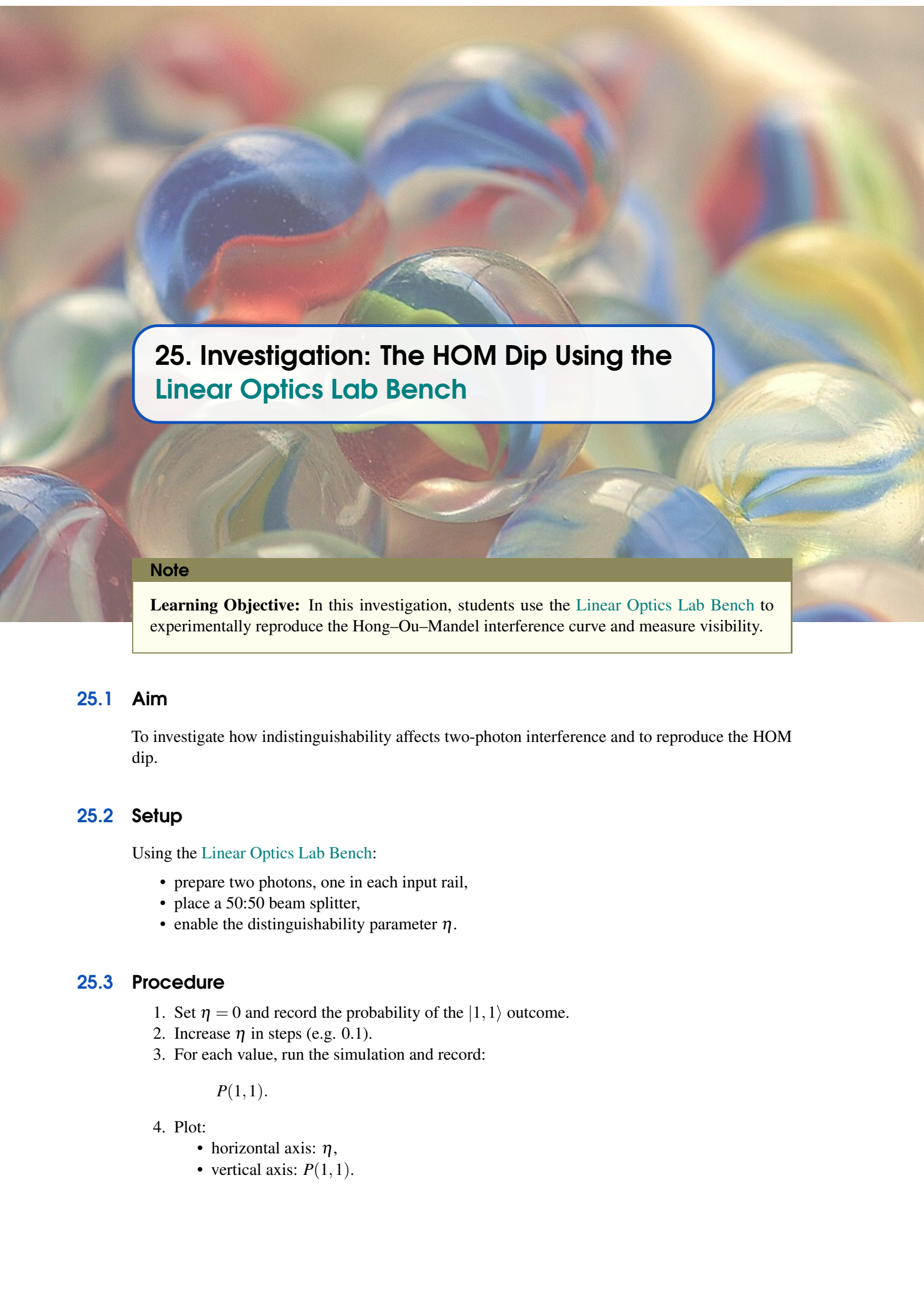
Exercise 24.1. Explain what η represents physically.

Exercise

Exercise 24.2. What value of $P(1,1)$ would you expect for completely distinguishable photons?

Summary

Partial indistinguishability leads to partial interference. The transition between classical and quantum behaviour can be controlled experimentally.



25. Investigation: The HOM Dip Using the Linear Optics Lab Bench

Note

Learning Objective: In this investigation, students use the [Linear Optics Lab Bench](#) to experimentally reproduce the Hong–Ou–Mandel interference curve and measure visibility.

25.1 Aim

To investigate how indistinguishability affects two-photon interference and to reproduce the HOM dip.

25.2 Setup

Using the [Linear Optics Lab Bench](#):

- prepare two photons, one in each input rail,
- place a 50:50 beam splitter,
- enable the distinguishability parameter η .

25.3 Procedure

1. Set $\eta = 0$ and record the probability of the $|1, 1\rangle$ outcome.
2. Increase η in steps (e.g. 0.1).
3. For each value, run the simulation and record:

$$P(1, 1).$$

4. Plot:
 - horizontal axis: η ,
 - vertical axis: $P(1, 1)$.

25.4 Expected Result

You should observe:

- a maximum at $\eta = 0$,
- a dip to zero at $\eta = 1$.

This is known as the **HOM dip**.

25.5 Analysis Questions

Exercise

Exercise 25.1. Explain why $P(1,1)$ is largest when the photons are distinguishable.

Exercise

Exercise 25.2. Explain why $P(1,1)$ goes to zero for indistinguishable photons.

Exercise

Exercise 25.3. Estimate the visibility of your curve.

Exercise

Exercise 25.4. Discuss possible experimental reasons why a real system might not reach zero at $\eta = 1$.

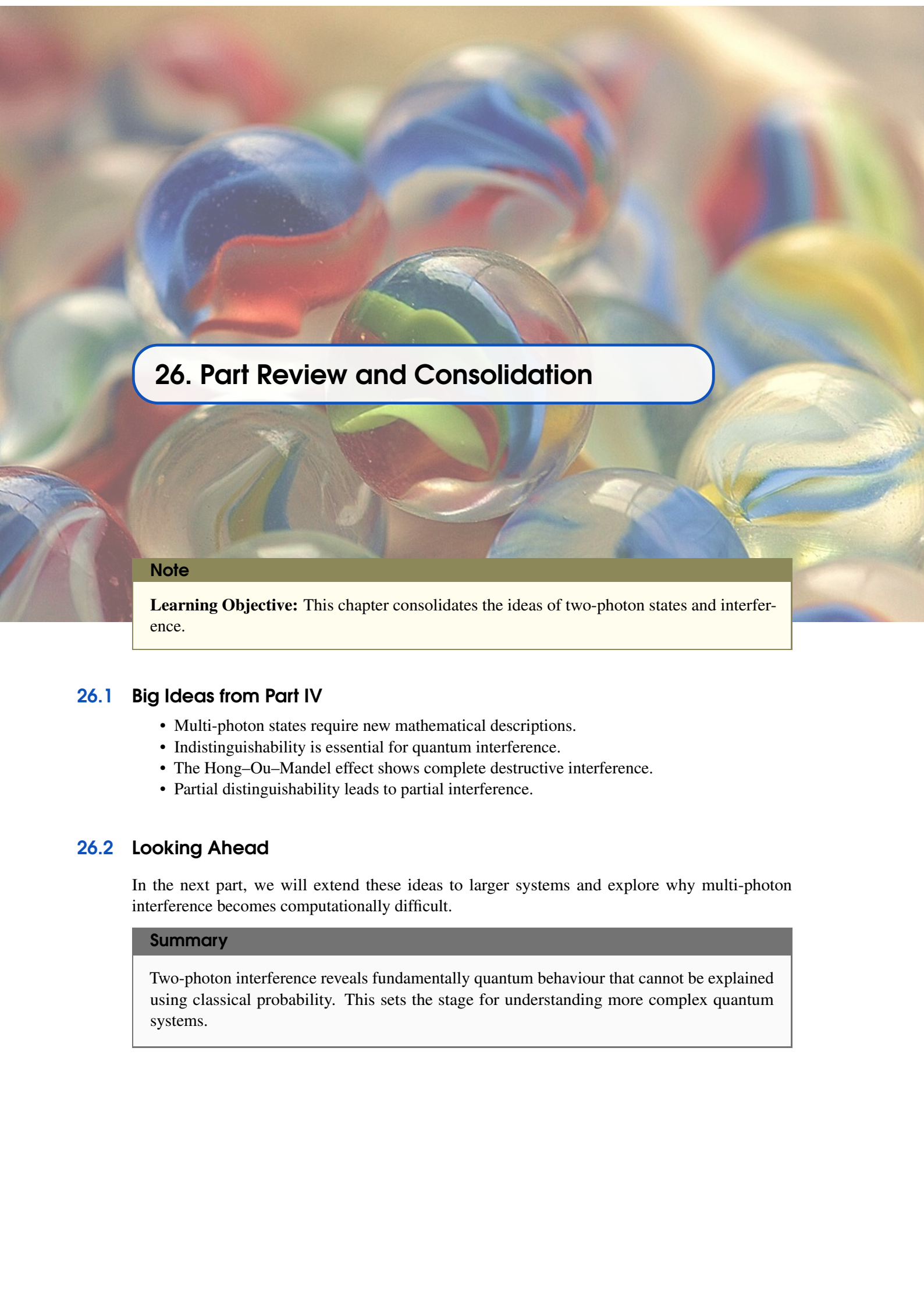
25.6 Extension

Investigation

Repeat the experiment using a finite number of shots. Compare the noisy results with the ideal curve. How does increasing the number of shots affect the smoothness of the curve?

Summary

The HOM dip is a direct demonstration of quantum interference between indistinguishable particles. By varying indistinguishability, we can continuously move between classical and quantum behaviour.

A close-up, shallow depth-of-field photograph of numerous colorful marbles. The marbles are in various colors including blue, red, green, yellow, and white, with some showing internal patterns or swirls. They are scattered across the frame, with some in sharp focus and others blurred in the background.

26. Part Review and Consolidation

Note

Learning Objective: This chapter consolidates the ideas of two-photon states and interference.

26.1 Big Ideas from Part IV

- Multi-photon states require new mathematical descriptions.
- Indistinguishability is essential for quantum interference.
- The Hong–Ou–Mandel effect shows complete destructive interference.
- Partial distinguishability leads to partial interference.

26.2 Looking Ahead

In the next part, we will extend these ideas to larger systems and explore why multi-photon interference becomes computationally difficult.

Summary

Two-photon interference reveals fundamentally quantum behaviour that cannot be explained using classical probability. This sets the stage for understanding more complex quantum systems.

Many Photons, Many Modes, and Why It Gets Hard

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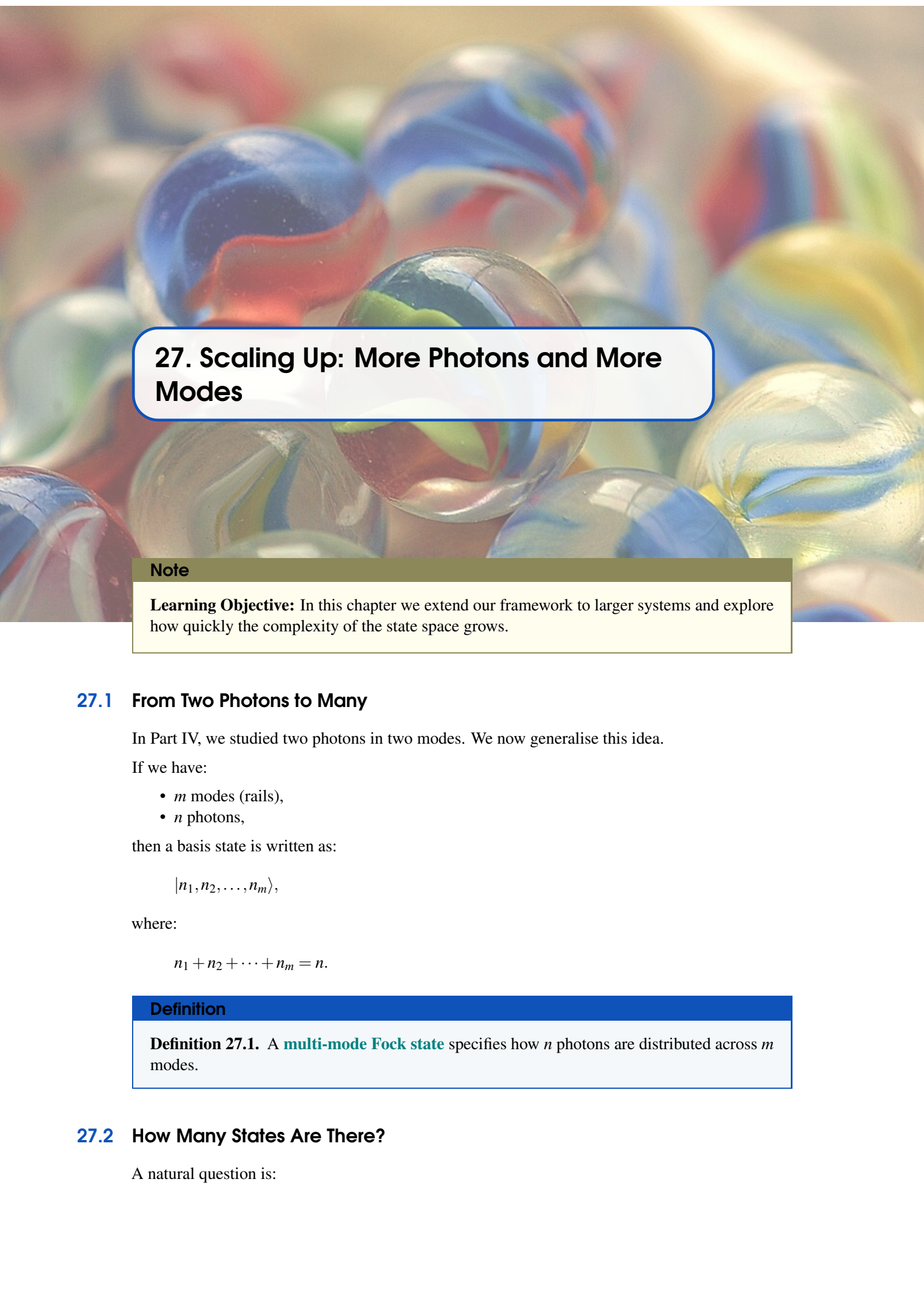
- 29.1 Aim
- 29.2 The Optical Circuit
- 29.3 Step 1: Construct the Unitary
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- 31.3 Looking Ahead



27. Scaling Up: More Photons and More Modes

Note

Learning Objective: In this chapter we extend our framework to larger systems and explore how quickly the complexity of the state space grows.

27.1 From Two Photons to Many

In Part IV, we studied two photons in two modes. We now generalise this idea.

If we have:

- m modes (rails),
- n photons,

then a basis state is written as:

$$|n_1, n_2, \dots, n_m\rangle,$$

where:

$$n_1 + n_2 + \dots + n_m = n.$$

Definition

Definition 27.1. A **multi-mode Fock state** specifies how n photons are distributed across m modes.

27.2 How Many States Are There?

A natural question is:

How many different basis states are possible?

The answer grows very quickly.

For n photons in m modes, the number of basis states is:

$$\binom{n+m-1}{n}.$$

Worked Example

For 3 photons in 3 modes:

$$\binom{5}{3} = \frac{5!}{(5-3)!3!} = 10.$$

R Even for small systems, the number of possible states becomes large. This is the first sign that multi-photon systems are complex.

27.3 State Vectors Become Large

A general state is now:

$$|\psi\rangle = \sum_{\text{all configurations}} c_i |n_1, n_2, \dots, n_m\rangle.$$

So instead of 2 components (single photon, 2 modes), we may have:

- 10 components,
- 20 components,
- or far more.

Note

The size of the state vector grows rapidly as we increase the number of photons and modes.

27.4 Exercises

Exercise

Exercise 27.1. List all Fock states for 3 photons in 2 modes.

Exercise

Exercise 27.2. Calculate the number of basis states for 4 photons in 3 modes.

Summary

As we increase the number of photons and modes, the size of the state space grows rapidly, making the system more complex to describe and compute.

Note

Learning Objective: In this chapter we extend the idea of interference to many photons and explain why multiple paths must be considered simultaneously.

27.5 Multiple Ways to Reach an Outcome

In a multi-photon system, a given output configuration can arise in many different ways.

For example, suppose:

- 3 photons enter a network,
- we observe a particular output distribution.

There may be many different ways the photons could have taken paths through the network to produce that outcome.

27.6 Adding Amplitudes Over Paths

For indistinguishable photons:

- we cannot distinguish which photon took which path,
- so we must add amplitudes for all possible assignments.

Definition

Definition 27.2. The total amplitude for an outcome is the sum of amplitudes over all indistinguishable paths leading to that outcome.

27.7 Contrast with Classical Case

If photons were distinguishable:

- we would treat each path separately,
- probabilities would add.

For indistinguishable photons:

- amplitudes add,
- interference occurs between many paths.

 The number of contributing paths grows very rapidly with the number of photons.

27.8 Exercises

Exercise

Exercise 27.3. Explain why the number of possible paths increases as the number of photons increases.

Exercise

Exercise 27.4. Why does indistinguishability force us to add amplitudes rather than probabilities?

Summary

Multi-photon interference arises from summing amplitudes over many indistinguishable paths, leading to highly complex interference patterns.

Note

Learning Objective: In this chapter we explain why calculating multi-photon outcomes is computationally difficult, even though the underlying rules are simple.

27.9 The Core Problem

To predict the probability of an outcome, we must:

1. find all ways the photons could produce that outcome,
2. calculate the amplitude for each way,
3. add them together,
4. square the result.

This becomes extremely difficult as the number of photons increases.

Each possible assignment of photons to outputs corresponds to a different contribution to the total amplitude.

The number of such assignments grows very rapidly:

- 2 photons → a few paths,
- 3 photons → many paths,
- 6 photons → enormous number of paths.

27.10 Introducing the Permanent (Conceptually)

Mathematically, the sum over all these paths is captured by a quantity called the **permanent** of a matrix.

Definition

Definition 27.3. The **permanent** of a matrix is similar to the determinant, but without alternating signs.

For a matrix A ,

$$\text{Per}(A) = \sum_{\text{all permutations}} \prod A_{i,\sigma(i)}.$$

- R** Unlike determinants, permanents do not have cancellation between positive and negative terms. This makes them much harder to compute.

27.11 Why This Is Hard

Computing a permanent requires:

- summing over all permutations,
- which grows factorially.

So:

- 3 photons $\rightarrow$ 6 terms,
- 4 photons $\rightarrow$ 24 terms,
- 10 photons $\rightarrow$ millions of terms.

Note

There is no known efficient general method for computing permanents of large matrices.

27.12 Physical Meaning

In linear optics:

- each term corresponds to one way photons could propagate,
- the permanent sums all those possibilities.

So the difficulty of computing permanents reflects the complexity of multi-photon interference.

27.13 Exercises

Exercise

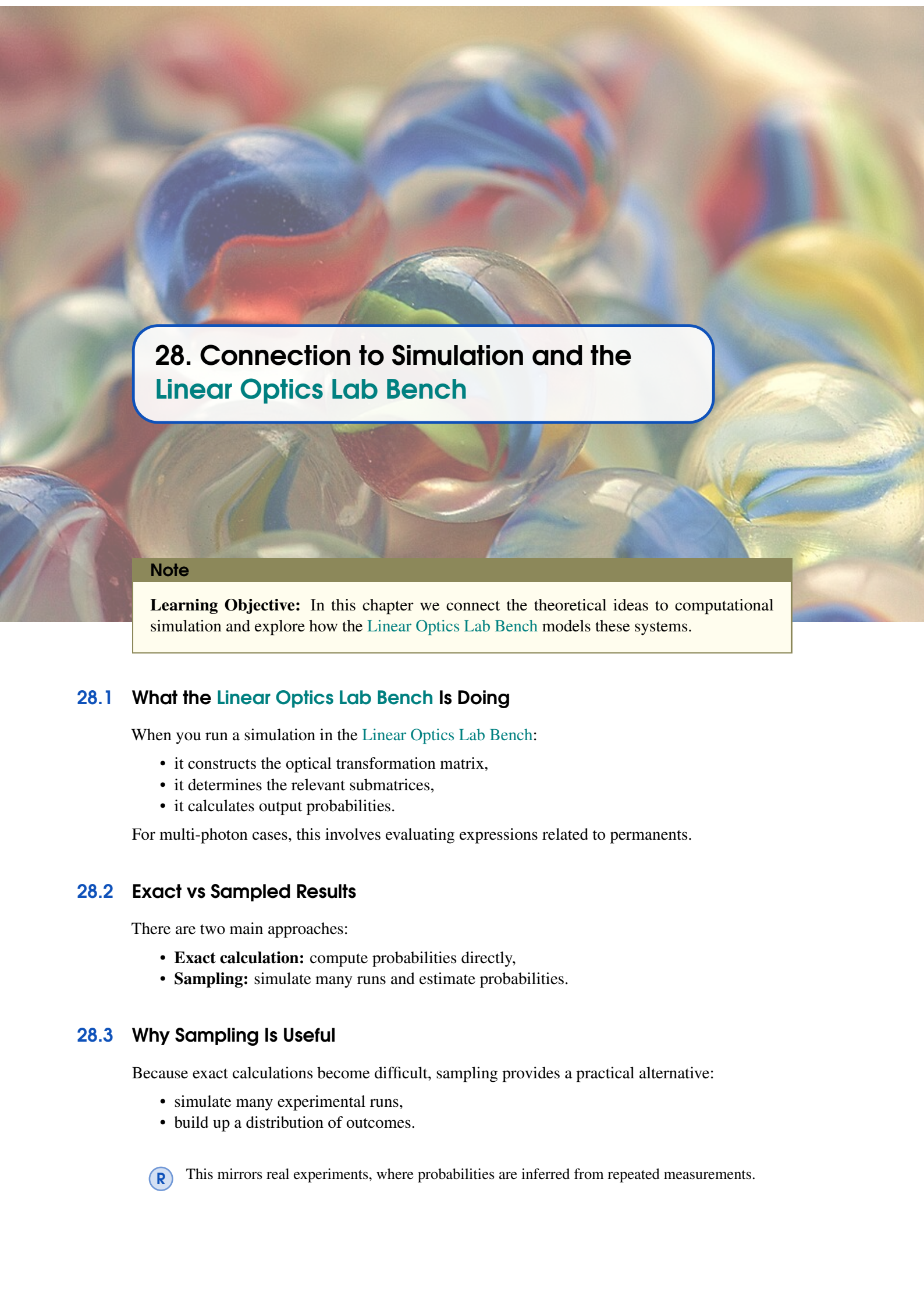
Exercise 27.5. Explain the difference between a determinant and a permanent.

Exercise

Exercise 27.6. Why does the number of permutations grow factorially?

Summary

The difficulty of multi-photon quantum systems arises because amplitudes must be summed over many permutations, leading to quantities known as matrix permanents, which are computationally hard to evaluate.



28. Connection to Simulation and the Linear Optics Lab Bench

Note

Learning Objective: In this chapter we connect the theoretical ideas to computational simulation and explore how the [Linear Optics Lab Bench](#) models these systems.

28.1 What the [Linear Optics Lab Bench](#) Is Doing

When you run a simulation in the [Linear Optics Lab Bench](#):

- it constructs the optical transformation matrix,
- it determines the relevant submatrices,
- it calculates output probabilities.

For multi-photon cases, this involves evaluating expressions related to permanents.

28.2 Exact vs Sampled Results

There are two main approaches:

- **Exact calculation:** compute probabilities directly,
- **Sampling:** simulate many runs and estimate probabilities.

28.3 Why Sampling Is Useful

Because exact calculations become difficult, sampling provides a practical alternative:

- simulate many experimental runs,
- build up a distribution of outcomes.

 This mirrors real experiments, where probabilities are inferred from repeated measurements.

28.4 Exercises

Exercise

Exercise 28.1. Explain the difference between exact probability calculation and sampling.

Exercise

Exercise 28.2. Why might sampling be preferred for large systems?

Summary

Simulation tools allow us to explore multi-photon systems even when exact calculations are difficult, by using sampling to approximate probability distributions.

Note

Learning Objective: In this investigation, students explore how multi-photon systems become complex and observe the growth of possible outcomes.

28.5 Aim

To explore how increasing the number of photons and modes affects the complexity of the system.

28.6 Procedure

1. Start with 2 photons in 2 modes.
2. Record the possible outputs and their probabilities.
3. Increase to 3 photons in 3 modes.
4. Observe:
 - number of outputs,
 - structure of the probability distribution.
5. Increase further if possible.

28.7 Analysis Questions

Exercise

Exercise 28.3. How does the number of possible outputs change as you increase photons and modes?

Exercise

Exercise 28.4. Does the distribution become more complex or more uniform?

Exercise

Exercise 28.5. Why do you think predicting the outcome becomes harder?

28.8 Extension**Investigation**

Compare distinguishable and indistinguishable cases for a multi-photon system. How does the distribution change?

Summary

As systems grow, both the number of possible outcomes and the complexity of interference increase rapidly. This is why multi-photon quantum systems are difficult to analyse.

Note

Learning Objective: In this chapter we develop a deep understanding of why multi-photon probabilities are calculated using permanents, using a complete worked example with a concrete optical transformation.

28.9 The Plan

We now answer the central question of this part:

How do we calculate the probability of a multi-photon outcome from a given optical circuit?

We will:

1. choose a concrete unitary matrix,
2. follow photons through all possible paths,
3. construct the submatrix,
4. compute the permanent,
5. compare with the physical interpretation.

28.10 A Concrete Optical Transformation

Consider a 3-mode optical system with unitary matrix:

$$U = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & e^{2\pi i/3} & e^{4\pi i/3} \\ 1 & e^{4\pi i/3} & e^{2\pi i/3} \end{pmatrix}.$$

This matrix represents a symmetric 3-mode mixing device.

 Each element U_{ij} is the amplitude for a photon to go from input mode j to output mode i .

28.11 The Input and Output States

Let the input state be:

$$|2, 1, 0\rangle$$

(two photons in mode 1, one photon in mode 2).

Let the desired output be:

$$|1, 2, 0\rangle.$$

28.12 Step 1: Think in Terms of Paths

To understand what is happening, we temporarily label the photons:

- Photon A and B start in mode 1,
- Photon C starts in mode 2.

We want:

- one photon in output mode 1,
- two photons in output mode 2.

Each possible way of assigning photons to outputs is a **path**.

28.13 Step 2: Why Permutations Appear

We must consider all ways of assigning the three input photons to the three output slots:

- which photon goes to output 1,
- which photons go to output 2.

There are $3! = 6$ such assignments.

Each assignment contributes a product of matrix elements.

28.14 Step 3: Constructing the Submatrix

Now we translate this into matrix language.

Input occupation

$$|2, 1, 0\rangle \Rightarrow \text{columns: } (1, 1, 2)$$

Output occupation

$$|1, 2, 0\rangle \Rightarrow \text{rows: } (1, 2, 2)$$

Submatrix

We repeat:

- column 1 twice,
- column 2 once,

- row 1 once,
- row 2 twice.

So:

$$U_{\text{sub}} = \frac{1}{\sqrt{3}} \begin{pmatrix} U_{1,1} & U_{1,1} & U_{1,2} \\ U_{2,1} & U_{2,1} & U_{2,2} \\ U_{2,1} & U_{2,1} & U_{2,2} \end{pmatrix}.$$

Substitute values:

$$= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & \omega \\ 1 & 1 & \omega \end{pmatrix},$$

where $\omega = e^{2\pi i/3}$.

28.15 Step 4: Computing the Permanent

Recall that

$$U_{\text{sub}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & \omega \\ 1 & 1 & \omega \end{pmatrix}, \quad \text{where } \omega = e^{2\pi i/3}.$$

To make the arithmetic clearer, write

$$U_{\text{sub}} = \frac{1}{\sqrt{3}} M, \quad M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & \omega \\ 1 & 1 & \omega \end{pmatrix}.$$

Since this is a 3×3 matrix,

$$\text{Per}(U_{\text{sub}}) = \left(\frac{1}{\sqrt{3}} \right)^3 \text{Per}(M) = \frac{1}{3\sqrt{3}} \text{Per}(M).$$

We now compute the permanent of M .

For a 3×3 matrix,

$$\text{Per}(M) = M_{11}M_{22}M_{33} + M_{11}M_{23}M_{32} + M_{12}M_{21}M_{33} + M_{12}M_{23}M_{31} + M_{13}M_{21}M_{32} + M_{13}M_{22}M_{31}.$$

Substituting the entries of M , we get:

$$\text{Per}(M) = (1)(1)(\omega) + (1)(\omega)(1) + (1)(1)(\omega) + (1)(\omega)(1) + (1)(1)(1) + (1)(1)(1).$$

So

$$\text{Per}(M) = \omega + \omega + \omega + \omega + 1 + 1 = 4\omega + 2.$$

Now use

$$\omega = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}.$$

Then

$$4\omega + 2 = 4 \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) + 2 = -2 + 2i\sqrt{3} + 2 = 2i\sqrt{3}.$$

Therefore,

$$\text{Per}(M) = 2i\sqrt{3},$$

and so

$$\text{Per}(U_{\text{sub}}) = \frac{1}{3\sqrt{3}} \cdot 2i\sqrt{3} = \frac{2i}{3}.$$

28.16 Step 5: Calculating the Probability

The transition probability is

$$P = \frac{|\text{Per}(U_{\text{sub}})|^2}{2! \cdot 1! \cdot 1! \cdot 2!}.$$

From the previous section,

$$\text{Per}(U_{\text{sub}}) = \frac{2i}{3},$$

so

$$|\text{Per}(U_{\text{sub}})|^2 = \left| \frac{2i}{3} \right|^2 = \frac{4}{9}.$$

The factorial factor is

$$2! \cdot 1! \cdot 1! \cdot 2! = 2 \cdot 1 \cdot 1 \cdot 2 = 4.$$

Therefore,

$$P = \frac{4/9}{4} = \frac{1}{9}.$$

So the probability of the transition

$$|2, 1, 0\rangle \longrightarrow |1, 2, 0\rangle$$

for this optical transformation is

$$\boxed{P = \frac{1}{9}}.$$

28.17 Step 6: What This Number Means Physically

This result means that if we repeatedly prepare the input state

$$|2, 1, 0\rangle$$

and send it through this 3-mode optical device, then the output pattern

$$|1, 2, 0\rangle$$

should occur with probability

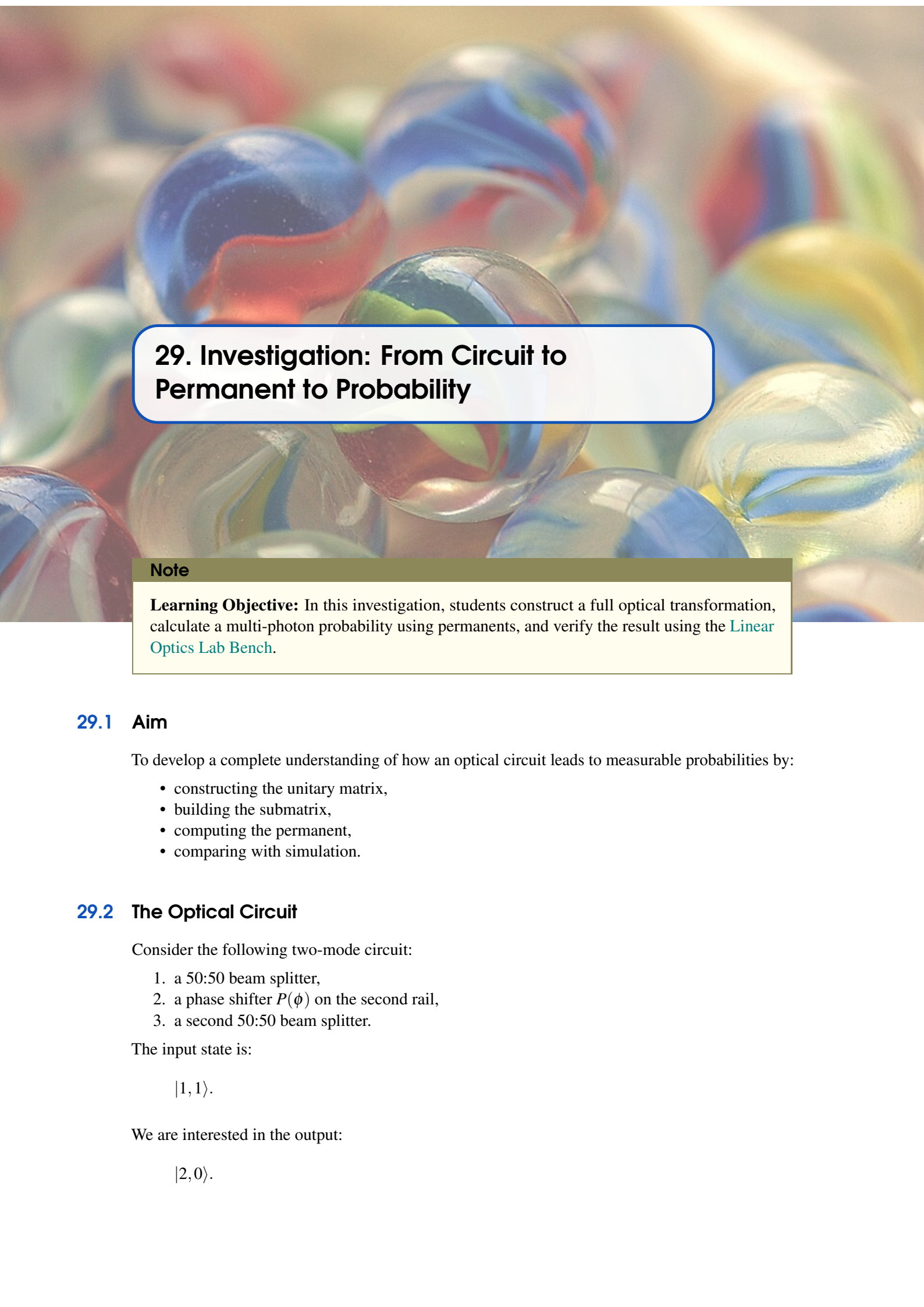
$$\frac{1}{9}.$$

So in a large number of runs, we would expect this output to appear about one ninth of the time.

This is an important moment in the course, because we have now completed the full chain:

physical circuit $\longrightarrow$ unitary matrix $\longrightarrow$ submatrix $\longrightarrow$ permanent $\longrightarrow$ probability

The permanent is therefore not just a formal algebraic object. It is the quantity that turns the optical transformation matrix into an experimentally testable prediction.



29. Investigation: From Circuit to Permanent to Probability

Note

Learning Objective: In this investigation, students construct a full optical transformation, calculate a multi-photon probability using permanents, and verify the result using the [Linear Optics Lab Bench](#).

29.1 Aim

To develop a complete understanding of how an optical circuit leads to measurable probabilities by:

- constructing the unitary matrix,
- building the submatrix,
- computing the permanent,
- comparing with simulation.

29.2 The Optical Circuit

Consider the following two-mode circuit:

1. a 50:50 beam splitter,
2. a phase shifter $P(\phi)$ on the second rail,
3. a second 50:50 beam splitter.

The input state is:

$$|1, 1\rangle.$$

We are interested in the output:

$$|2, 0\rangle.$$

29.3 Step 1: Construct the Unitary

The matrices are:

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}.$$

Exercise

Exercise 29.1. Show that the overall unitary is:

$$U = BP(\phi)B.$$

Exercise

Exercise 29.2. Compute U explicitly and show that:

$$U = \frac{1}{2} \begin{pmatrix} 1 + e^{i\phi} & 1 - e^{i\phi} \\ 1 - e^{i\phi} & 1 + e^{i\phi} \end{pmatrix}.$$

29.4 Step 2: Construct the Submatrix

The input is:

$$|1, 1\rangle \Rightarrow \text{columns } (1, 2).$$

The output is:

$$|2, 0\rangle \Rightarrow \text{rows } (1, 1).$$

Exercise

Exercise 29.3. Construct the submatrix U_{sub} by:

- repeating column 1 once and column 2 once,
- repeating row 1 twice.

Show that:

$$U_{\text{sub}} = \begin{pmatrix} U_{11} & U_{12} \\ U_{11} & U_{12} \end{pmatrix}.$$

29.5 Step 3: Compute the Permanent

Exercise

Exercise 29.4. Compute:

$$\text{Per}(U_{\text{sub}}).$$

(Hint: for a 2×2 matrix,

$$\text{Per} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad + bc.)$$

Worked Example

Using the expression for U ,

$$U_{11} = \frac{1 + e^{i\phi}}{2}, \quad U_{12} = \frac{1 - e^{i\phi}}{2}.$$

Then:

$$\text{Per}(U_{\text{sub}}) = U_{11}U_{12} + U_{12}U_{11} = 2U_{11}U_{12}.$$

So:

$$\text{Per}(U_{\text{sub}}) = 2 \cdot \frac{1 + e^{i\phi}}{2} \cdot \frac{1 - e^{i\phi}}{2} = \frac{1 - e^{2i\phi}}{2}.$$

29.6 Step 4: Calculate the Probability

The probability is:

$$P(2, 0) = \frac{|\text{Per}(U_{\text{sub}})|^2}{2!}.$$

Exercise

Exercise 29.5. Show that:

$$|\text{Per}(U_{\text{sub}})|^2 = \frac{|1 - e^{2i\phi}|^2}{4}.$$

Exercise

Exercise 29.6. Use the identity

$$|1 - e^{i\theta}|^2 = 2(1 - \cos \theta)$$

to show that:

$$P(2, 0) = \frac{1 - \cos(2\phi)}{4}.$$

29.7 Step 5: Physical Interpretation

Exercise

Exercise 29.7. Explain why the probability depends on 2ϕ rather than ϕ .

Exercise

Exercise 29.8. What values of ϕ maximise and minimise $P(2,0)$?

29.8 Step 6: Linear Optics Lab Bench Verification

Investigation

Use the [Linear Optics Lab Bench](#) to reproduce this result:

1. Construct the circuit:
 - beam splitter,
 - phase shifter on one rail,
 - second beam splitter.
2. Set the input state to two photons, one in each rail.
3. Vary the phase ϕ .
4. Record the probability of the output $|2,0\rangle$.
5. Plot:
 - horizontal axis: ϕ ,
 - vertical axis: $P(2,0)$.

Compare your graph with the theoretical prediction:

$$P(2,0) = \frac{1 - \cos(2\phi)}{4}.$$

29.9 Extension

Investigation

Repeat the experiment for the output $|1,1\rangle$. How does this compare with the Hong–Ou–Mandel effect studied earlier?

29.10 Final Reflection

Exercise

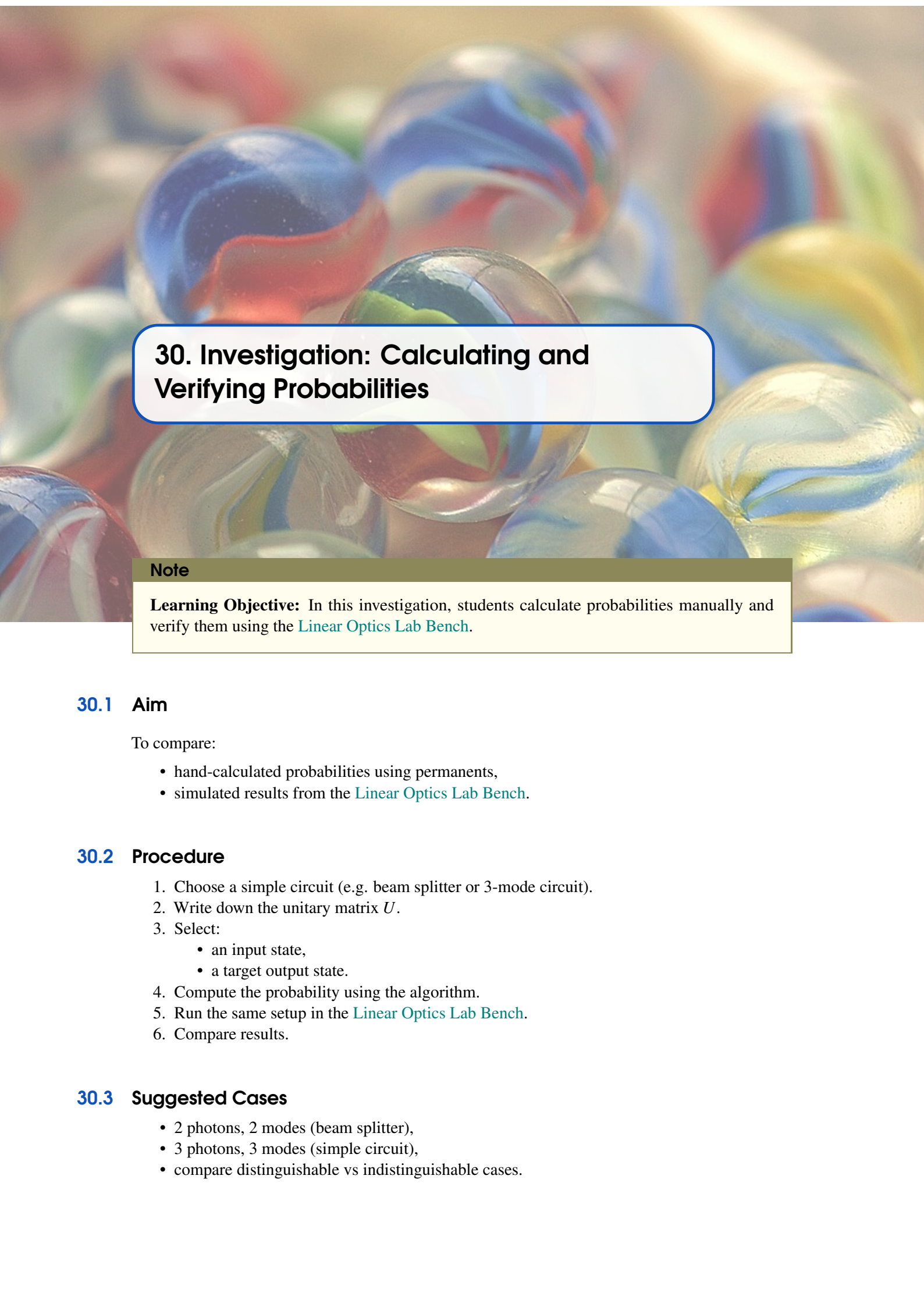
Exercise 29.9. Describe the full chain of reasoning from:

- optical circuit,
- to unitary matrix,
- to submatrix,
- to permanent,

- to probability.

Summary

This investigation shows that multi-photon probabilities can be calculated from first principles and verified experimentally. The permanent is the key mathematical object that connects the optical transformation to observable outcomes.



30. Investigation: Calculating and Verifying Probabilities

Note

Learning Objective: In this investigation, students calculate probabilities manually and verify them using the [Linear Optics Lab Bench](#).

30.1 Aim

To compare:

- hand-calculated probabilities using permanents,
- simulated results from the [Linear Optics Lab Bench](#).

30.2 Procedure

1. Choose a simple circuit (e.g. beam splitter or 3-mode circuit).
2. Write down the unitary matrix U .
3. Select:
 - an input state,
 - a target output state.
4. Compute the probability using the algorithm.
5. Run the same setup in the [Linear Optics Lab Bench](#).
6. Compare results.

30.3 Suggested Cases

- 2 photons, 2 modes (beam splitter),
- 3 photons, 3 modes (simple circuit),
- compare distinguishable vs indistinguishable cases.

30.4 Analysis Questions

Exercise

Exercise 30.1. How closely do the simulated results match your calculated values?

Exercise

Exercise 30.2. What sources of discrepancy might appear in finite-shot simulations?

Exercise

Exercise 30.3. How does the difficulty of calculation change as the system size increases?

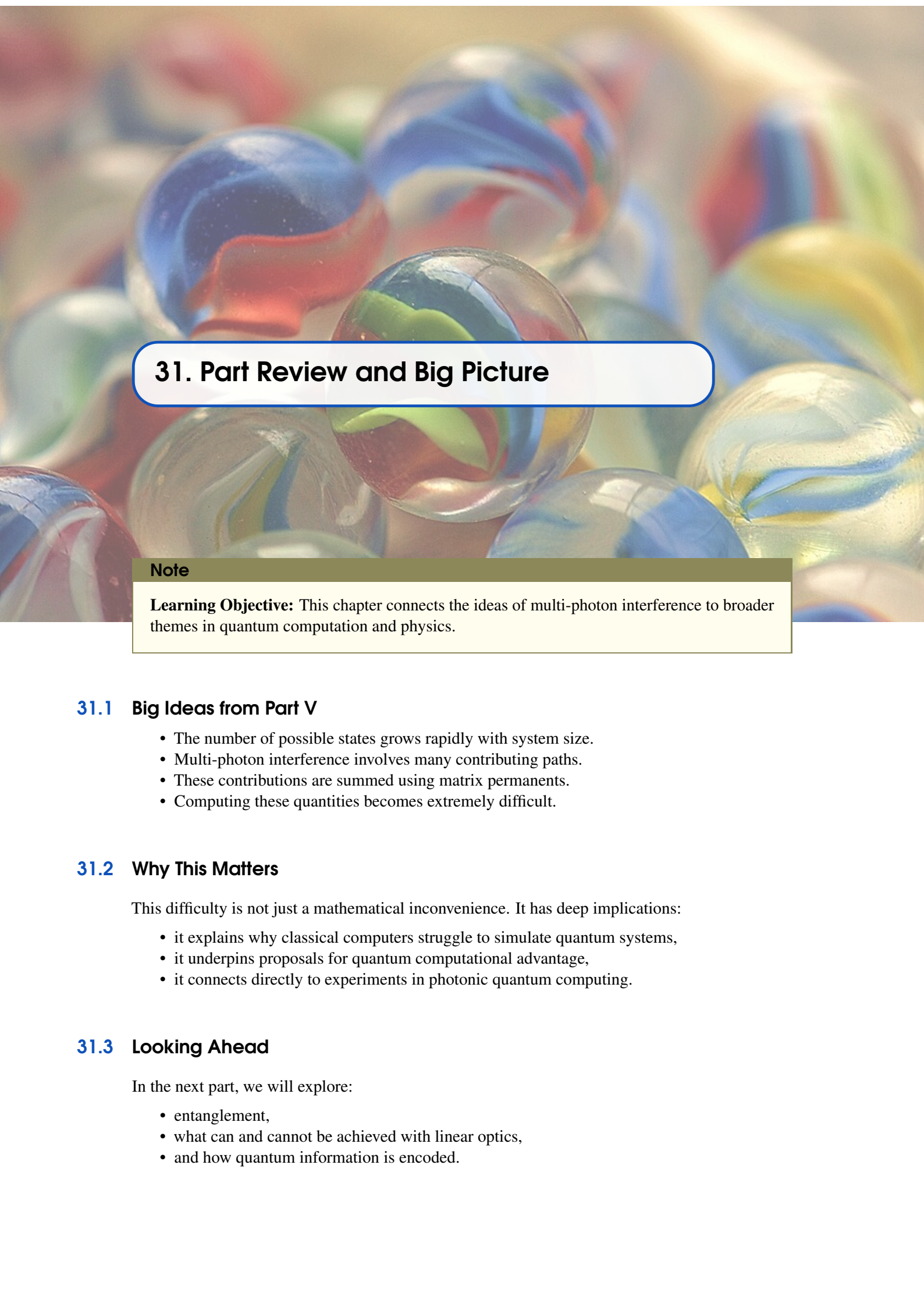
30.5 Extension

Investigation

Attempt a 3-photon calculation. How many terms are required in the permanent? How practical is it to do by hand?

Summary

This investigation connects theory and experiment. It shows that the abstract mathematics of permanents directly predicts observable outcomes in quantum optical systems.

A close-up, slightly blurred image of many colorful marbles in shades of blue, red, green, and yellow, creating a vibrant, textured background.

31. Part Review and Big Picture

Note

Learning Objective: This chapter connects the ideas of multi-photon interference to broader themes in quantum computation and physics.

31.1 Big Ideas from Part V

- The number of possible states grows rapidly with system size.
- Multi-photon interference involves many contributing paths.
- These contributions are summed using matrix permanents.
- Computing these quantities becomes extremely difficult.

31.2 Why This Matters

This difficulty is not just a mathematical inconvenience. It has deep implications:

- it explains why classical computers struggle to simulate quantum systems,
- it underpins proposals for quantum computational advantage,
- it connects directly to experiments in photonic quantum computing.

31.3 Looking Ahead

In the next part, we will explore:

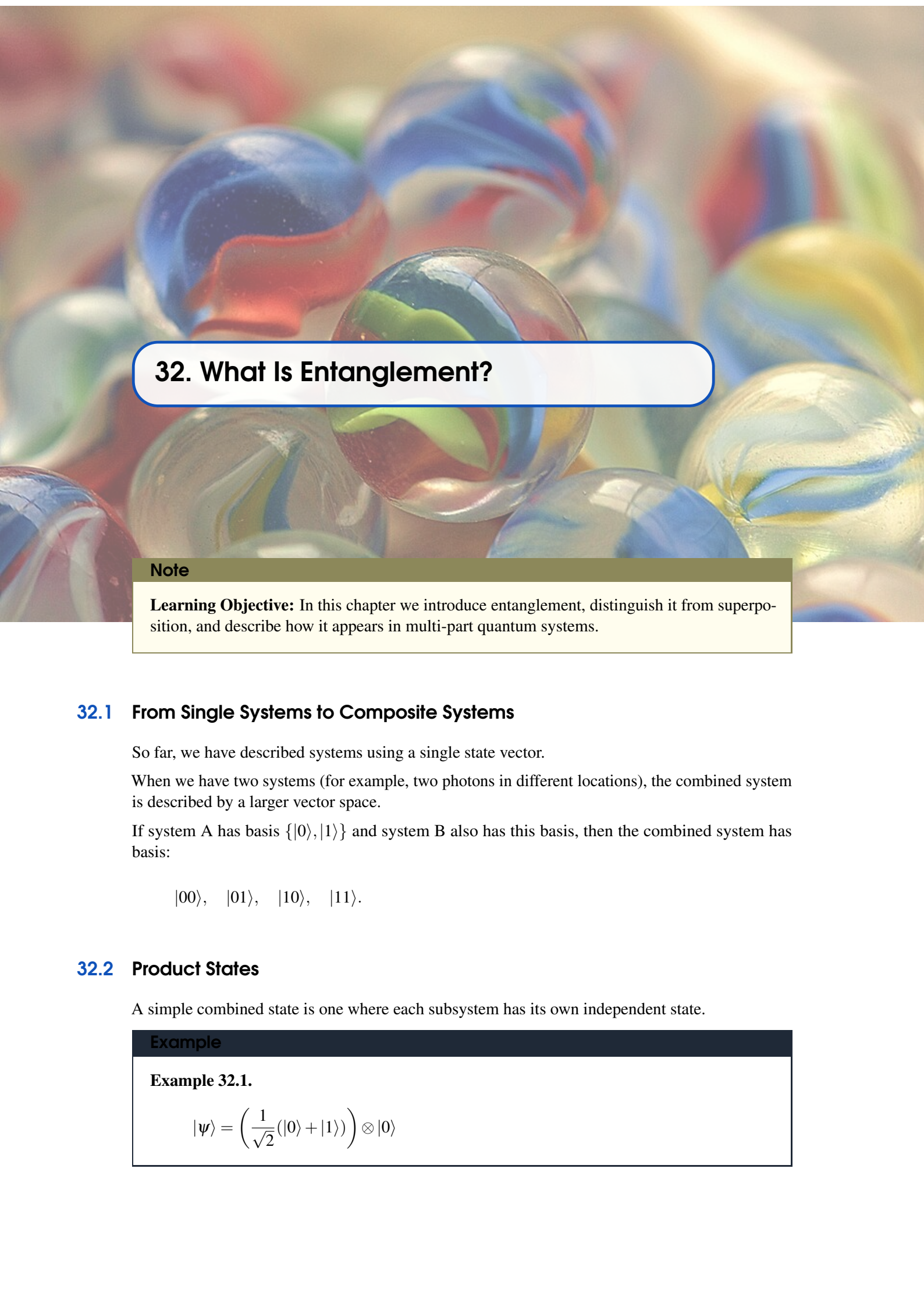
- entanglement,
- what can and cannot be achieved with linear optics,
- and how quantum information is encoded.

Summary

Multi-photon quantum systems reveal a fundamental increase in complexity. This is not just a technical challenge, but a defining feature of quantum mechanics that leads to new computational possibilities.

Entanglement, Encoding, and the Limits of Linear Optics

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32. What Is Entanglement?

Note

Learning Objective: In this chapter we introduce entanglement, distinguish it from superposition, and describe how it appears in multi-part quantum systems.

32.1 From Single Systems to Composite Systems

So far, we have described systems using a single state vector.

When we have two systems (for example, two photons in different locations), the combined system is described by a larger vector space.

If system A has basis $\{|0\rangle, |1\rangle\}$ and system B also has this basis, then the combined system has basis:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

32.2 Product States

A simple combined state is one where each subsystem has its own independent state.

Example

Example 32.1.

$$|\psi\rangle = \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) \otimes |0\rangle$$

Definition

Definition 32.2. A **product state** is a state that can be written as a tensor product of two individual states.

32.3 Entangled States

Now consider:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

This state cannot be written as a product of two individual states.

Definition

Definition 32.3. An **entangled state** is a state that cannot be written as a product of states of its subsystems.

32.4 Physical Meaning

In this state:

- if we measure one system and get 0, the other is also 0,
- if we measure one system and get 1, the other is also 1.

The outcomes are correlated in a way that cannot be explained by independent states.

 Entanglement is a property of the *joint system*, not of the individual parts.

32.5 Exercises**Exercise**

Exercise 32.1. Show that the state

$$|00\rangle$$

is not entangled.

Exercise

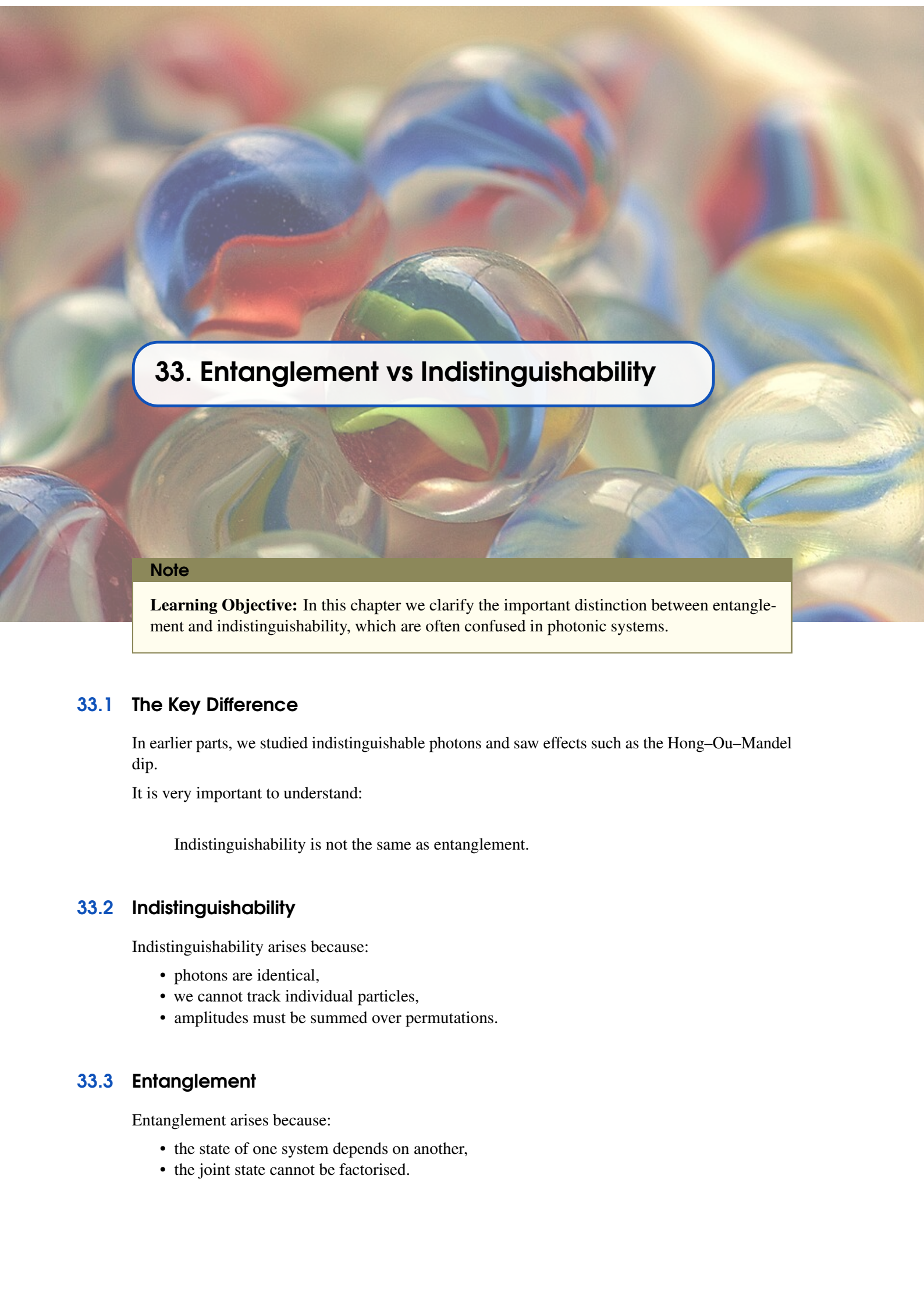
Exercise 32.2. Explain why

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

cannot be written as a product state.

Summary

Entanglement occurs when a composite quantum system cannot be separated into independent parts. It is one of the most distinctive features of quantum mechanics.

A close-up, shallow depth-of-field photograph of numerous colorful marbles. The marbles are in various colors including blue, red, green, yellow, and white, with some showing internal patterns or swirls. They are scattered across the frame, with some in sharp focus and others blurred in the background.

33. Entanglement vs Indistinguishability

Note

Learning Objective: In this chapter we clarify the important distinction between entanglement and indistinguishability, which are often confused in photonic systems.

33.1 The Key Difference

In earlier parts, we studied indistinguishable photons and saw effects such as the Hong–Ou–Mandel dip.

It is very important to understand:

Indistinguishability is not the same as entanglement.

33.2 Indistinguishability

Indistinguishability arises because:

- photons are identical,
- we cannot track individual particles,
- amplitudes must be summed over permutations.

33.3 Entanglement

Entanglement arises because:

- the state of one system depends on another,
- the joint state cannot be factorised.

33.4 Example Comparison

Indistinguishable photons:

$$|1, 1\rangle \rightarrow \frac{1}{\sqrt{2}}(|2, 0\rangle - |0, 2\rangle)$$

Entangled qubits:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

These look similar mathematically, but represent different physical situations.

 Indistinguishability is about particle identity. Entanglement is about correlations between subsystems.

33.5 Exercises

Exercise

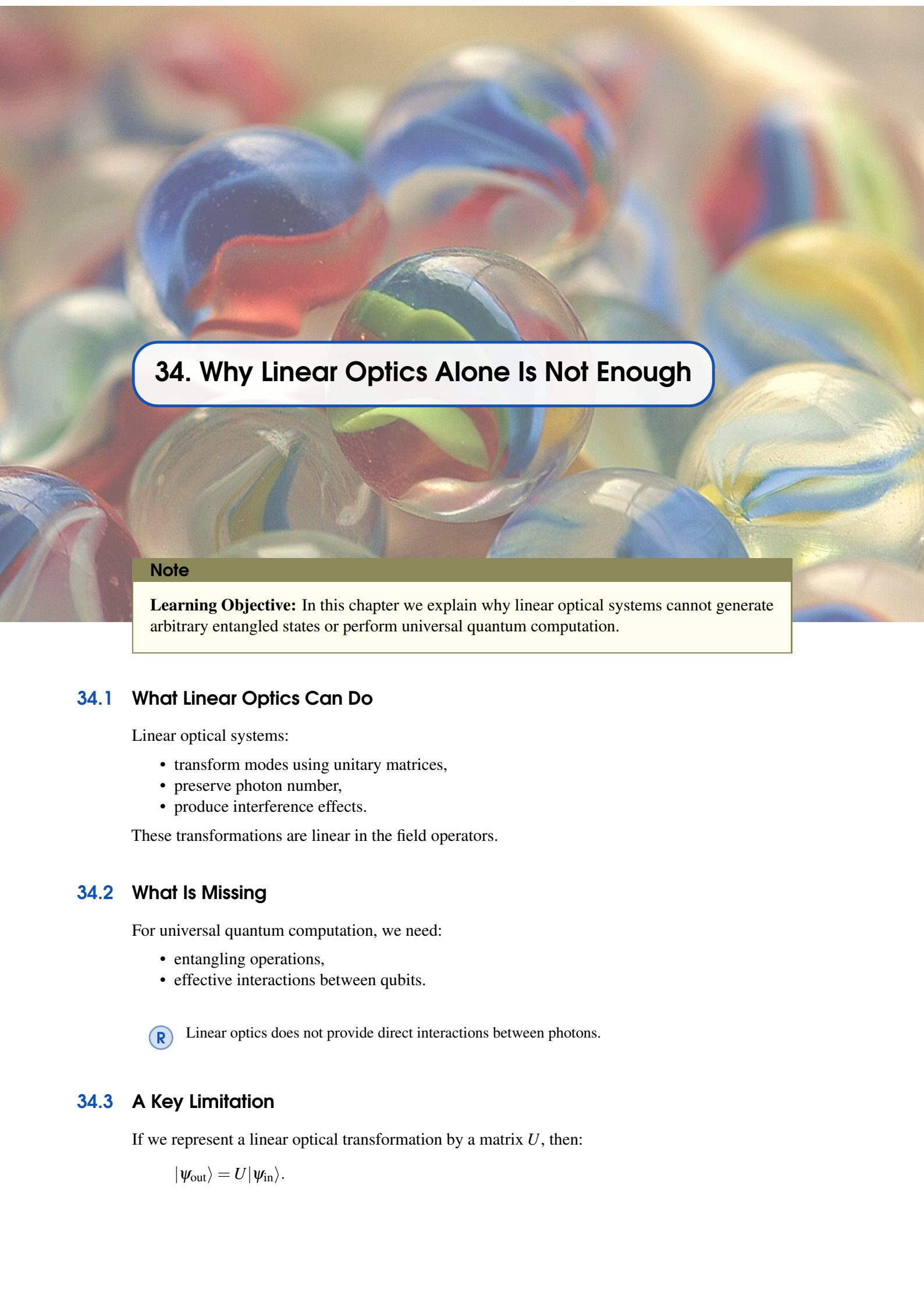
Exercise 33.1. Explain why the Hong–Ou–Mandel effect does not require entanglement.

Exercise

Exercise 33.2. Give an example of a system that is entangled but does not involve indistinguishable particles.

Summary

Indistinguishability and entanglement are distinct concepts. Both lead to non-classical behaviour, but they arise from different physical principles.



34. Why Linear Optics Alone Is Not Enough

Note

Learning Objective: In this chapter we explain why linear optical systems cannot generate arbitrary entangled states or perform universal quantum computation.

34.1 What Linear Optics Can Do

Linear optical systems:

- transform modes using unitary matrices,
- preserve photon number,
- produce interference effects.

These transformations are linear in the field operators.

34.2 What Is Missing

For universal quantum computation, we need:

- entangling operations,
- effective interactions between qubits.

 Linear optics does not provide direct interactions between photons.

34.3 A Key Limitation

If we represent a linear optical transformation by a matrix U , then:

$$|\psi_{\text{out}}\rangle = U|\psi_{\text{in}}\rangle.$$

This transformation:

- mixes amplitudes,
- but does not introduce the type of nonlinearity needed for general entangling gates.

34.4 Intuition

Photons:

- do not interact strongly with each other,
- pass through each other without direct coupling.

So:

there is nothing in the system that forces one photon to change its behaviour based on another.

34.5 Exercises

Exercise

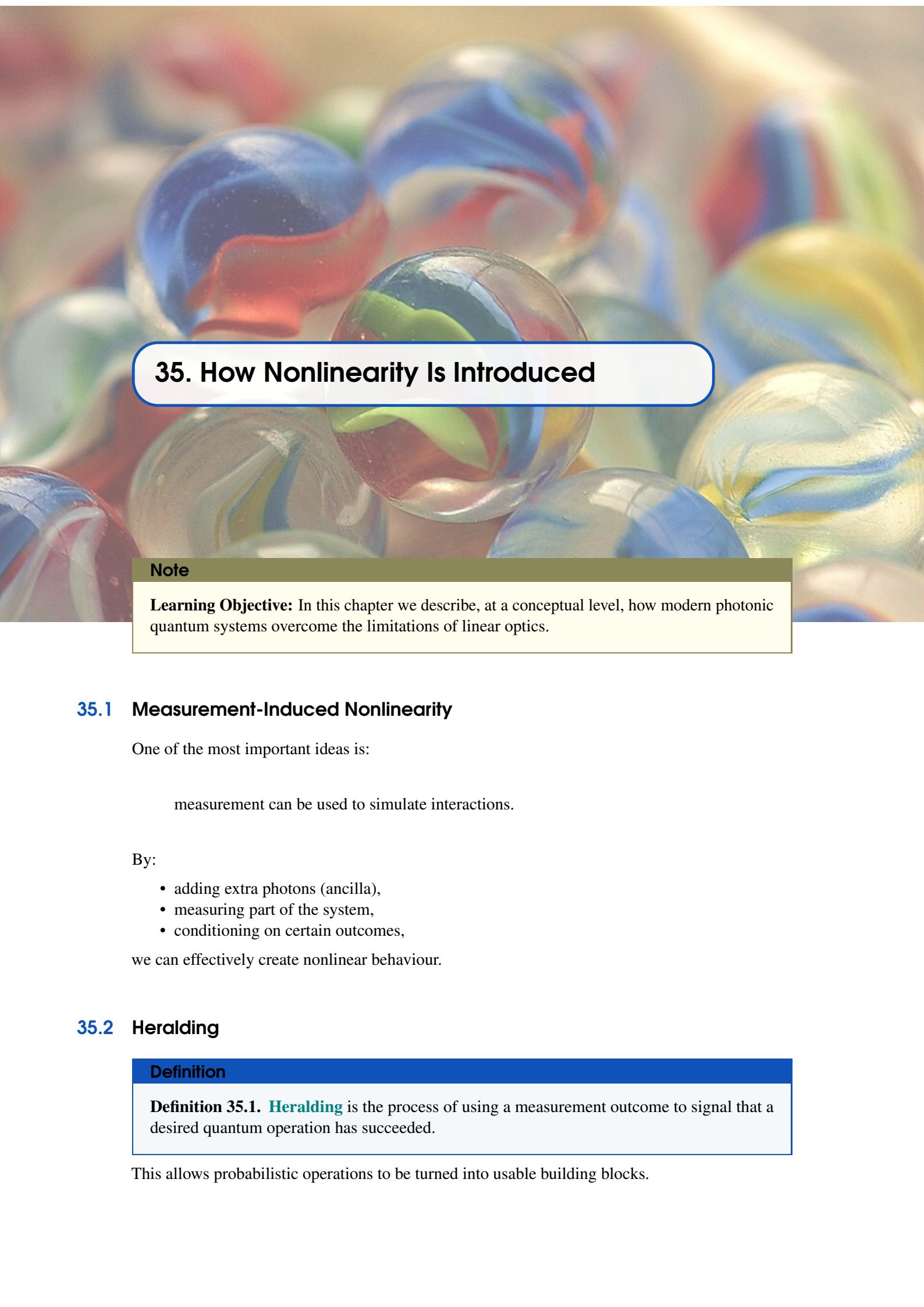
Exercise 34.1. Explain why a beam splitter cannot act as a controlled operation between photons.

Exercise

Exercise 34.2. Why is interaction between particles important for quantum computing?

Summary

Linear optics provides powerful interference effects but lacks the interactions needed to implement general entangling operations required for quantum computation.

A close-up, slightly blurred image of several colorful marbles in shades of blue, red, green, and yellow, creating a vibrant, abstract background.

35. How Nonlinearity Is Introduced

Note

Learning Objective: In this chapter we describe, at a conceptual level, how modern photonic quantum systems overcome the limitations of linear optics.

35.1 Measurement-Induced Nonlinearity

One of the most important ideas is:

measurement can be used to simulate interactions.

By:

- adding extra photons (ancilla),
- measuring part of the system,
- conditioning on certain outcomes,

we can effectively create nonlinear behaviour.

35.2 Heralding

Definition

Definition 35.1. **Heralding** is the process of using a measurement outcome to signal that a desired quantum operation has succeeded.

This allows probabilistic operations to be turned into usable building blocks.

35.3 The KLM Idea

A major breakthrough showed that:

- linear optics + measurement + ancilla
- is sufficient for quantum computation.

This is known as the KLM protocol.

35.4 Other Approaches

Modern photonic systems use:

- integrated photonic circuits,
- fast optical switching,
- time-bin encoding,
- cluster states and measurement-based computation.

35.5 Physical Picture

Instead of direct photon-photon interaction:

- we use interference,
- plus measurement,
- to create effective interactions.

 This is a fundamentally different approach from systems like trapped ions or superconducting qubits, where particles interact directly.

35.6 Exercises

Exercise

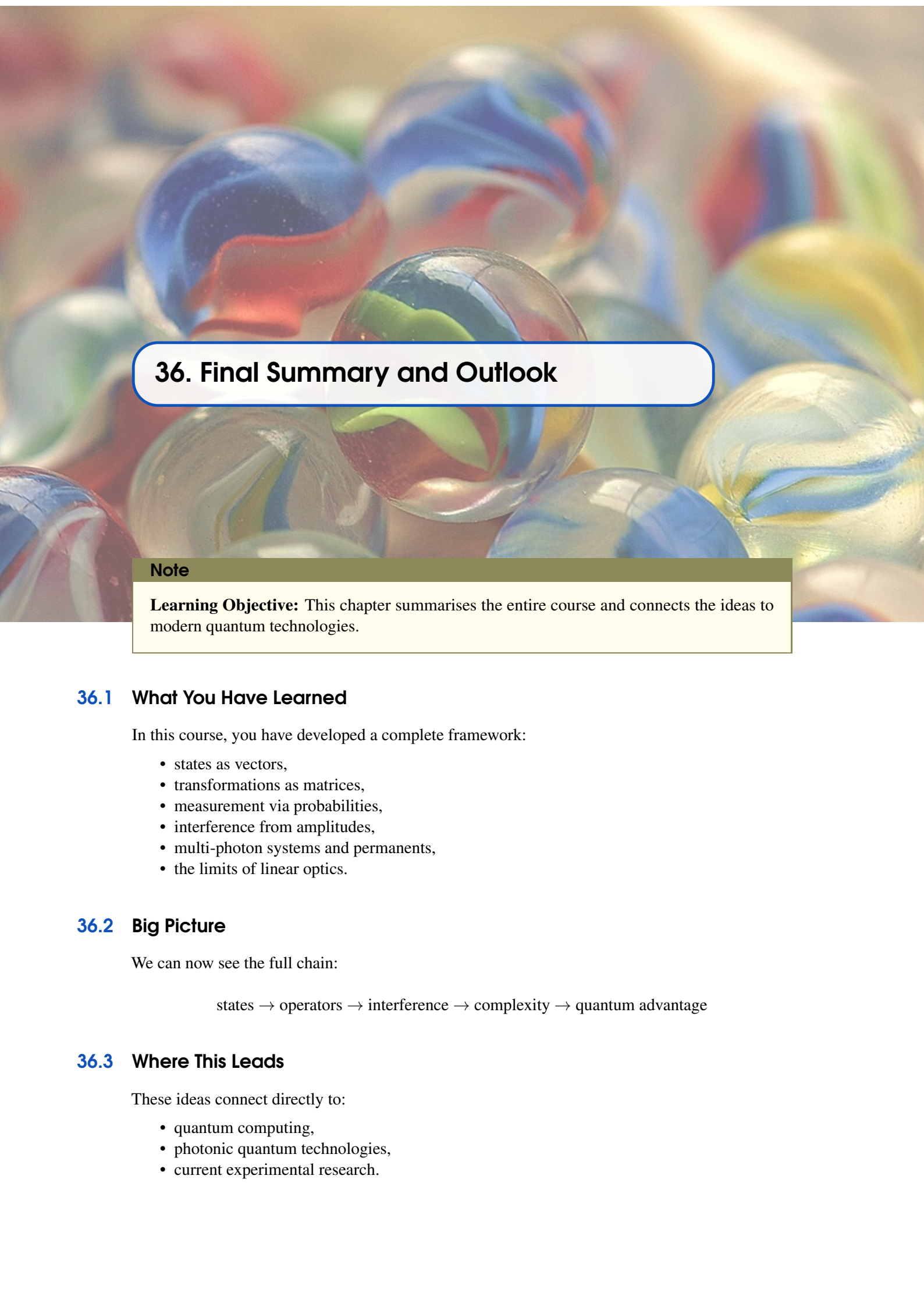
Exercise 35.1. Explain the idea of measurement-induced nonlinearity in your own words.

Exercise

Exercise 35.2. Why is heralding useful in photonic systems?

Summary

Although photons do not interact directly, measurement and additional resources can be used to simulate the nonlinear operations required for quantum computation.

A close-up, shallow depth-of-field photograph of numerous colorful marbles. The marbles are in various shades of blue, red, green, and yellow, with some showing internal patterns or swirls. They are scattered across the frame, with some in sharp focus and others blurred in the background.

36. Final Summary and Outlook

Note

Learning Objective: This chapter summarises the entire course and connects the ideas to modern quantum technologies.

36.1 What You Have Learned

In this course, you have developed a complete framework:

- states as vectors,
- transformations as matrices,
- measurement via probabilities,
- interference from amplitudes,
- multi-photon systems and permanents,
- the limits of linear optics.

36.2 Big Picture

We can now see the full chain:

states $\rightarrow$ operators $\rightarrow$ interference $\rightarrow$ complexity $\rightarrow$ quantum advantage

36.3 Where This Leads

These ideas connect directly to:

- quantum computing,
- photonic quantum technologies,
- current experimental research.

36.4 Final Thought

Quantum mechanics is not just about strange particles — it is a new way of calculating and predicting the behaviour of physical systems.

Summary

This course has shown how linear algebra, combined with physical insight, provides a powerful framework for understanding quantum systems. Photonic systems offer a uniquely intuitive and experimentally relevant way to explore these ideas.